









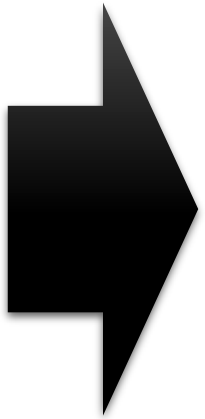
Point Operation

Intensity Transformation

Image Processing (310-3311)


Image in real world
Big, wide and large variety







What is intensity transformation?

OK, Before doing that thing.

We need to answer these questions.

When we need intensity transformation?

What method we should choose?

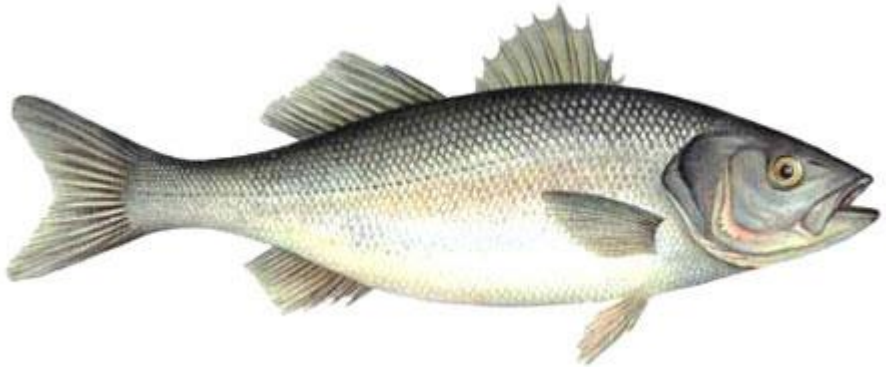
It depend on your problem.

- What **Goal** do you aim for?
- What **Feature** your algorithm used?

It depend on your problem.

- What **Goal** do you aim for?
- What **Feature** your algorithm used?

For example,



Sea Bass (ปลากะพงขาว)

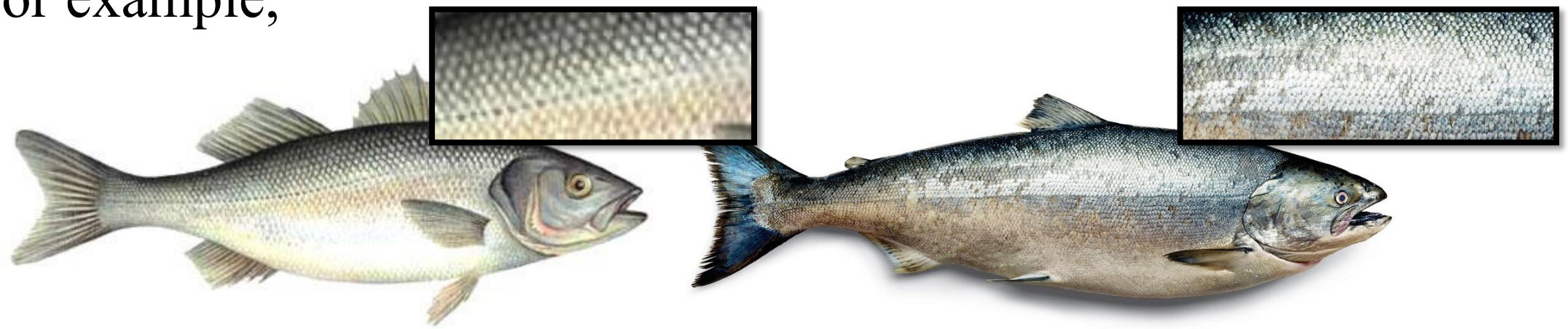


Salmon

It depend on your problem.

- What **Goal** do you aim for?
- What **Feature** your algorithm used?

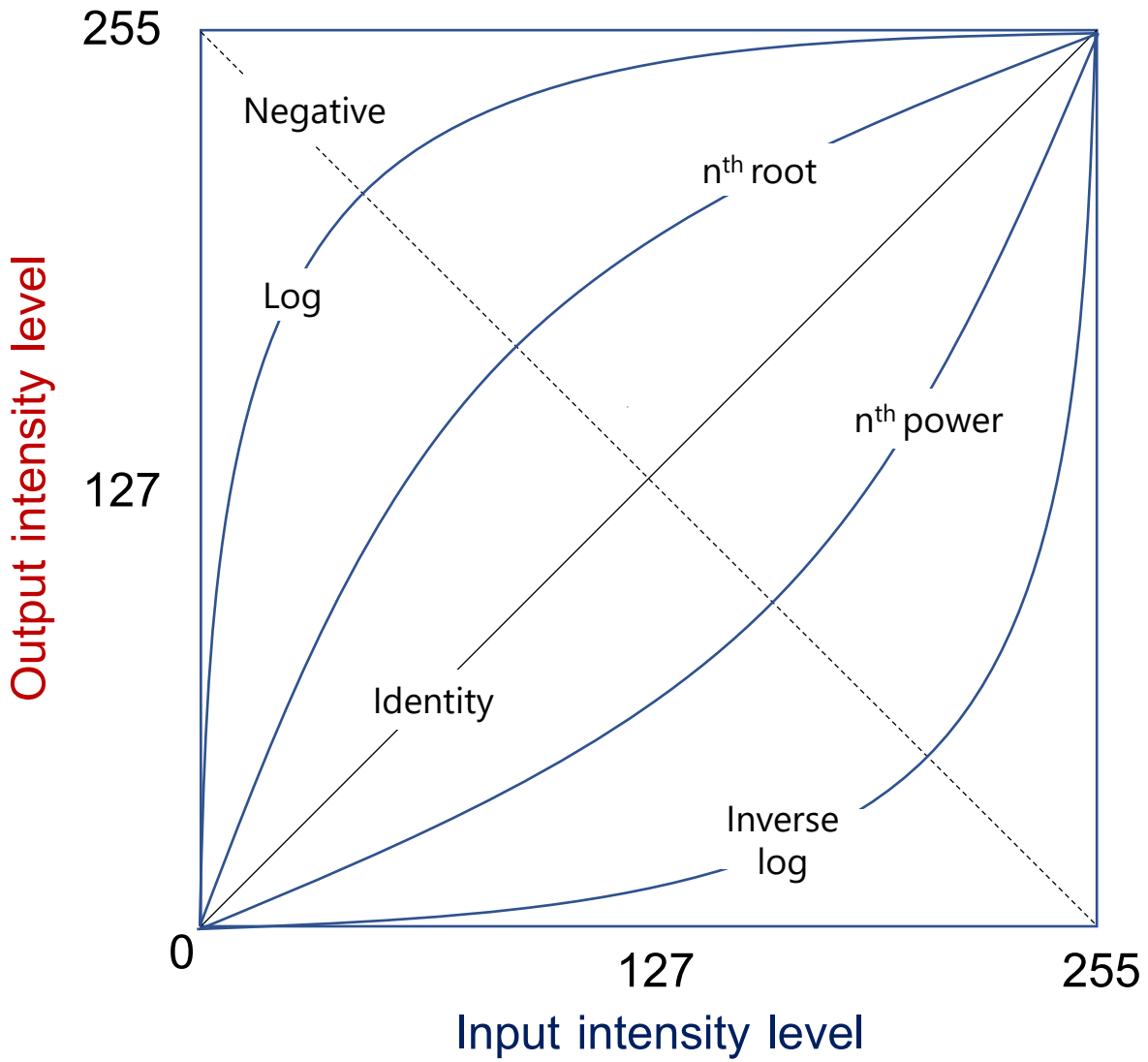
For example,



Sea Bass (ปลากะพงขาว)

Salmon

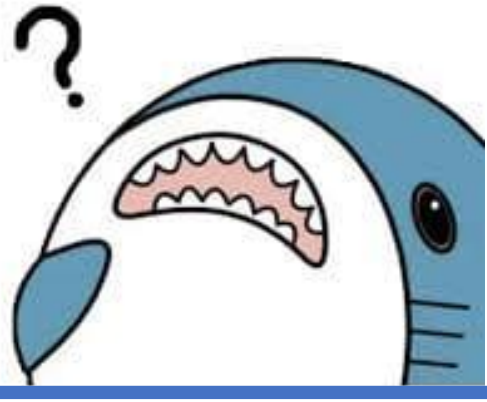
So,... How we do that?



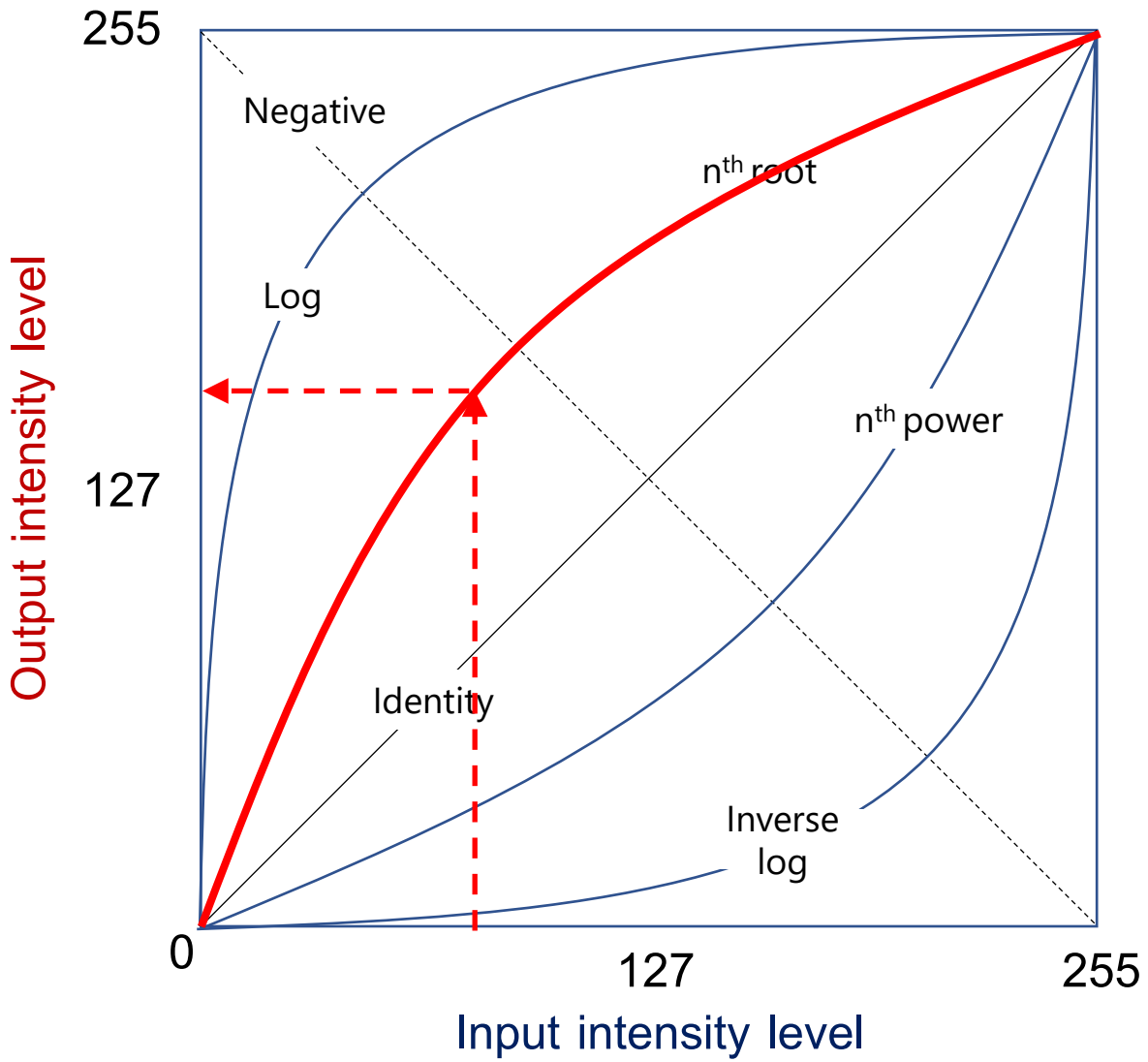
$$g(x,y) = T[f(x,y)]$$

f(x,y) : input image

g(x,y) : output image



So,... How we do that?



Mapping Table

Input	Output
0	0
1	25
2	33
3	56
255	255



After



Before

Demo03 - example1 - DIY1.py - C:\Users\Krisada\Desktop\DIP 2020\Co...

File Edit Format Run Options Window Help

```python
import cv2 as cv
import numpy as np

# Open the image.
img_input = cv.imread('pic\Venice.jpg', cv.IMREAD_GRAYSCALE)

# Apply Intensity transformation
img_output = np.log(img_input)

# Scaling Image intensity to [ 0 - 255 ]
img_max = np.max(img_output)
img_output = img_output * (255 / img_max)

# Specify the data type.
img_output = np.array(img_output, dtype = np.uint8)

# Save the output.
cv.imwrite('Demo03_DIY_img_input.png',img_input)
cv.imwrite('Demo03_DIY_img_output.png', img_output)

```

Ln: 20 Col: 0

ปรับระดับ Intensity Level ของภาพโดยการใช้ Log function

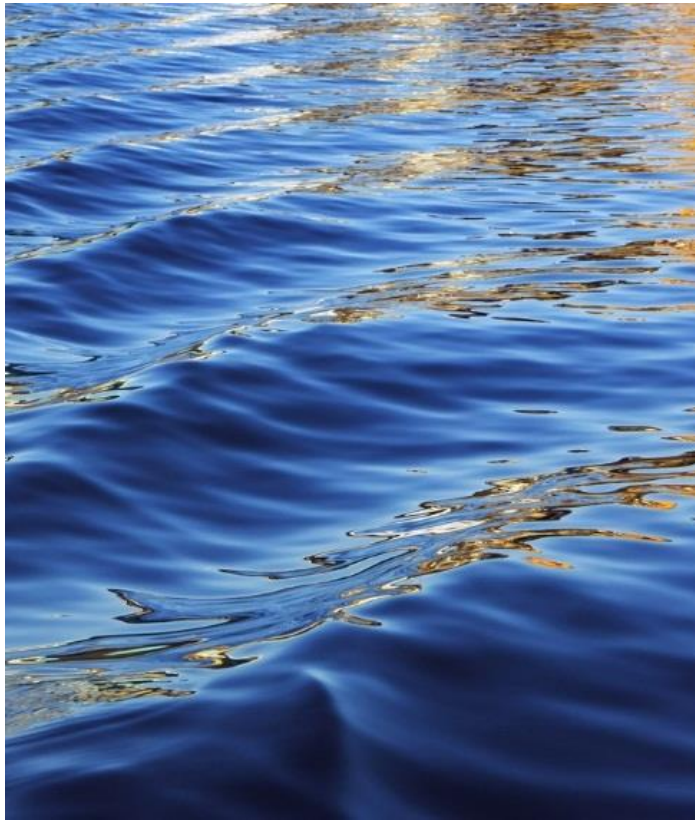
ถ้าไม่เปลี่ยนเป็น uint8 จะเกิดอะไรขึ้น ?

ถ้า data type เป็น float ทำงานได้มั้ย ?



Tools

Intensity Transformation



Power-law Transformation

$$s = cr^{\gamma}$$

where c and γ are positive constants.

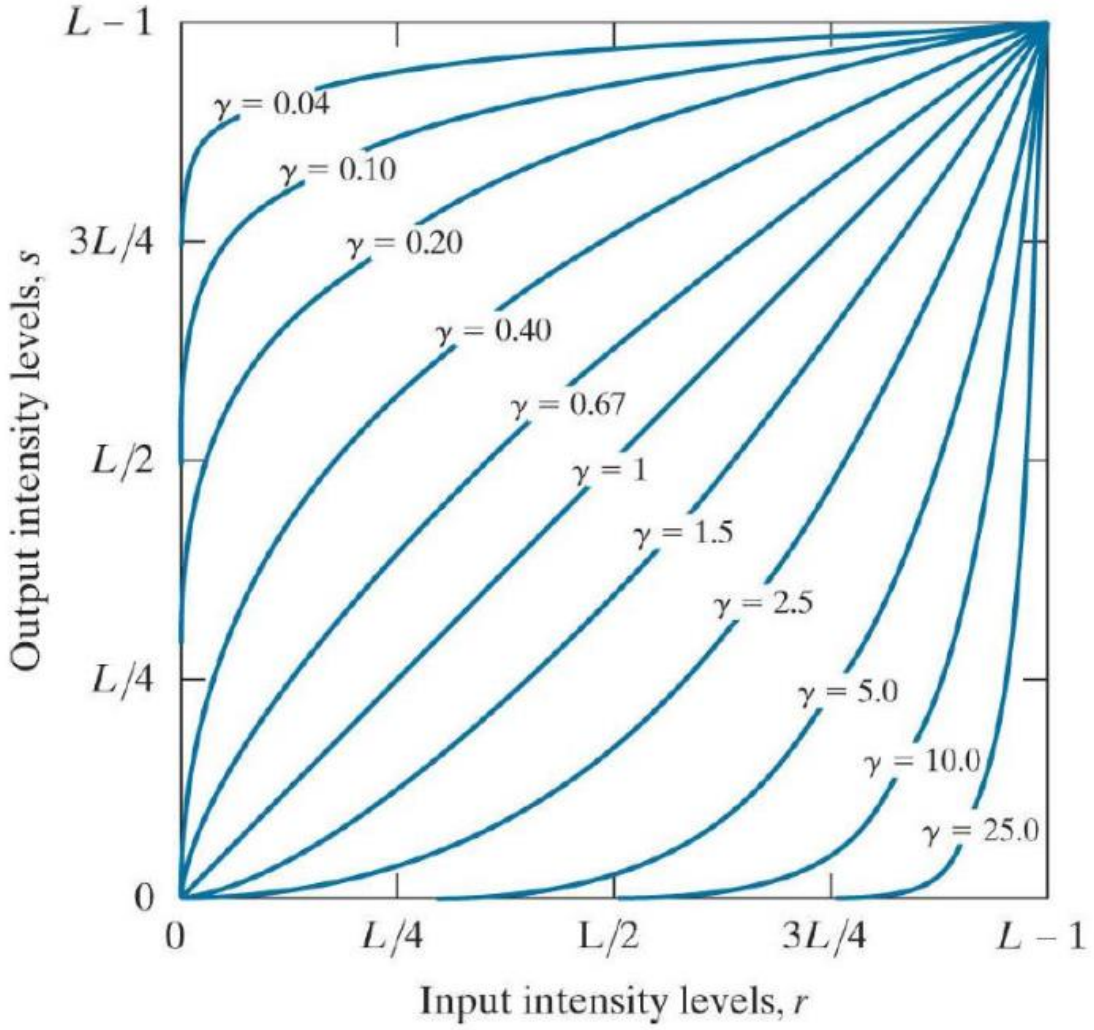
r input intensity levels

s output intensity levels

Sometimes this equation is written as

$$s = c(r + \varepsilon)^{\gamma}$$

to account for offsets (that is, a measurable output when the input is zero).



Power-law Transformation

The response of many devices used for image capture, printing, and display obey a power law.

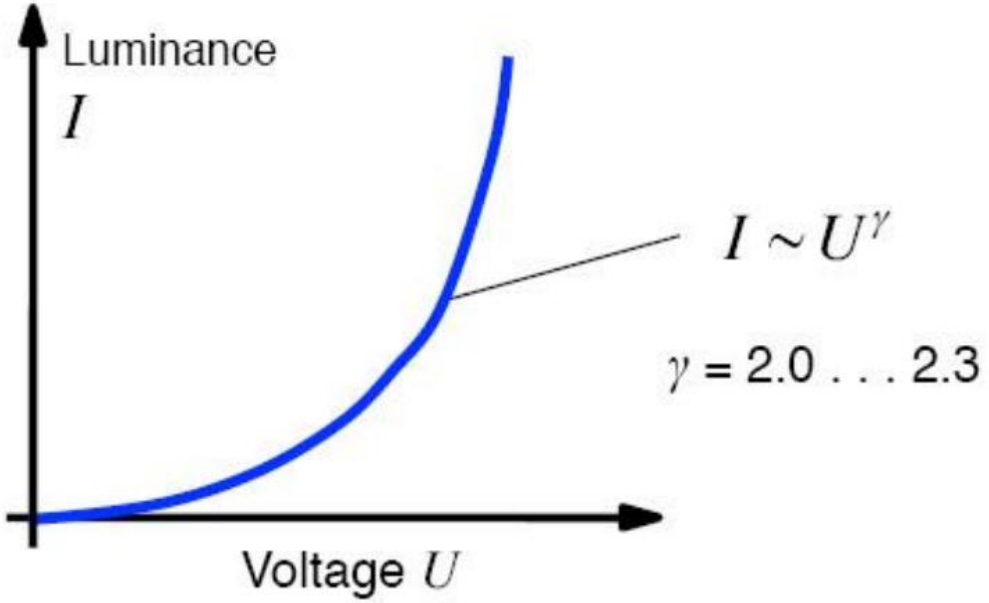
By convention, the exponent in a power-law equation is referred to as gamma (γ).

The process used to correct these power-law response phenomena is called **gamma correction**.





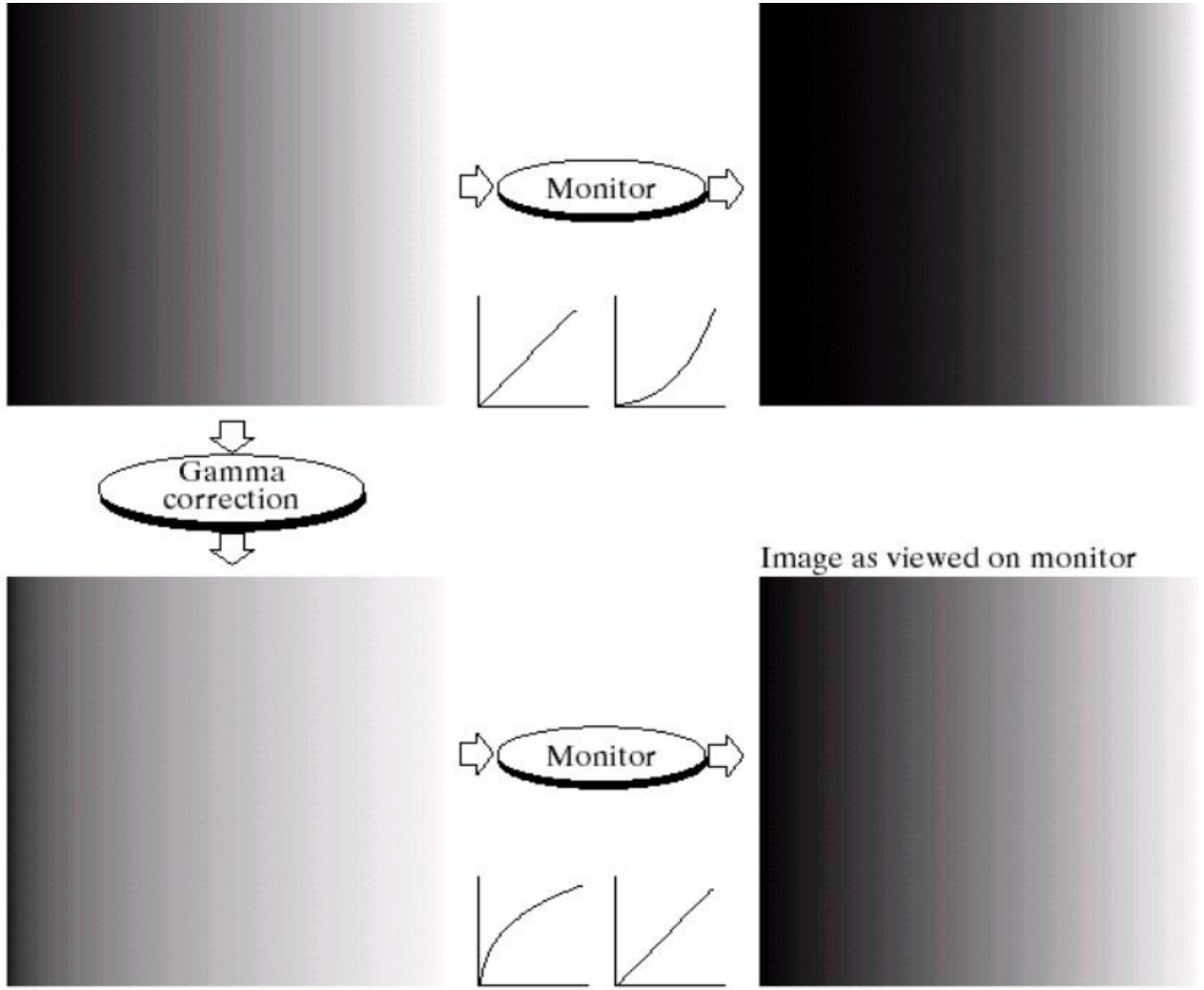
Cathode Ray Tubes (CRTs) are Nonlinear



Power-law Transformation

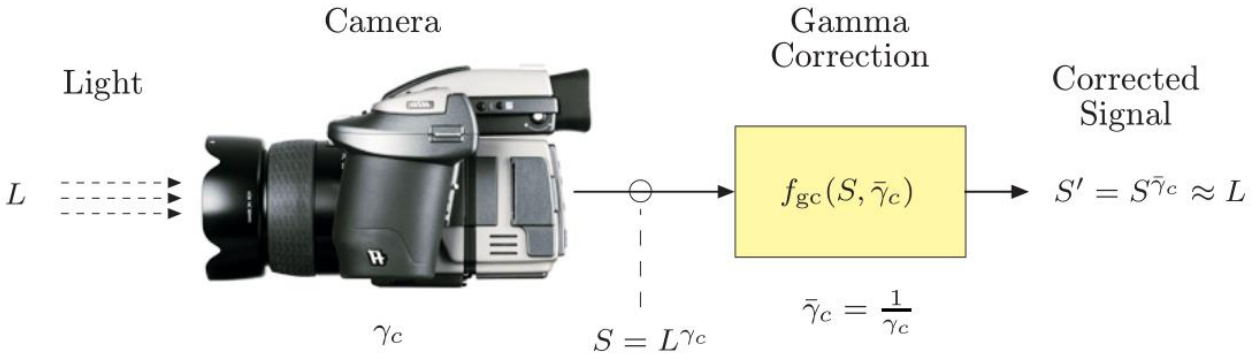


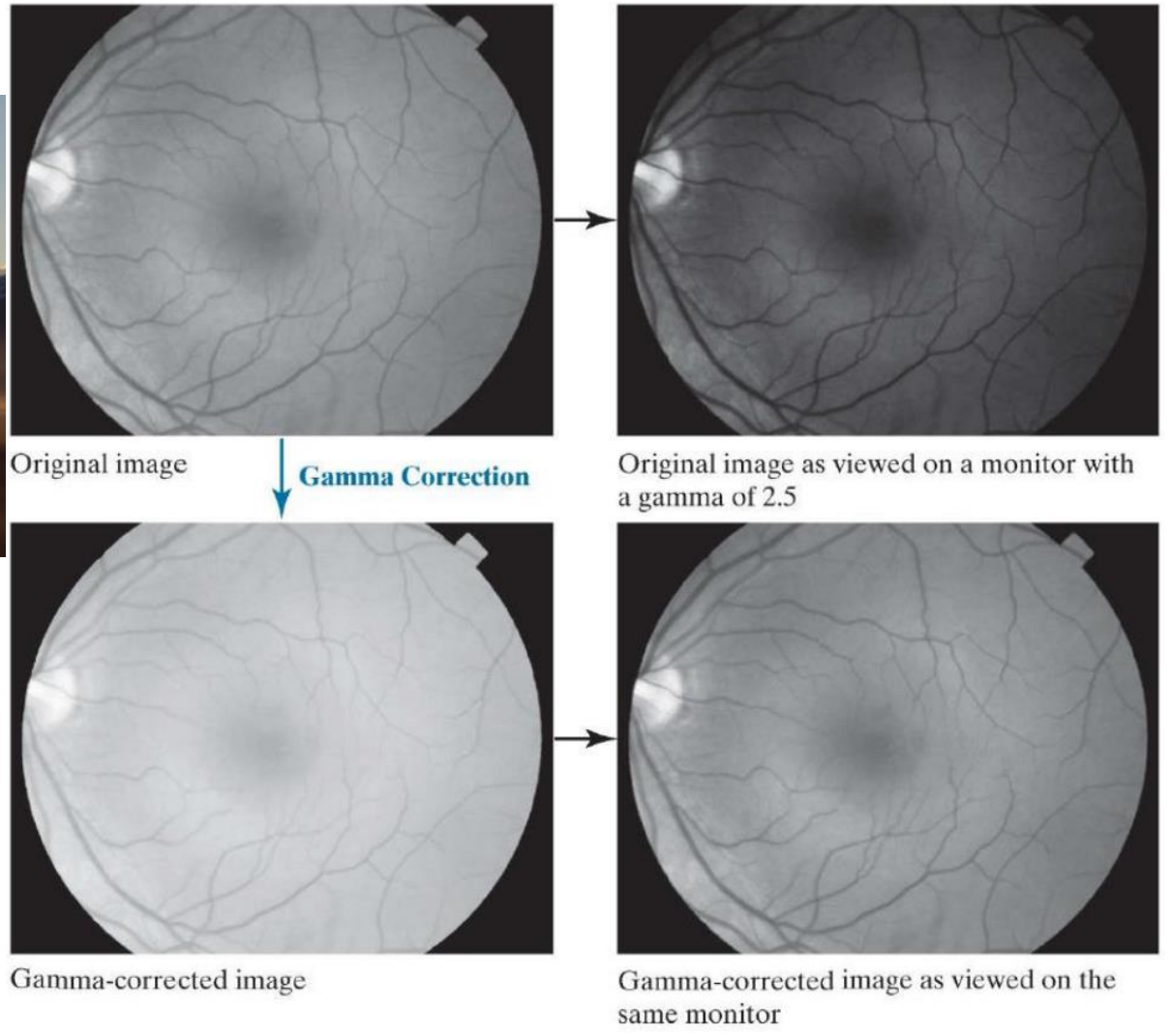
https://www.wikihow.com/Calibrate-Your-Monitor



Power-law Transformation









$\gamma_c = 1.3$

Storage

$\gamma_m = 2.6$



$\bar{\gamma}_c = \frac{1}{1.3}$

$\bar{\gamma}_m = \frac{1}{2.6}$

Processing

$\bar{\gamma}_s = \frac{1}{1.9}$

$\bar{\gamma}_p = \frac{1}{3.0}$



$\gamma_s = 1.9$

$\gamma_p = 3.0$

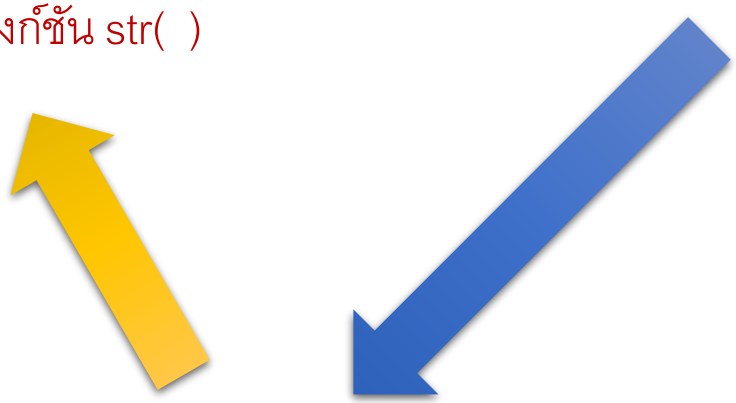




Power-law

Convert [ตัวเลข] ➔ [ตัวอักษร]

แนะนำให้ใช้ฟังก์ชั่น str( )



ถ้าสามารถทำได้แล้วให้ทดลองเขียน

For loop ค่า Gamma เพื่อ Save

ผลลัพธ์หลายๆ ภาพตามค่า Gamma

Demo03 - example2 - PowerLaw.py - C:\Users\Krisada\Desktop\...

File Edit Format Run Options Window Help

```python
import cv2 as cv
import numpy as np

# Open the image.
img_in = cv.imread('pic\Venice.jpg', cv.IMREAD_GRAYSCALE)

# Apply Intensity transformation
gamma = 2
gamma_corrected = (img_in / 255)**gamma

# Scaling output
gamma_corrected = gamma_corrected*255

# Convert data type
img_out = np.array(gamma_corrected, dtype = 'uint8')

# Show image
cv.imshow('Power-law', img_out)

# Save the output.
cv.imwrite('Demo03_Pow_img_input.png',img_in)
cv.imwrite('Demo03_Pow_img_output.png', img_out)

```

Ln: 23 Col: 0

Tran





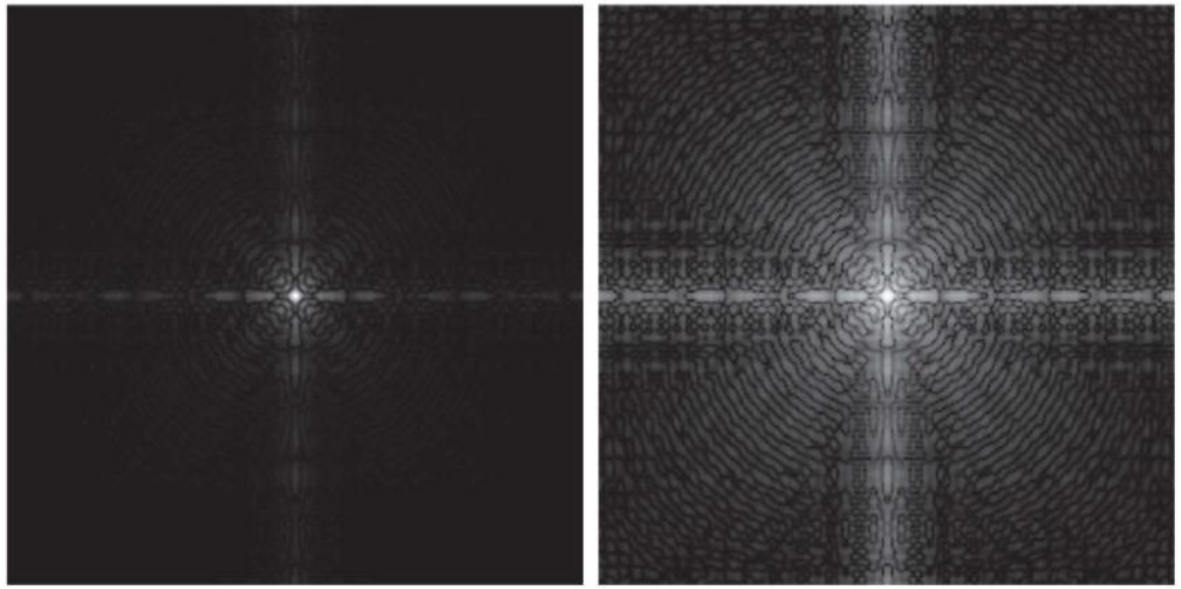




Logarithmic Transformations

$$s = c\log(1+r)$$

where c is a constant, and $r \geq 0$.



Fourier spectrum display

Demo03 - example3 - Log.py - C:\Users\Krisada\Desktop\DIP 2020...

File Edit Format Run Options Window Help

```
1  import cv2 as cv
2  import numpy as np
3
4  # Read an image
5  image = cv.imread('pic\Venice.jpg', cv.IMREAD_GRAYSCALE)
6
7  # Apply log transformation method
8  c = 255 / np.log(1 + np.max(image))
9  log_image = c * (np.log(1+ image))
10
11 # Specify the data type so that
12 # float value will be converted to int
13 log_image = np.array(log_image, dtype = np.uint8)
14
15 # Save images
16 cv.imwrite("Demo03_Log_output.png", log_image)
17
```

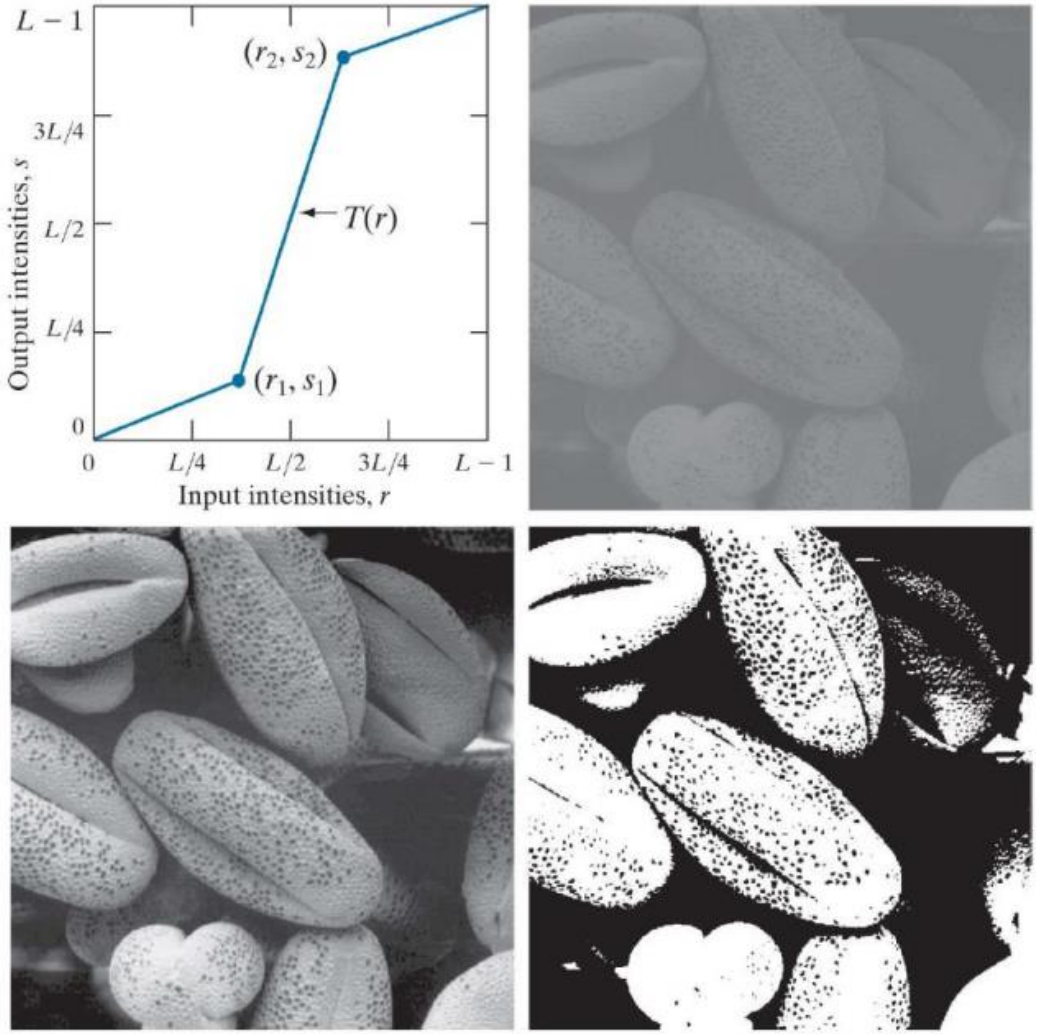
Ln: 17 Col: 0

Piecewise-Linear Transformation

Contrast stretching

Contrast can be defined as:

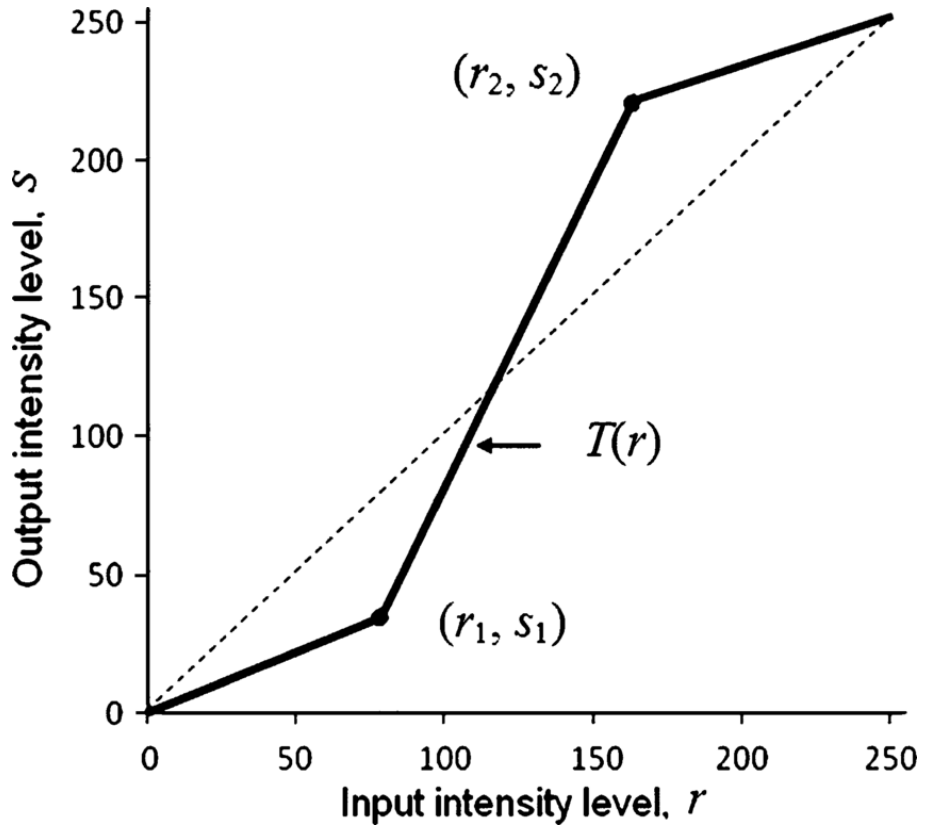
$$\text{contrast} = \frac{(I_{\max} - I_{\min})}{(I_{\max} + I_{\min})}$$





Piecewise-Lin

Contrast stretching




```python
import cv2
import numpy as np

# Function to map each intensity level to output intensity level.
def pixelVal(pix, r1, s1, r2, s2):
    if (0 <= pix and pix <= r1):
        return (s1 / r1)*pix
    elif (r1 < pix and pix <= r2):
        return ((s2 - s1)/(r2 - r1)) * (pix - r1) + s1
    else:
        return ((255 - s2)/(255 - r2)) * (pix - r2) + s2

# Open the image.
img = cv2.imread('sample.jpg')

# Define parameters.
r1 = 70
s1 = 0
r2 = 140
s2 = 255

# Vectorize the function to apply it to each value in the Numpy array.
pixelVal_vec = np.vectorize(pixelVal)

# Apply contrast stretching.
contrast_stretched = pixelVal_vec(img, r1, s1, r2, s2)
```

Piecewise-Linear Transformation

Contrast stretching





From previous slide…

When we need intensity transformation?

How can we know that this image need
Intensity Transformation or not?

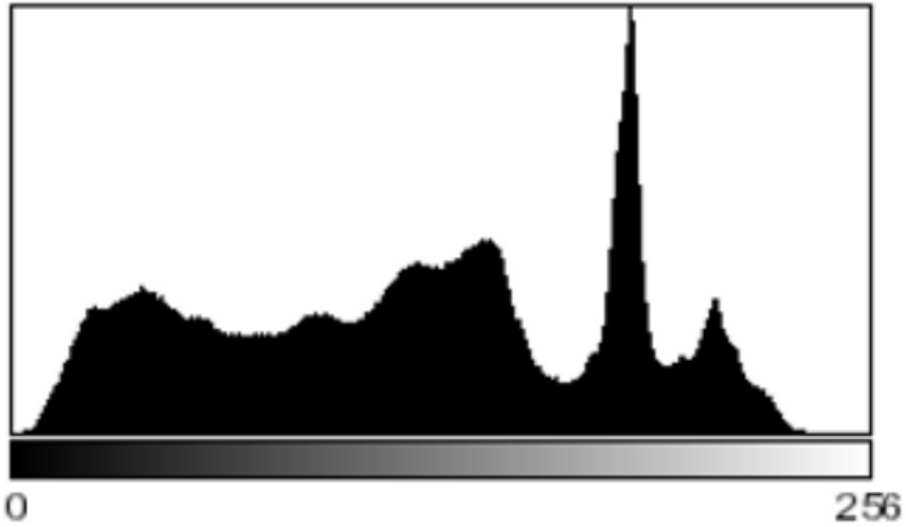


We know, this image too dark right?
But how can we tell computer?

Let's analysis image. Is it need to adjust intensity or not?

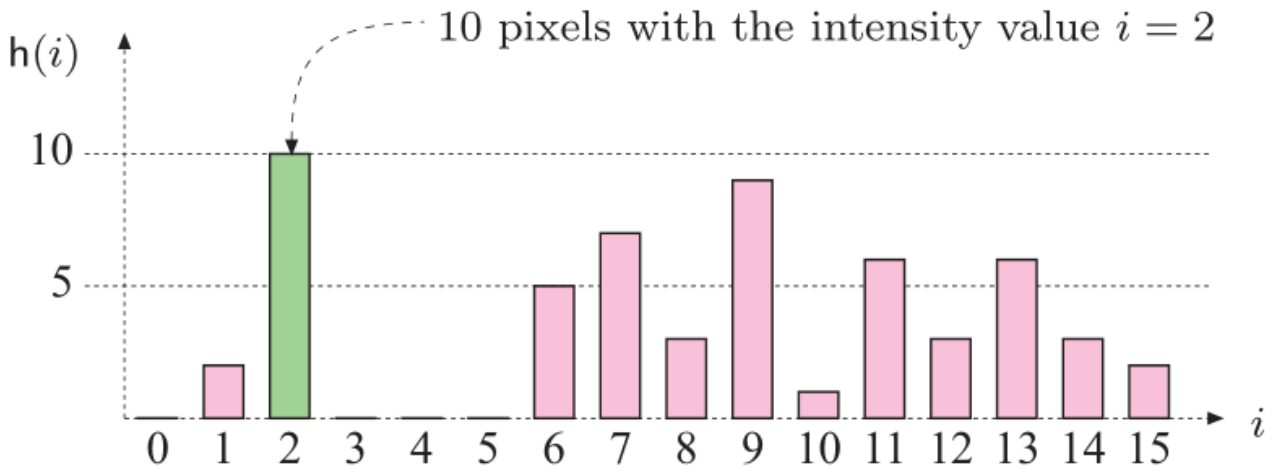
We will begin with a basic tool. **Histogram**





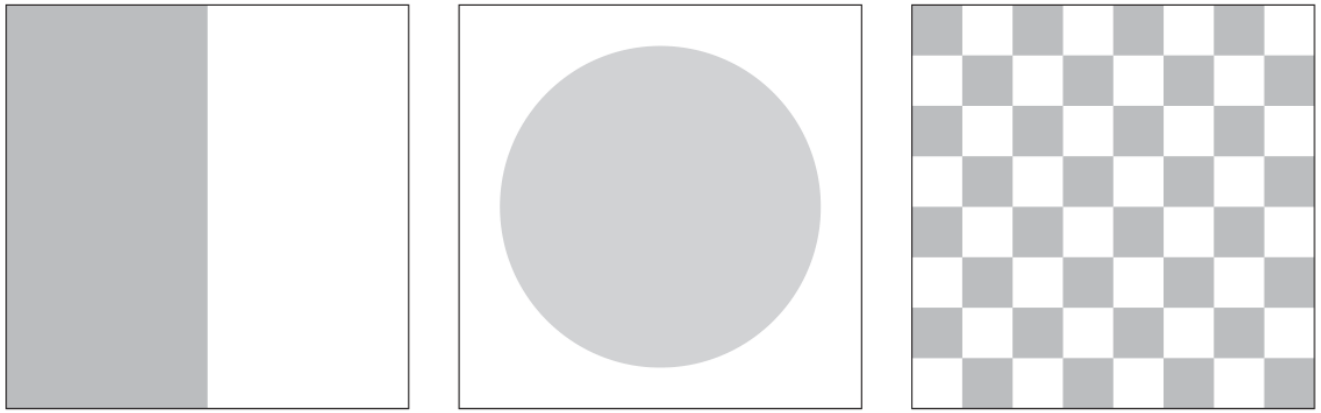
It is simple to conclude if an image is properly exposed by visual inspection of its **histogram**.

$$\mathsf{h}(i) = \text{ the } \textit{number} \text{ of pixels in } I \text{ with the intensity value } i,$$

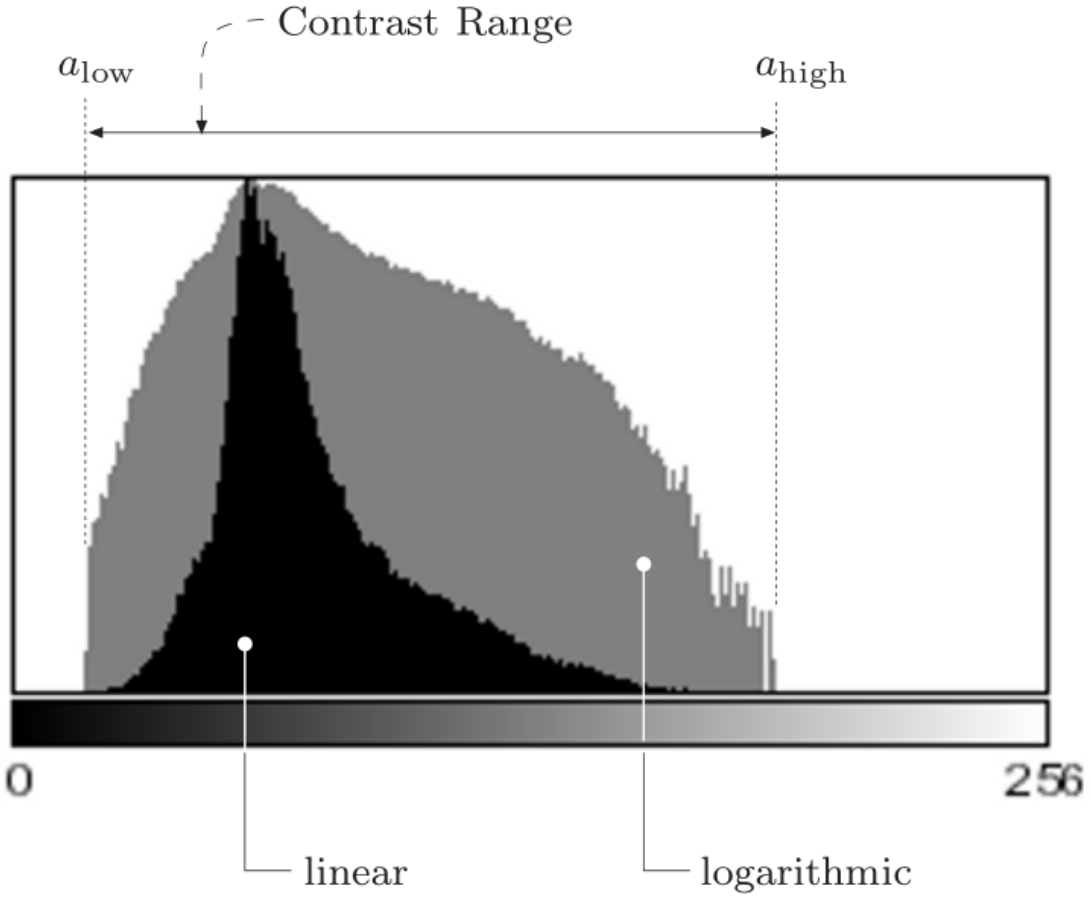


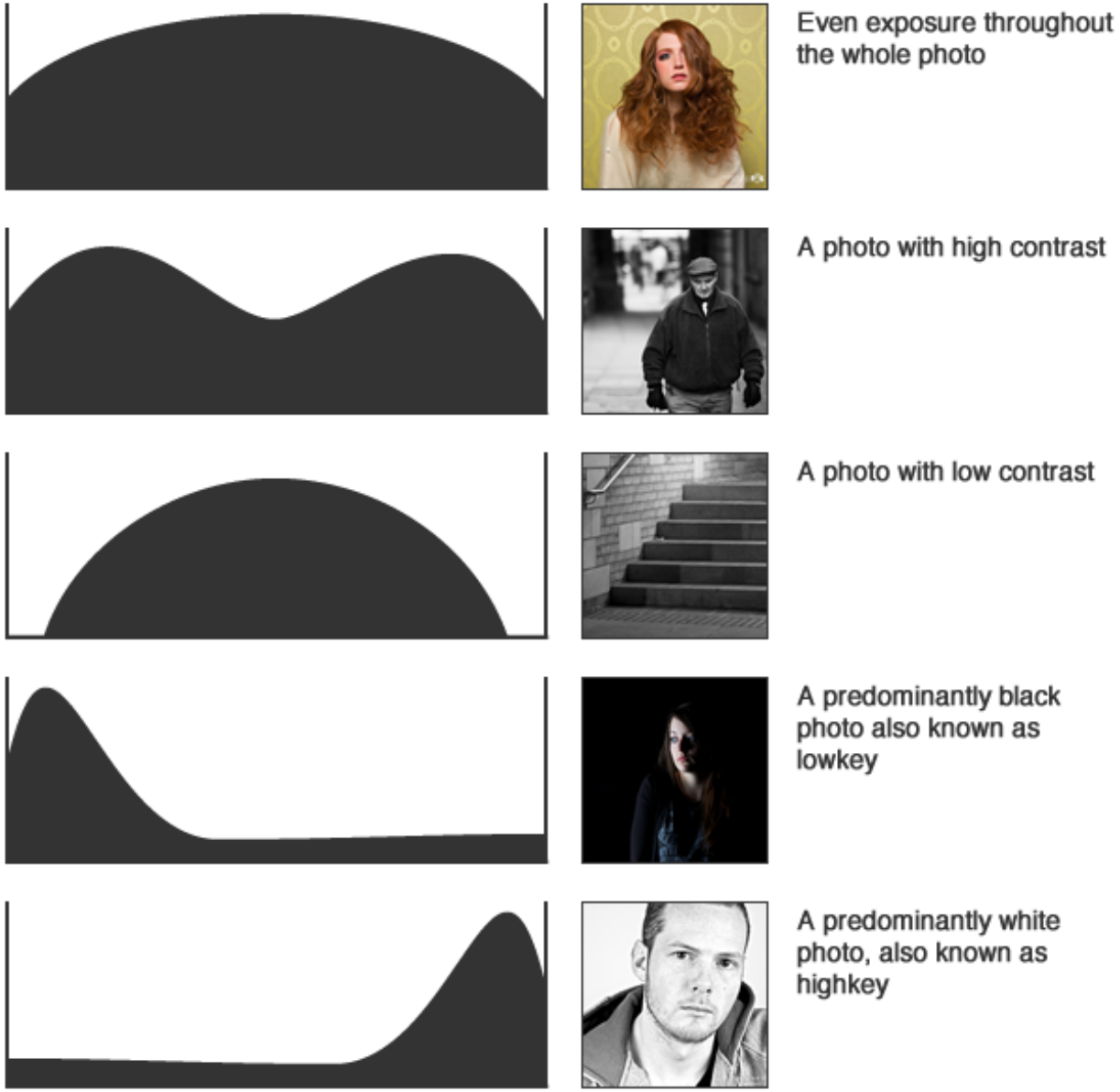
$h(i)$	0	2	10	0	0	0	5	7	3	9	1	6	3	6	3	2
i	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15

Three very different images with identical histograms

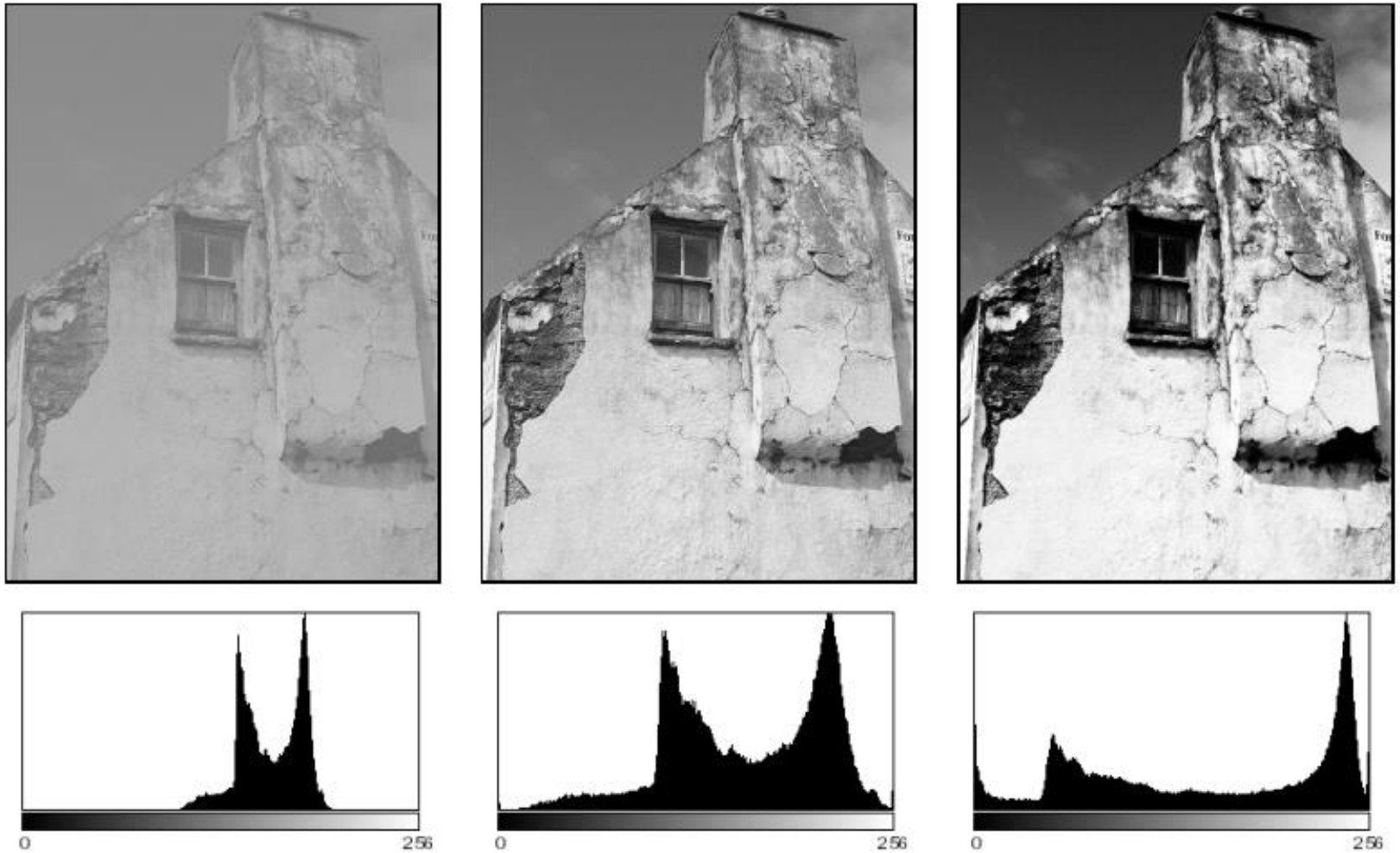


Interpreting Histograms





Interpreting Histograms



It's 1D Signal, So…

- Mean
- Variance
- Skewness
- Kurtosis
- etc.,.

Demo04_01_Histogram0.py - C:\Users\Krisada\Desktop\DIP 2020\C...

File Edit Format Run Options Window Help

```python
import numpy as np
import cv2 as cv

import matplotlib.pyplot as plt

img = cv.imread('sample.jpg')

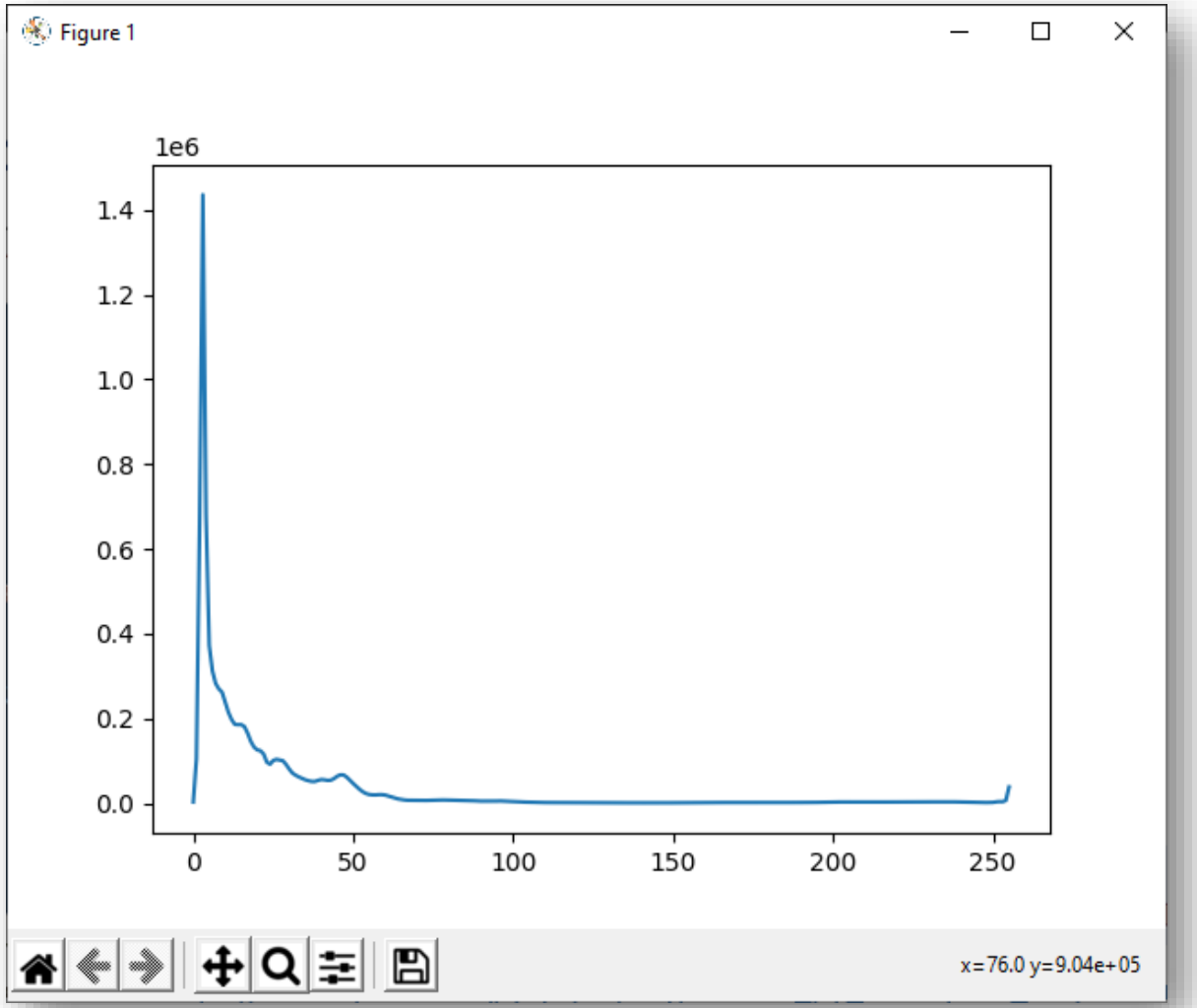
# Convert color image to gray-scale image
img_gray = cv.cvtColor(img, cv.COLOR_BGR2GRAY)

# Calculate histogram of image
histogram = cv.calcHist([img_gray], [0], None, [256], [0, 256])

# Plot histogram with matplotlib
plt.plot(histogram)
plt.show()

```

Ln: 1 Col: 0



Reference: https://docs.opencv.org/4.6.0/d1/db7/tutorial_py_histogram_begins.html

Demo04_01_Histogram1.py - C:\Users\Krisada\Desktop\DIP 2020\...

File Edit Format Run Options Window Help

```python
import numpy as np
import cv2 as cv

import matplotlib.pyplot as plt

img = cv.imread('sample.jpg')

img_gray = cv.cvtColor(img, cv.COLOR_BGR2GRAY)

histogram = cv.calcHist([img_gray], [0], None, [256], [0, 256])

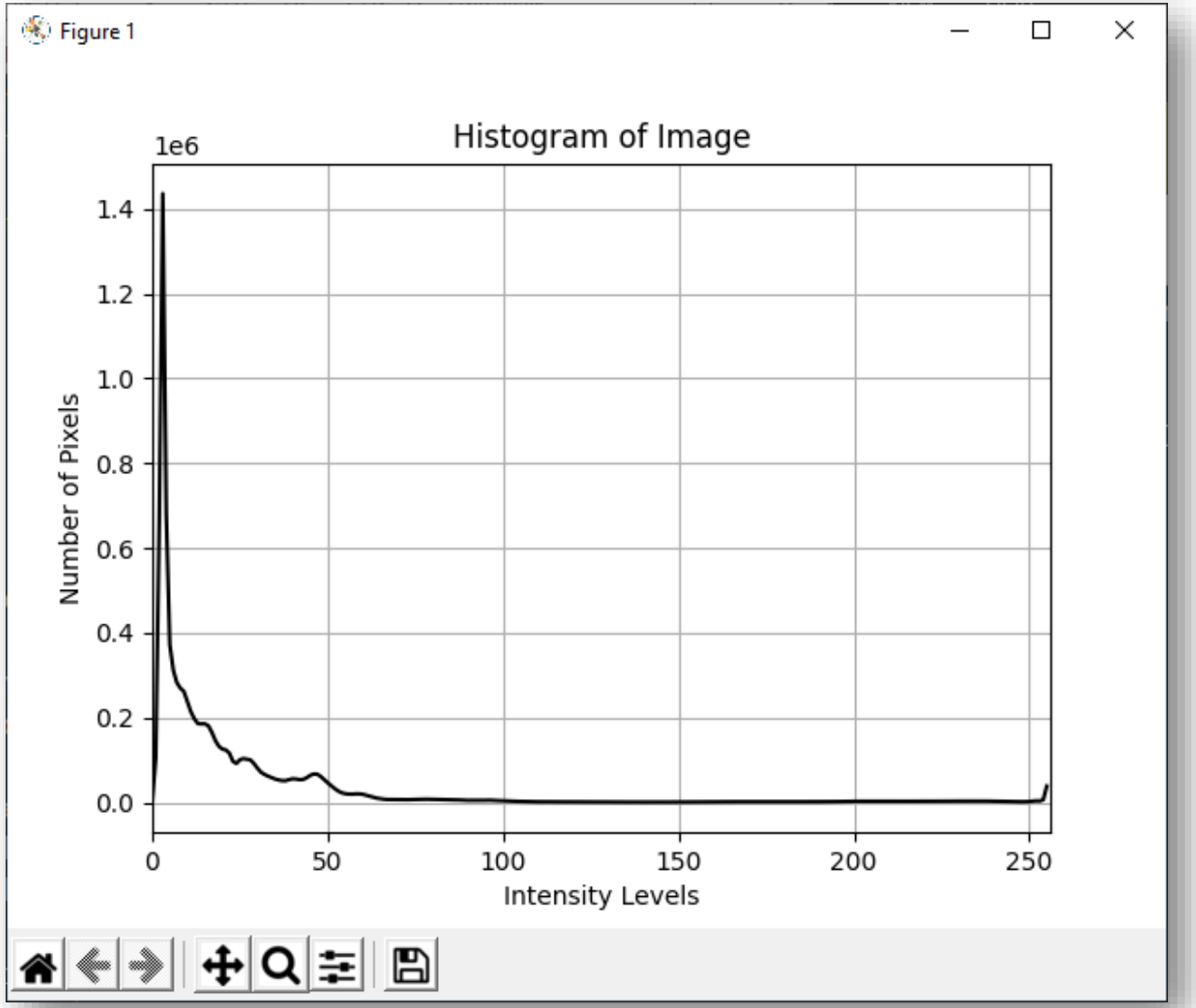
# Plot histogram with matplotlib
plt.plot(histogram, color='k')

plt.xlim([0, 256])
plt.grid(True)

plt.title('Histogram of Image')
plt.ylabel('Number of Pixels')
plt.xlabel('Intensity Levels')

plt.show()
```

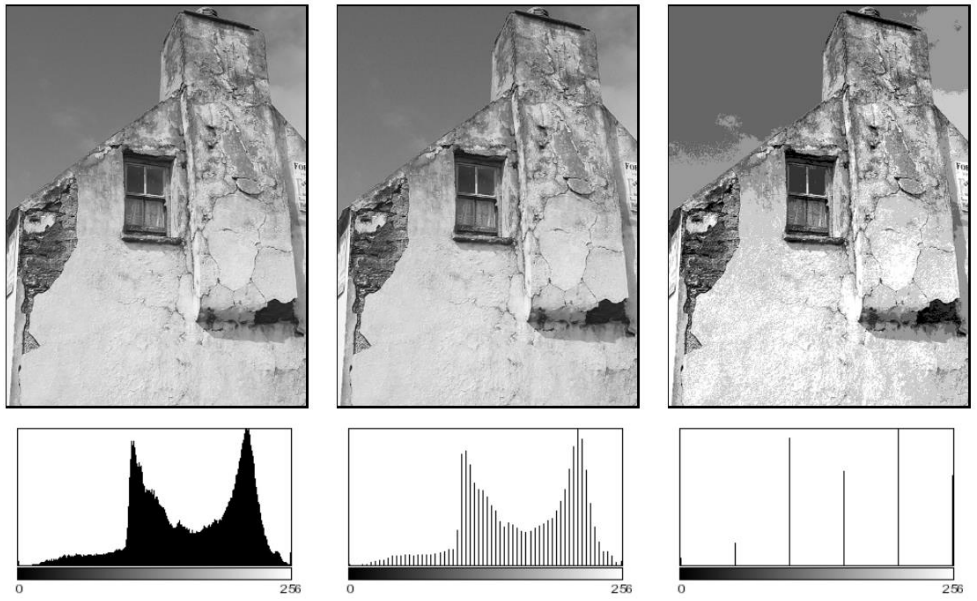
Ln: 23 Col: 0



แนะนำให้ทดลองปรับรูปแบบกราฟ ใส่ label, grid, ฯลฯ

Dynamic Range

The dynamic range of an image is, in principle, understood as the number of distinct pixel values in an image. In the ideal case, the dynamic range encompasses all K usable pixel values, in which case the value range is completely utilized.



High dynamic range

Low dynamic range

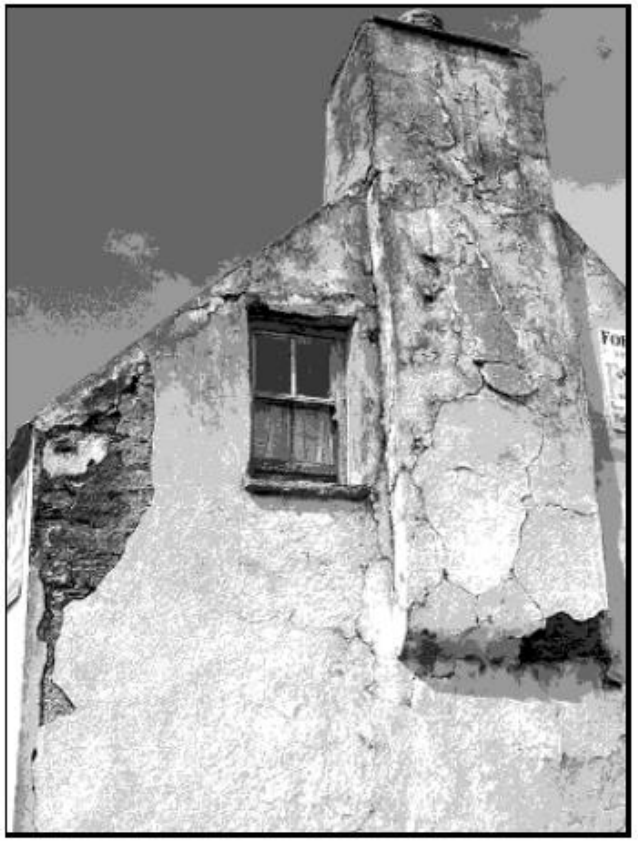


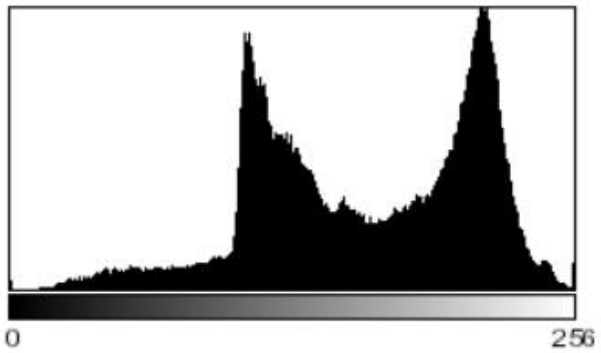
Dyn

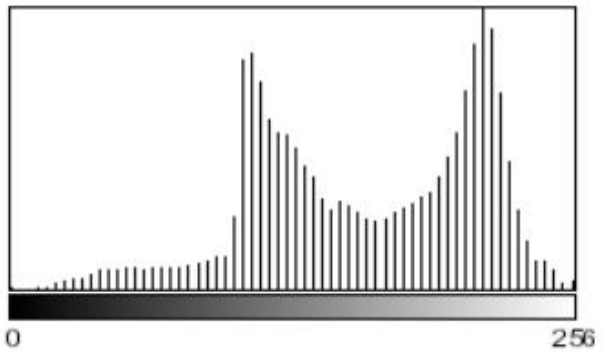
Th in
ar se
th

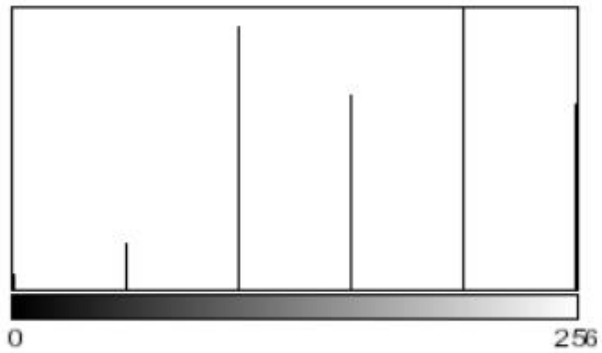














Modifying Image Intensity

Contrast and Brightness

$$f_{\text{contr}}(a) = a \cdot 1.5$$

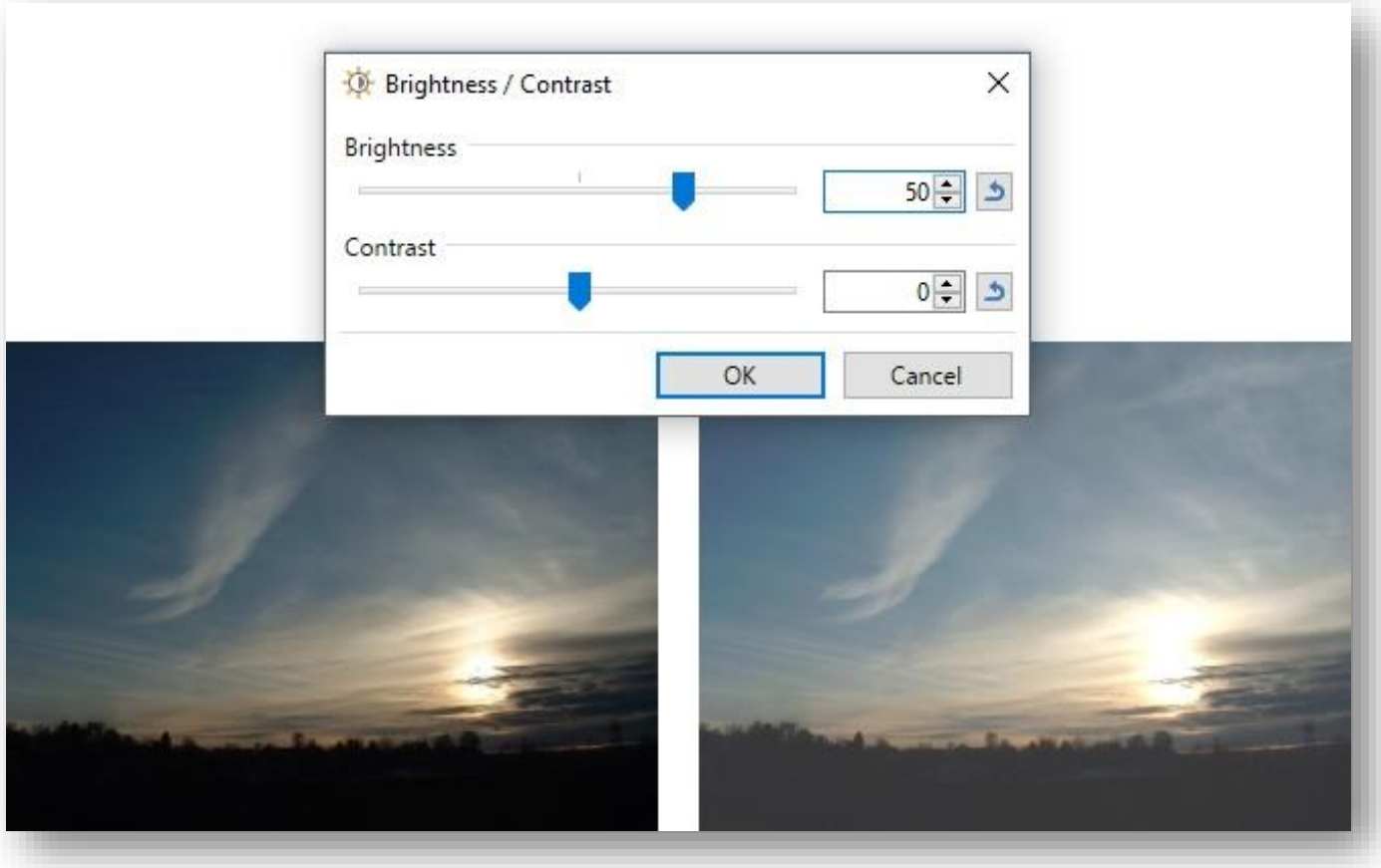
$$f_{\text{bright}}(a) = a + 10$$

Limiting the Results by Clamping

```
if (a > 255)  a = 255;
if (a < 0)  a = 0;
```

Automatic Contrast Adjustment

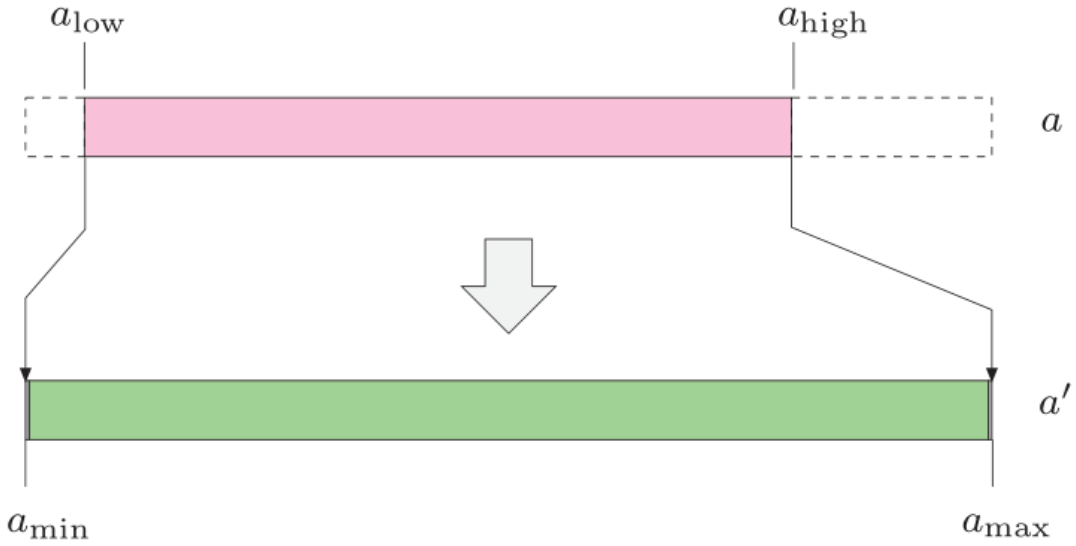
$$f_{\text{ac}}(a) = a_{\min} + (a - a_{\text{low}}) \cdot \frac{a_{\max} - a_{\min}}{a_{\text{high}} - a_{\text{low}}}$$





Modifying Image Intensity

Contrast and Brightness







Try to plot and see Histogram by yourself :)

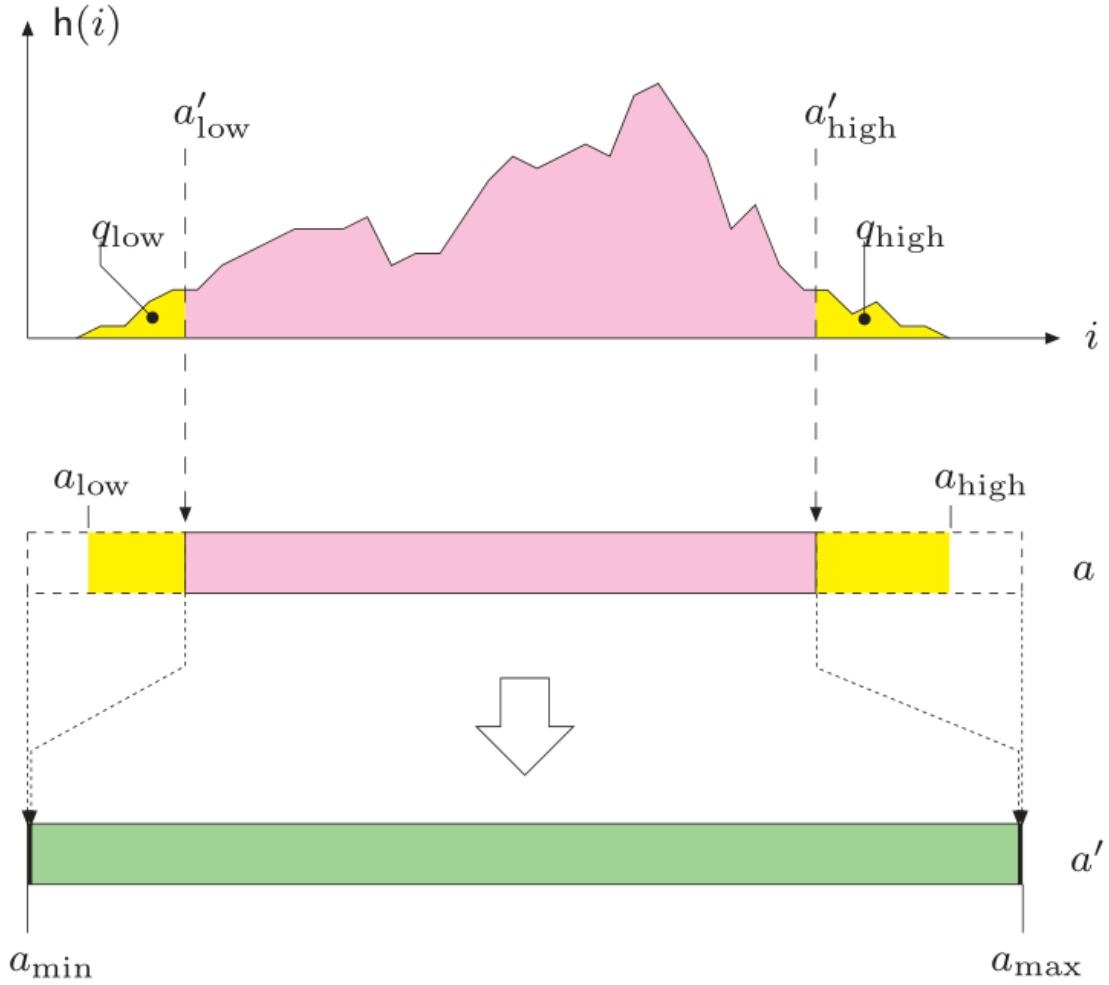


Modifying Image Intensity

Contrast and Brightness

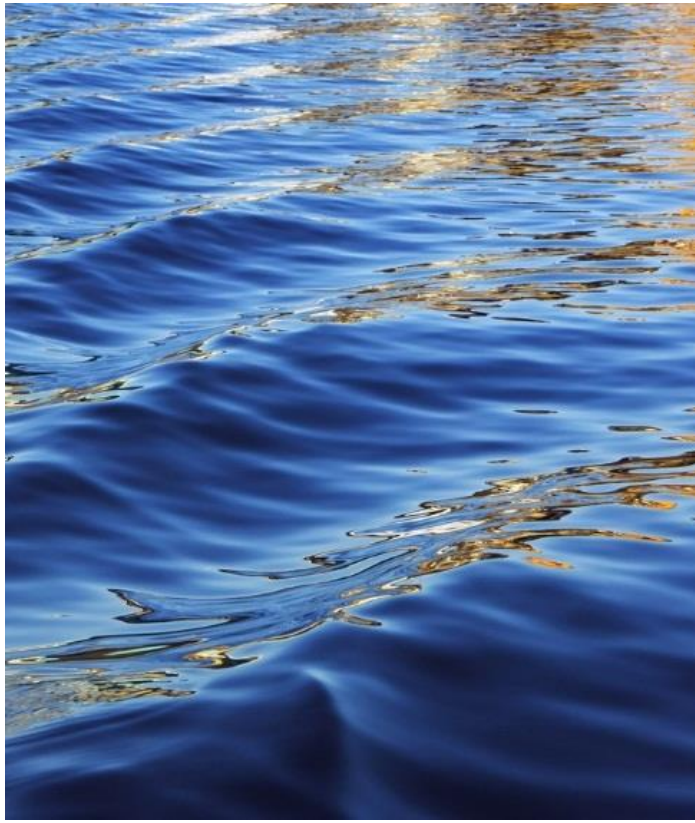
Modified Automatic Contrast Adjustment

$$f_{\text{mac}}(a) = \begin{cases} a_{\min} & \text{for } a \le a'_{\text{low}} \\ a_{\min} + \left(a - a'_{\text{low}}\right) \cdot \dfrac{a_{\max} - a_{\min}}{a'_{\text{high}} - a'_{\text{low}}} & \text{for } a'_{\text{low}} < a < a'_{\text{high}} \\ a_{\max} & \text{for } a \ge a'_{\text{high}}. \end{cases}$$





Take a break

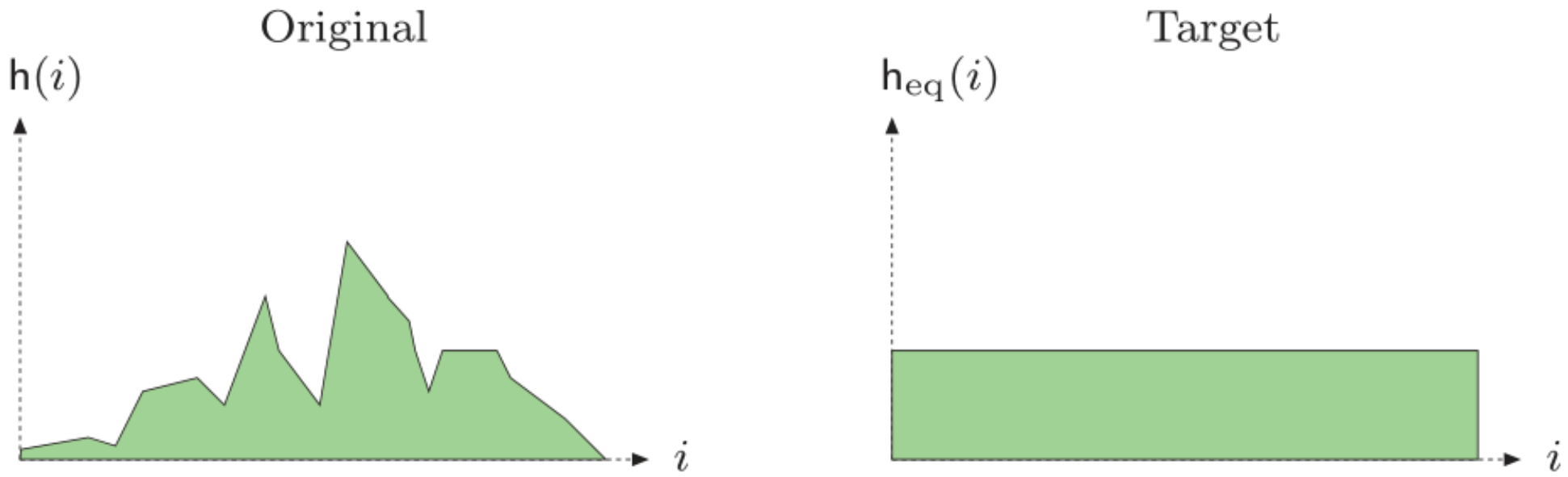




Modifying Image Intensity

Histogram Equalization

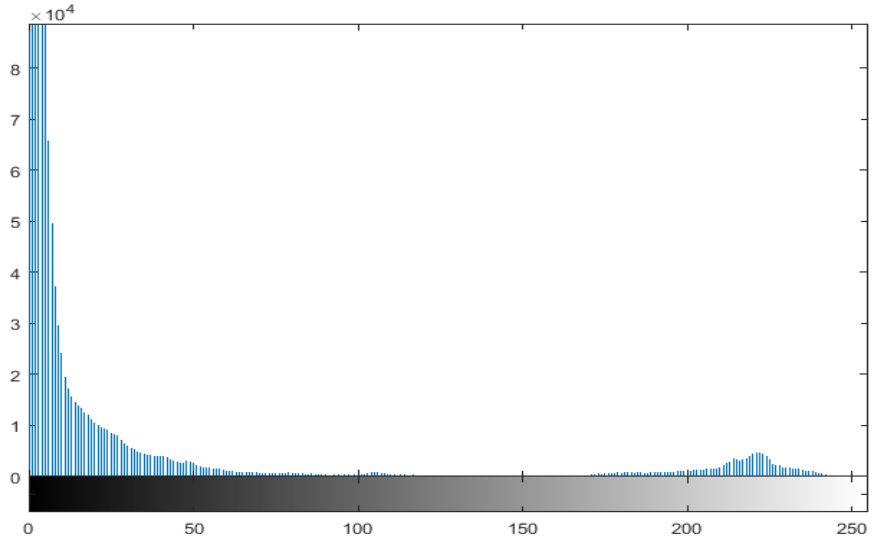
A frequent task is to adjust two different images in such a way that their resulting intensity distributions are similar, for example to use them in a print publication or to make them easier to compare.

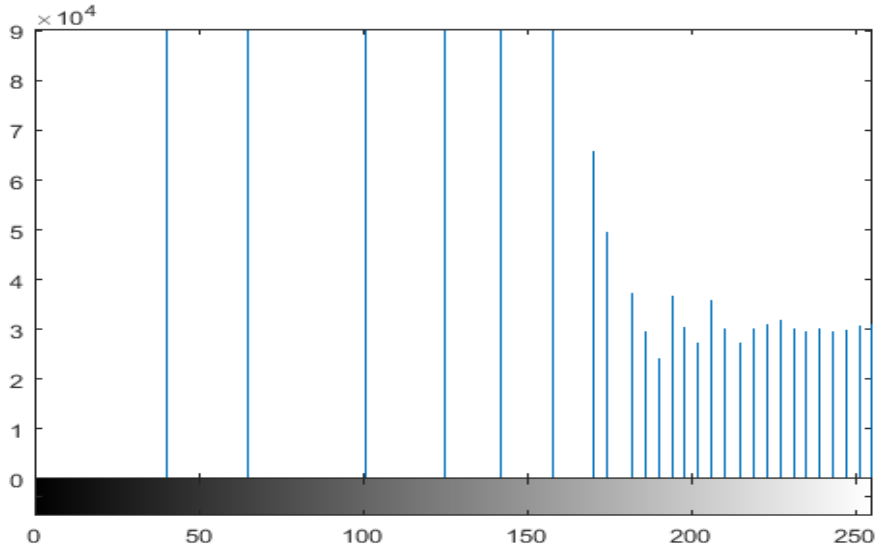


Basically, we might want to spread the intensity equally for each value.











Modifying Image Intensity

Histogram Equalization

Demo04_02_HistogramEQ.py - C:\Users\Krisada\Desktop\...

File Edit Format Run Options Window Help

```
1  import numpy as np
2  import cv2 as cv
3
4  img = cv.imread('Venice.jpg', cv.IMREAD_GRAYSCALE)
5
6  img_out = cv.equalizeHist(img)
7
8  cv.imwrite('HistEQ_Before.png', img)
9  cv.imwrite('HistEQ_After.png', img_out)
10
```

Ln: 10 Col: 0

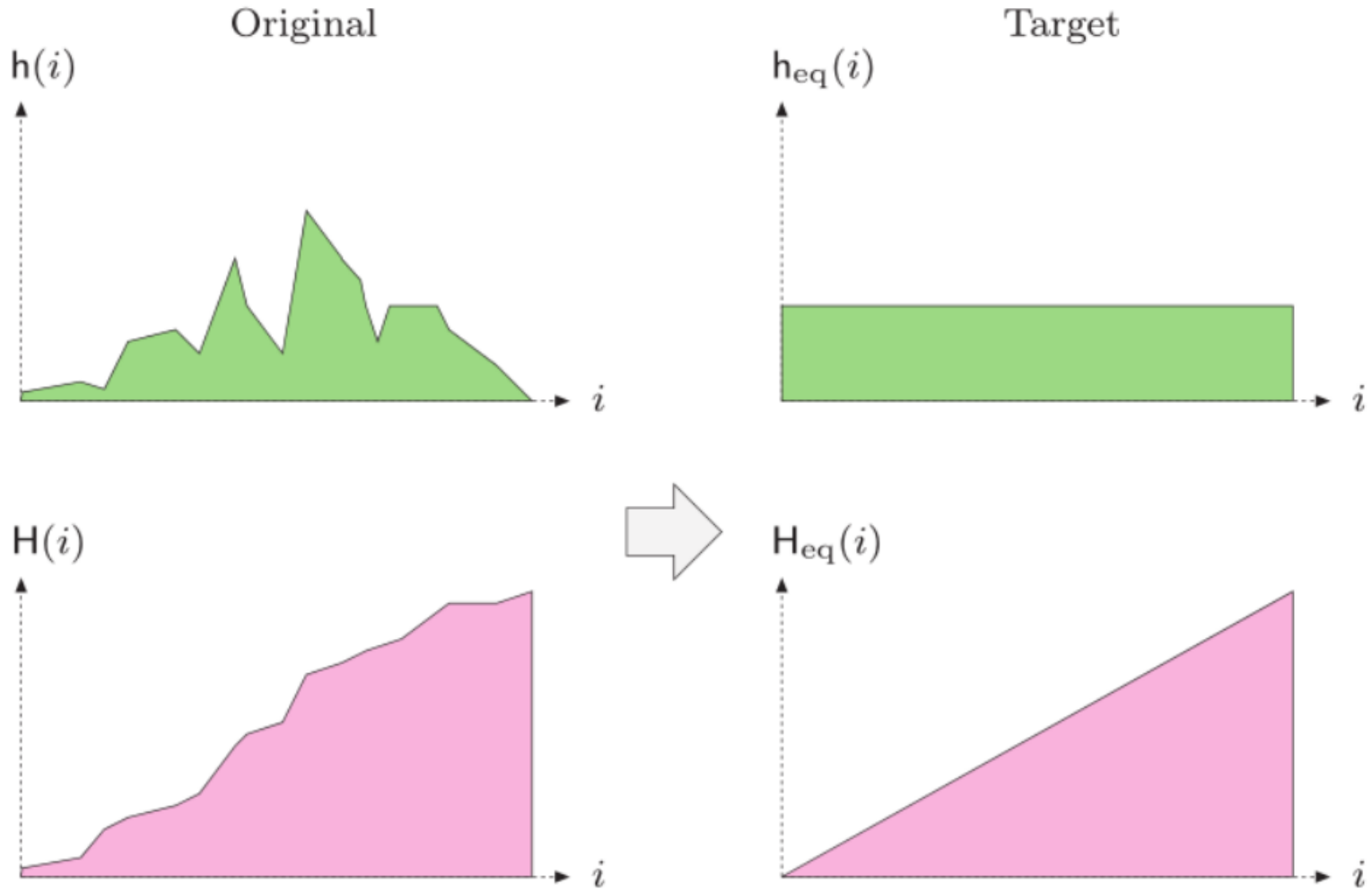




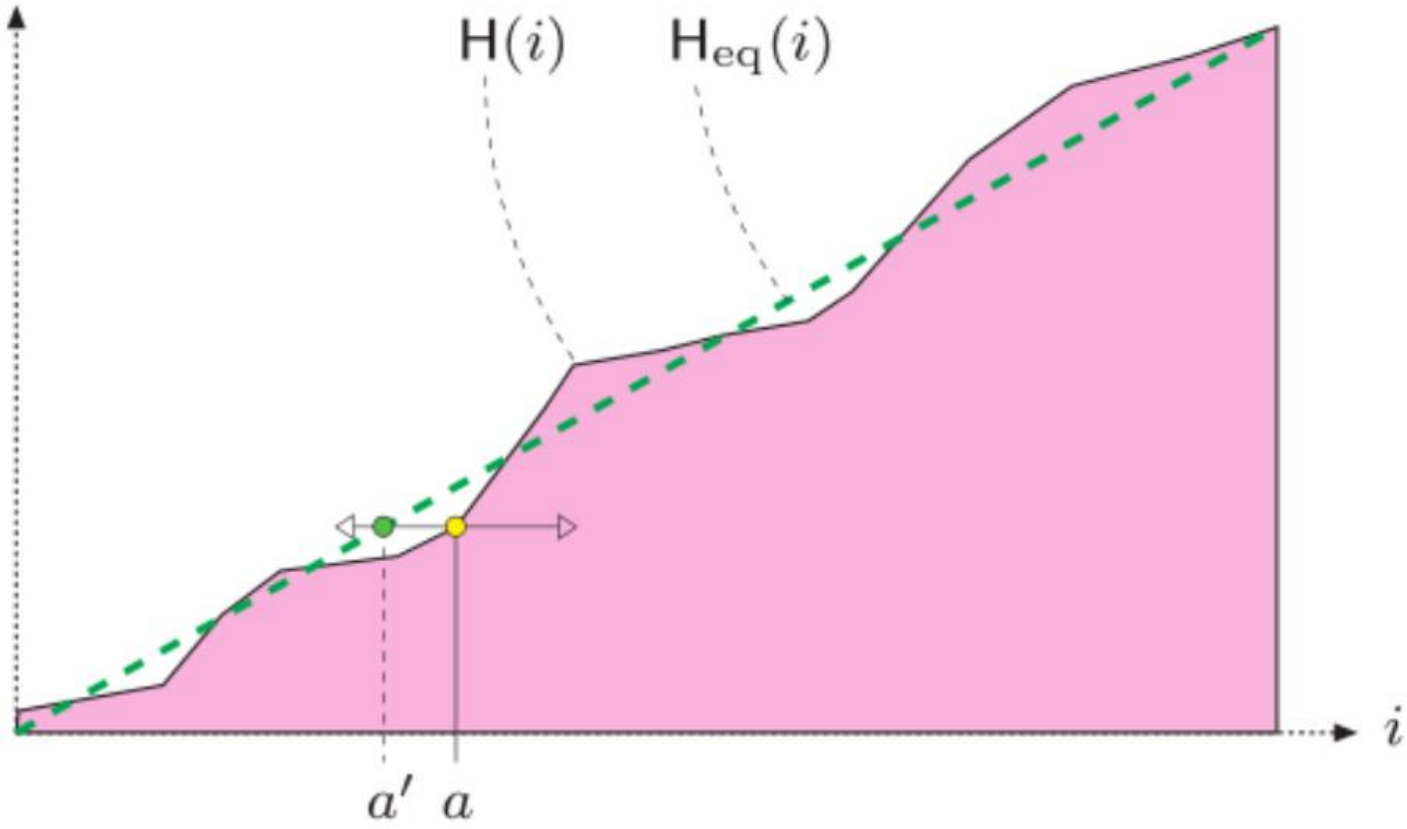
Concept Idea of Histogram Equalization

Histogram

Cumulative Histogram



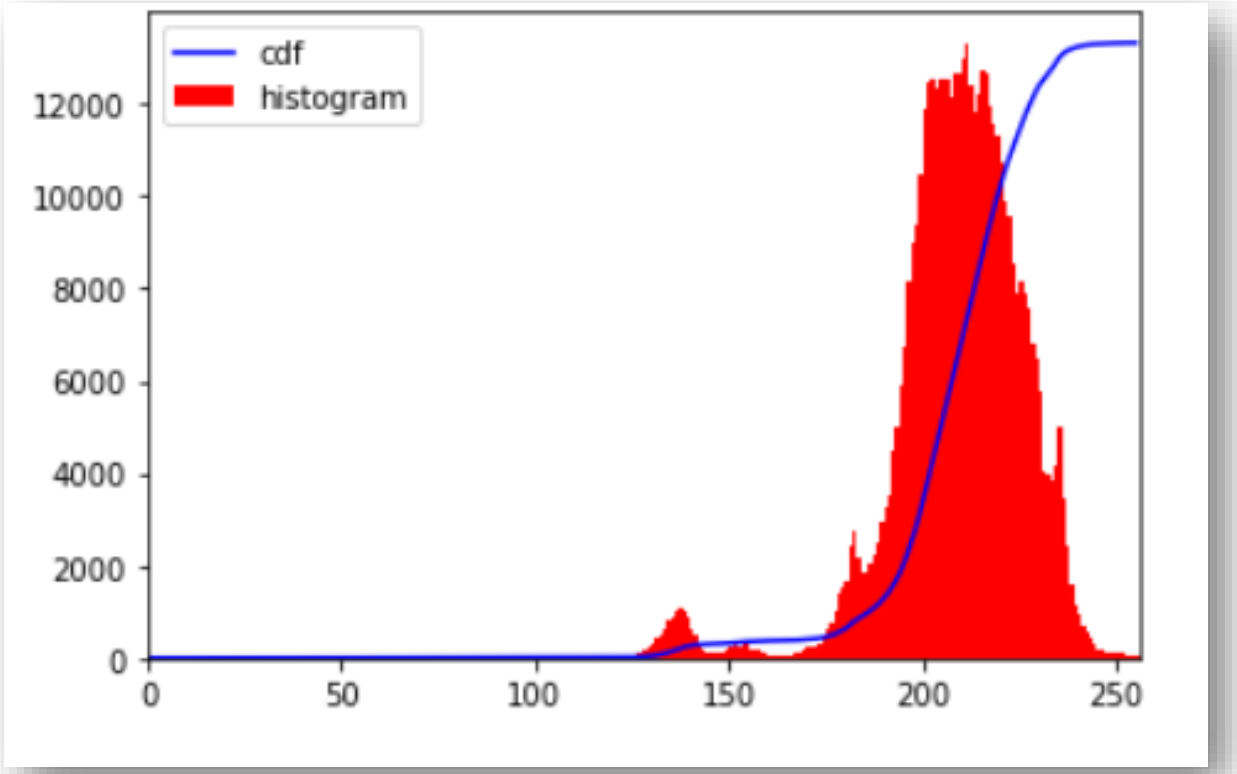
Concept Idea of Histogram Equalization



สร้าง mapping function จาก Cumulative Histogram ไปหา “เส้นตรง”

Concept Idea of Histogram Equalization

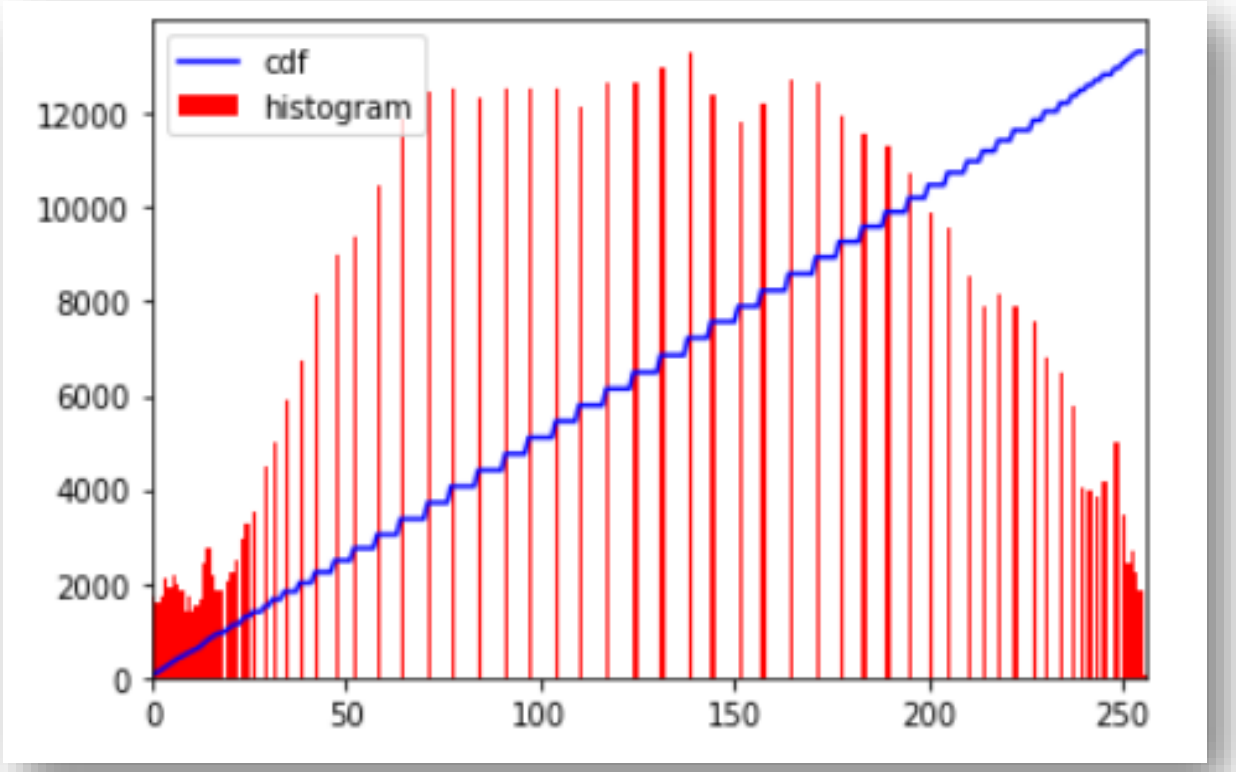




ภาพ Low Contrast (Histogram เกาะกลุ่มกันช่วงค่าสีหนึ่งมาก ๆ)

Concept Idea of Histogram Equalization

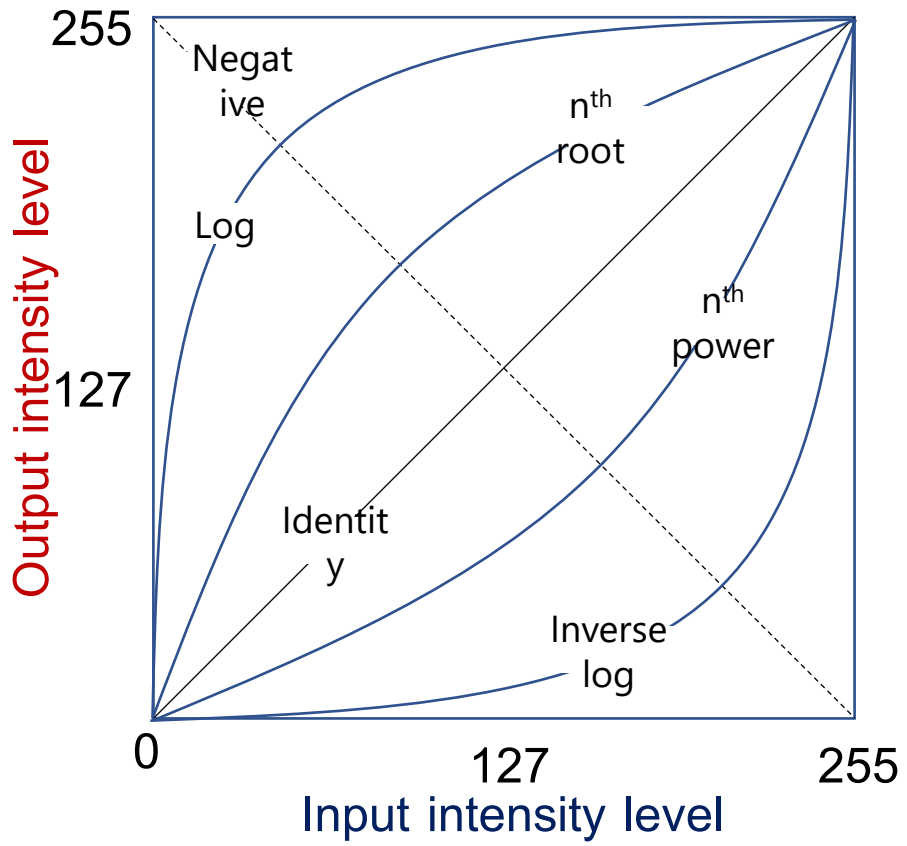




ภาพ High Contrast (จากการทำ Histogram Equalization)

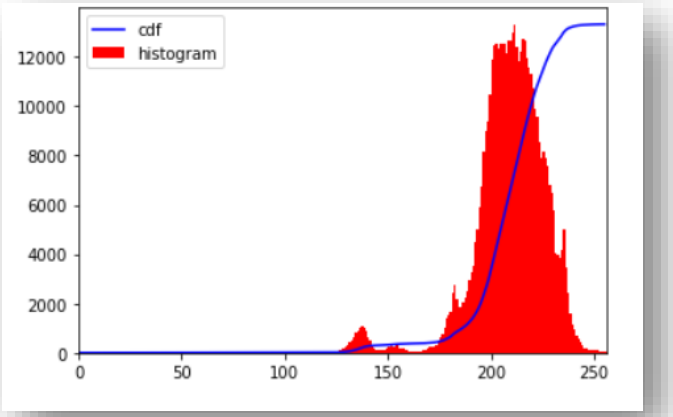
What we know now

[1] Change intensity based on known equation.

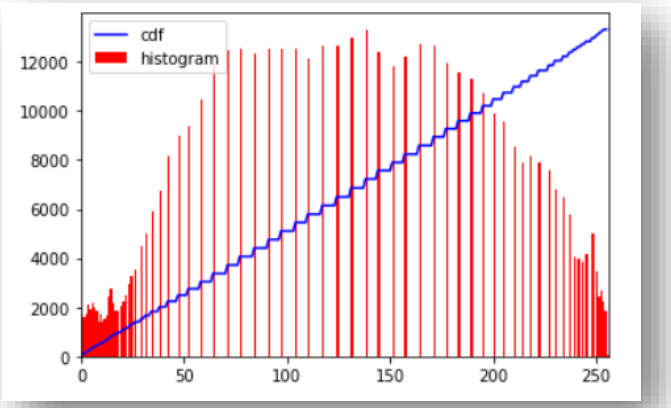


[2] Spread it out, uniformity.









What if, we don't know equation

&

We don't want to spread it



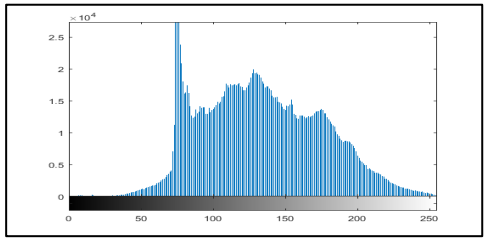


Wanna make left image look like a right image

Histogram Specification

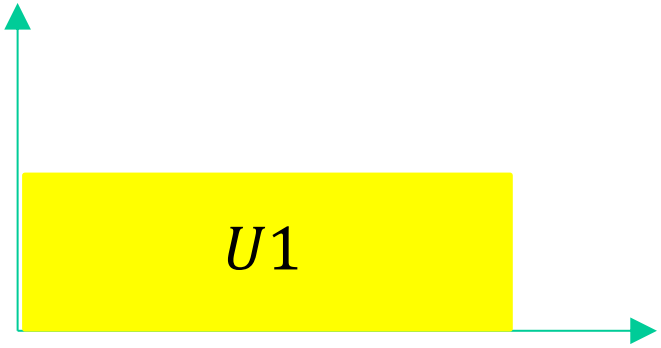
Input





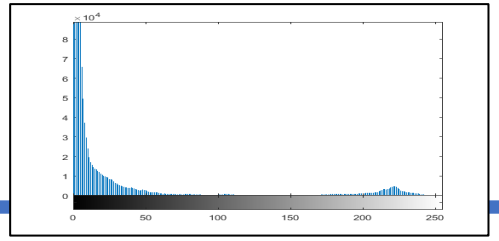
H_1 (Histogram Equalization)

A	U1



Goal





H_2 (Histogram Equalization)

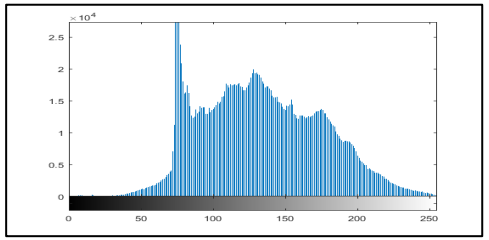
B	U2



Histogram Specification

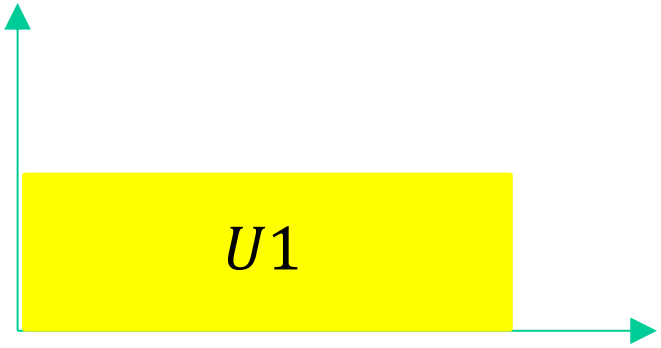
Input





H_1 (Histogram Equalization)

A	U1



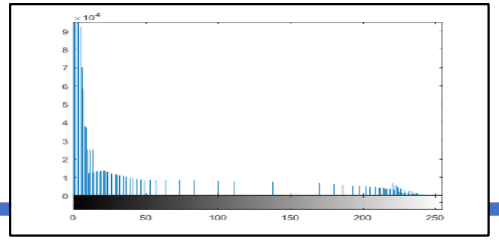
$U1 \approx U2$

A_B	U2

H_2^{-1} (Inverse Histogram Equalization)

Goal





Histogram Specification

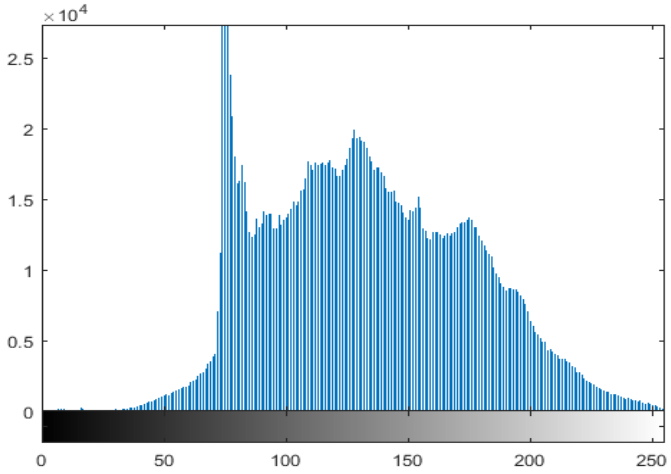


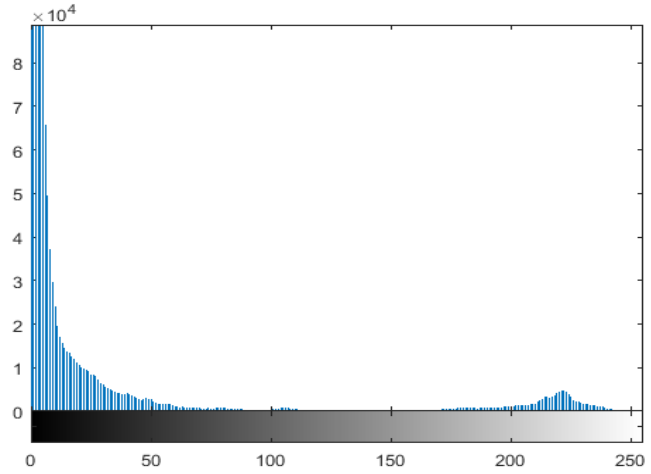


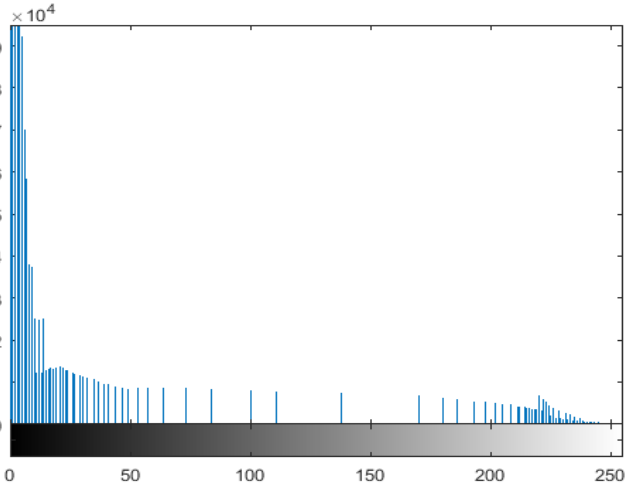


input

output



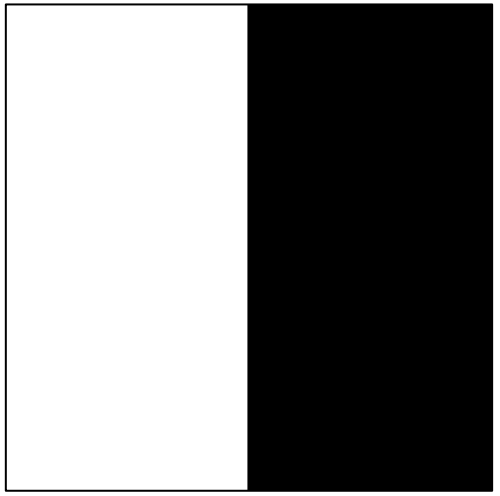




Let's play with some experiment



input



Draw something



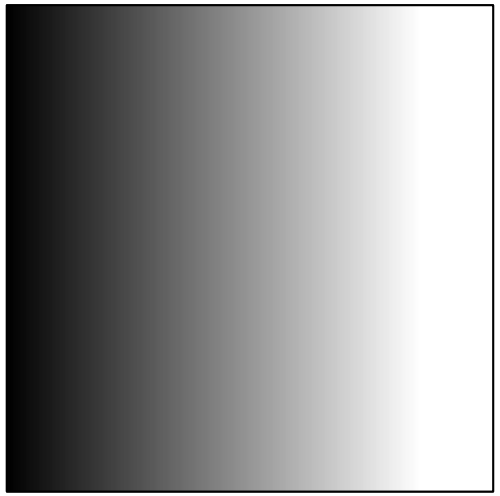
output

With this concept you can **specify** what you want.

Let's play with some experiment



input

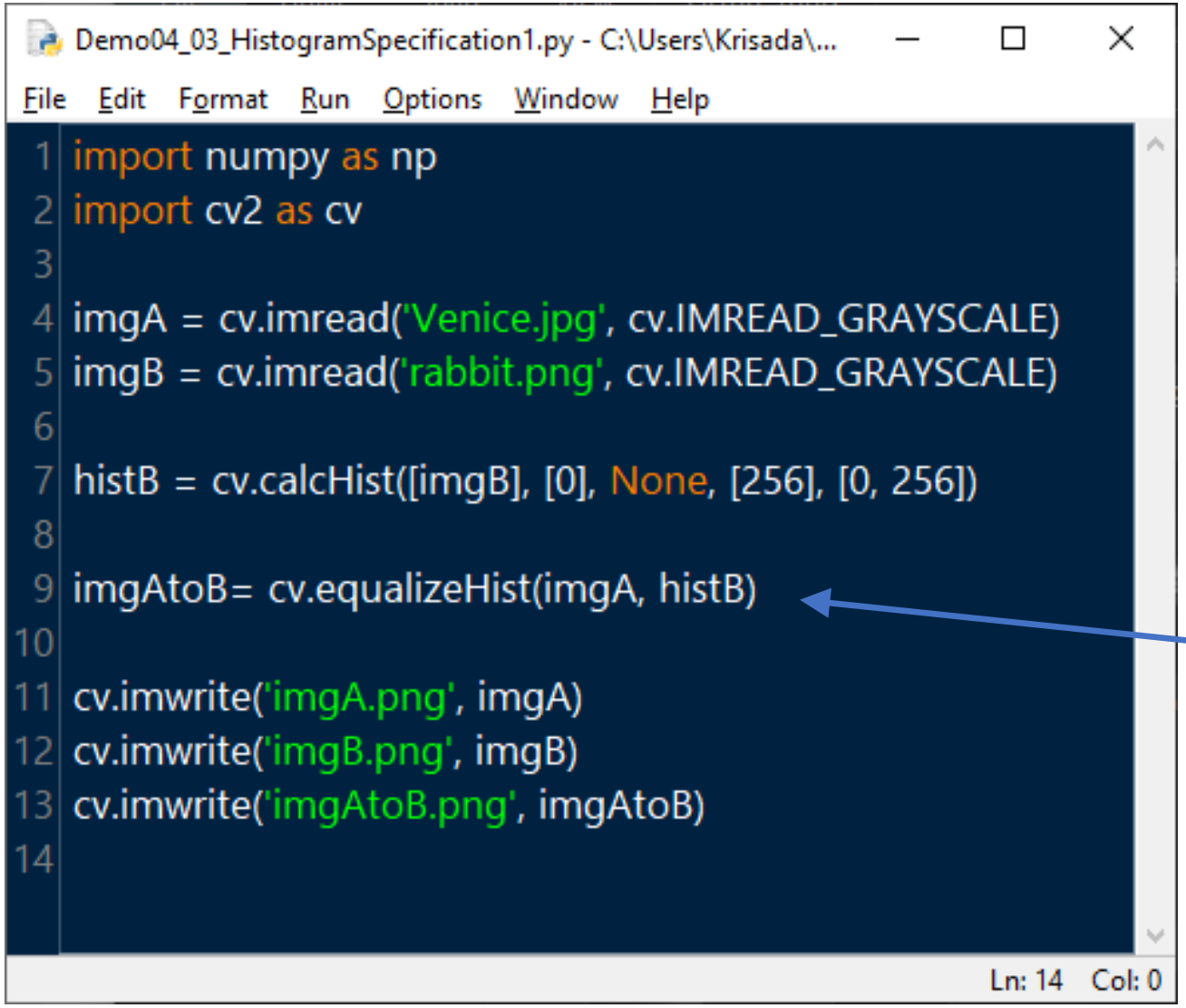


Draw something



output

With this concept you can **specify** what you want.



ใช้ฟังก์ชั่น equalizeHist( ) โดย
กำหนดเป้าหมายเหมือนกัน
(ใส่รูป หรือ histogram ที่ต้องการ)

Class Project

(LUMA)

Learning-based Universal Media Artist (LUMA)

ปรับแต่ง Style ให้เหมาะสมกับผู้ใช้งานแต่ละคน (User Account Control)

ฟีเจอร์เด่นภายในแอปพลิเคชัน (Key Features)

- AI Generation:
  - ❖ Text / Image-to-Image: พิมพ์คำสั่ง (Prompts) เจนฯ ภาพนิ่ง หรือ แก้ไขภาพนิ่งได้
  - ❖ (Optional) ใช้ LLM ขนาดเล็กแปลงข้อความเป็น Prompts ที่เหมาะสม
- Smart Canvas:
  - ❖ การจัดวาง Layout, การจับคู่สี (Color Palettes)
  - ❖ การลบ Background อัตโนมัติ (Background Removal)
  - ❖ การเลือกวัตถุในภาพอัตโนมัติ (Segmentation)
- Asset Hub:
  - ❖ คลังเก็บทรัพยากร สามารถค้นหา เช่น ใส่ Tag, Search with ....

Learning-based Universal Media Artist (LUMA)

กลุ่มเป้าหมายหลัก (Target Audience)

- CDTI CPE Student:
  - ❖ นักศึกษาชั้นปีที่ 2 (รุ่นน้อง) หรือ นักศึกษาชั้นปีที่ 4 (รุ่นพี่)
- Content Creators & YouTubers:
  - ❖ ผู้ที่ต้องการผลิต Content คุณภาพสูงแบบรวดเร็ว (ทำหน้าปก, ตัด , เจนฯ ภาพปลอดลิขสิทธิ์)
- Digital Artists & Designers:
  - ❖ ใช้เป็นเครื่องมือในด่านแรกเพื่อระดมไอเดีย (Brainstorming) และทำภาพ Mockup/Storyboard ส่งลูกค้า

หน้าตาและประสบการณ์การใช้งาน (UI/UX)

- ❖ Minimalist & Flexible Interface: หน้าตาแอปจะปรับเปลี่ยนตามโหมดที่ใช้ เพื่อไม่ให้รกสายตา

Distributed System

สมาชิกแต่ละคน ใช้ PC
แยก IP Address กัน



Browser (User)

ให้ใช้ความรู้จากรายวิชา Network
เพื่อทำให้ PCs เชื่อมโยงกัน
แต่ละเครื่องทำหน้าที่แตกต่างกัน

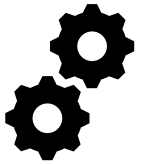


Nginx (Reverse Proxy)

Frontend
(HTML / CSS / JS)



192.168.1.10



Backend (Flask)

192.168.1.20



AI Server

192.168.1.30



Database
(SQLite)

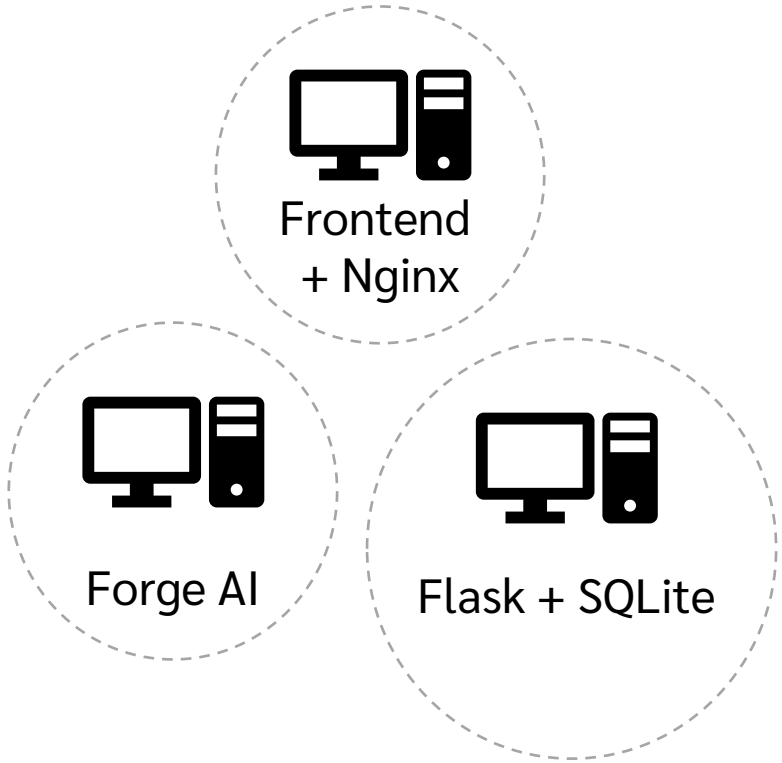
192.168.1.20

การแบ่งหน้าที่ในกลุ่ม (ตัวอย่าง)

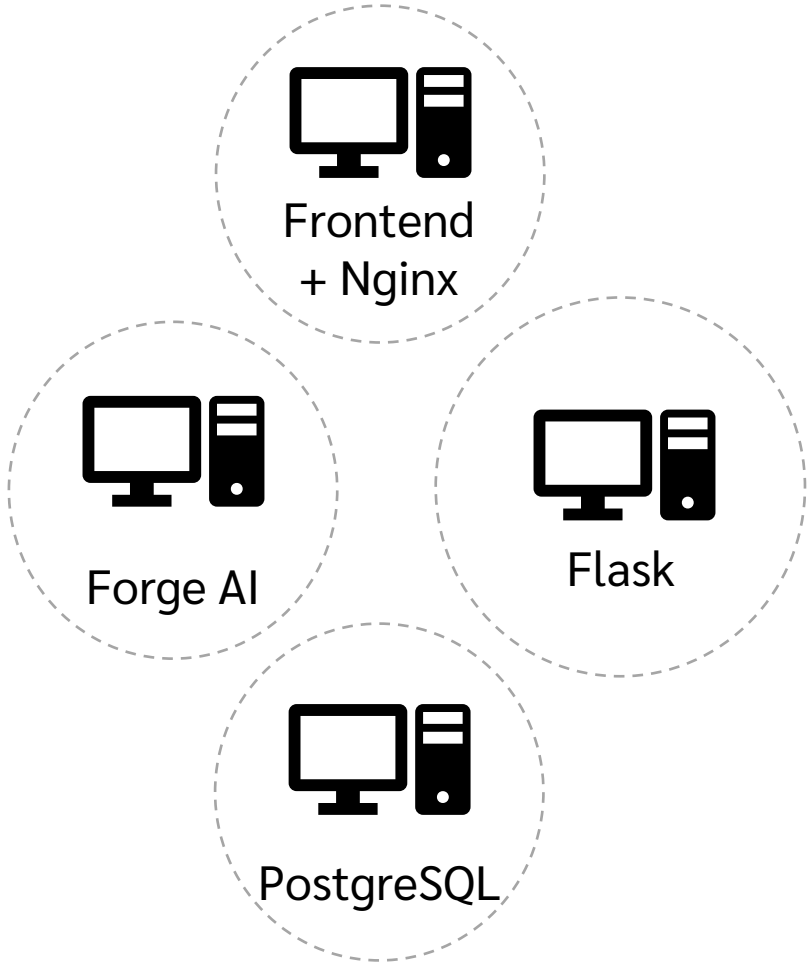
	สมาชิก	หน้าที่	สิ่งที่ส่งมอบ
	คนที่ 1	UX/UI Frontend	หน้าเว็บ, Bootstrap, JavaScript
	คนที่ 2	Flask Backend	Authentication, API, Database, Logging
	คนที่ 3	AI Engineer	Image Generation, Image Editing, Model/LoRA, ทดสอบ API, Queue
	คนที่ 4	QA / DevOps	ทดสอบระบบ, เขียนคู่มือ, Deployment, Dashboard, Backup
	คนที่ 5 (ถ้ามี)	Reverse Proxy, Routing	(ช่วยงานส่วนอื่น เพราะภาระงานต่ำ)

Distributed System

จำนวน PCs ที่ใช้ในระบบ



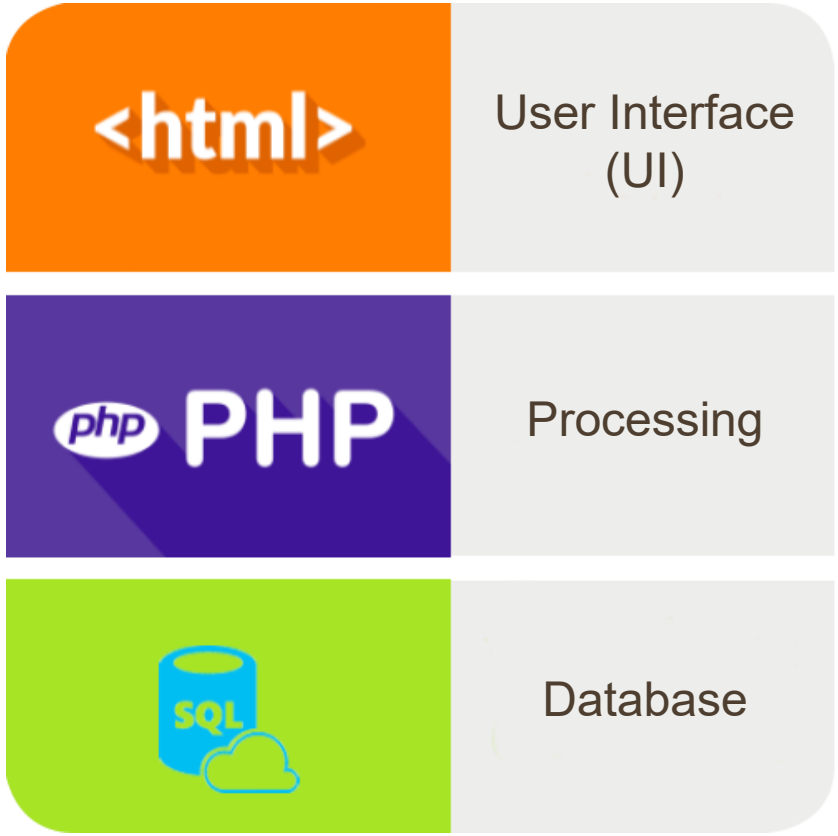
ตัวอย่างการใช้คอมพิวเตอร์ 3 เครื่อง

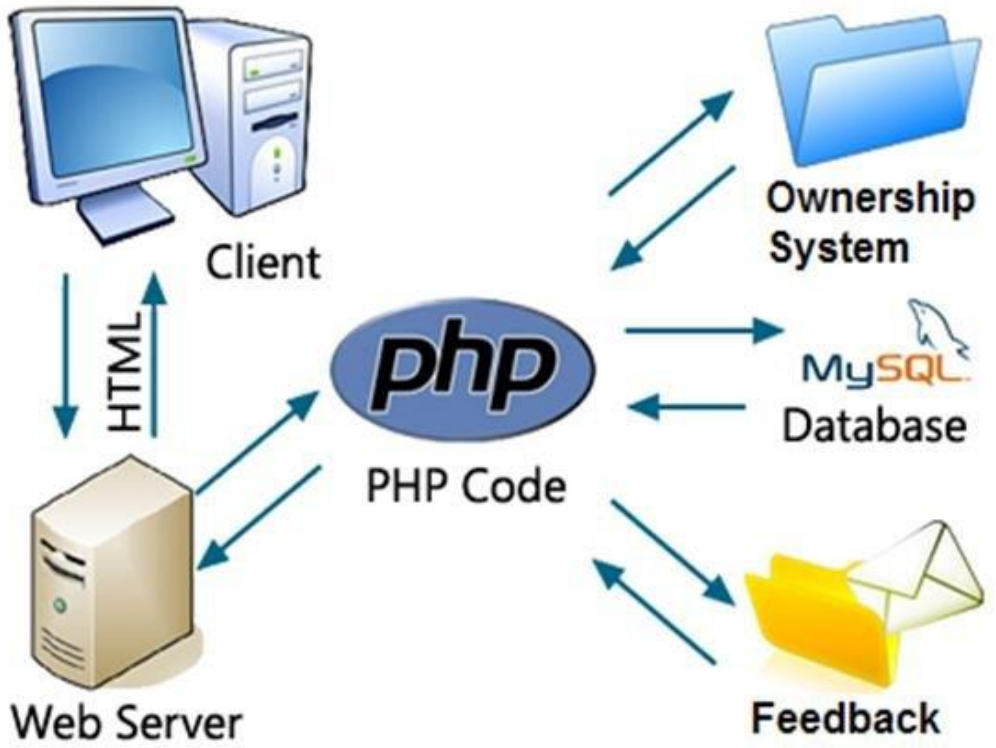


ตัวอย่างการใช้คอมพิวเตอร์ 4 เครื่อง

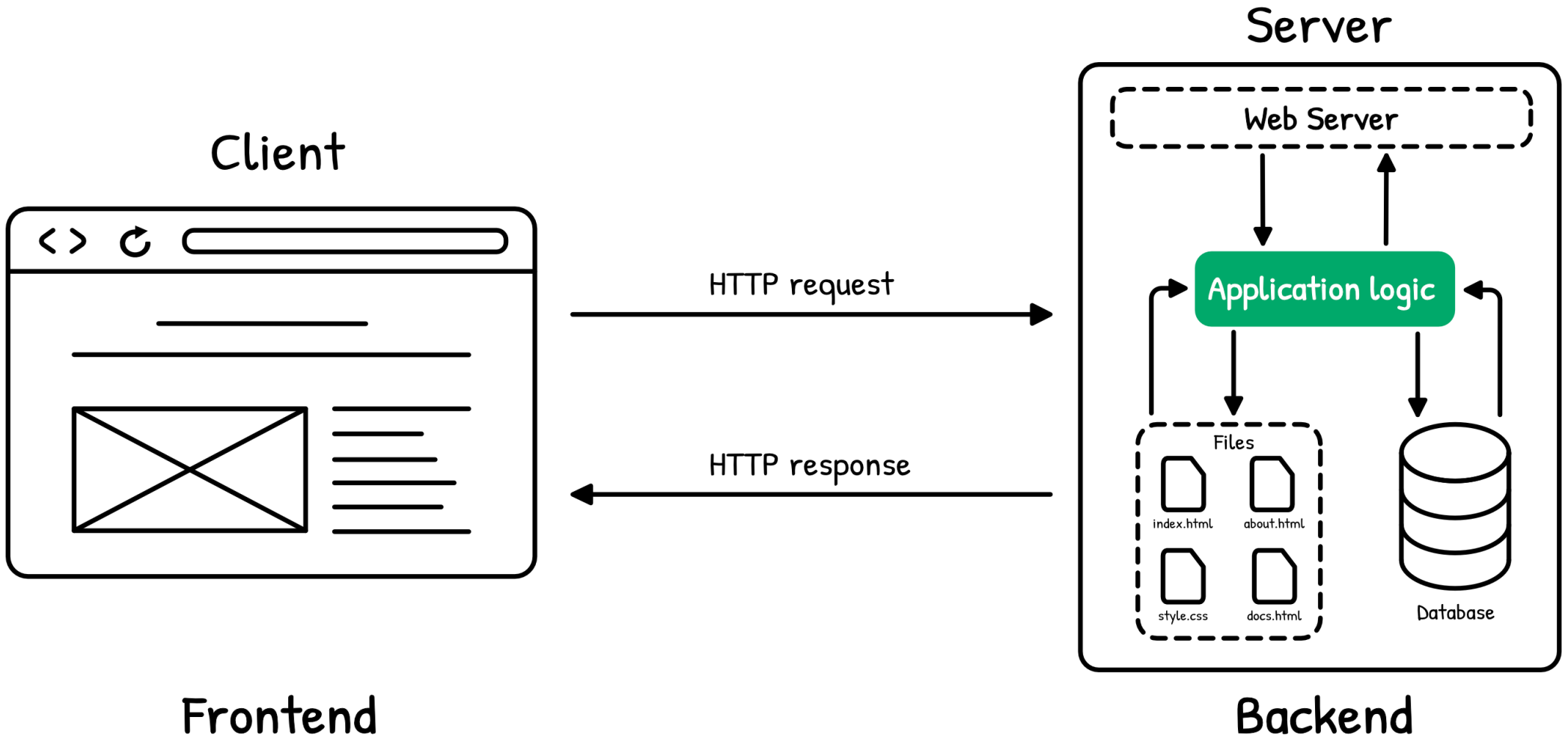
Overview Web Application

Old school web development





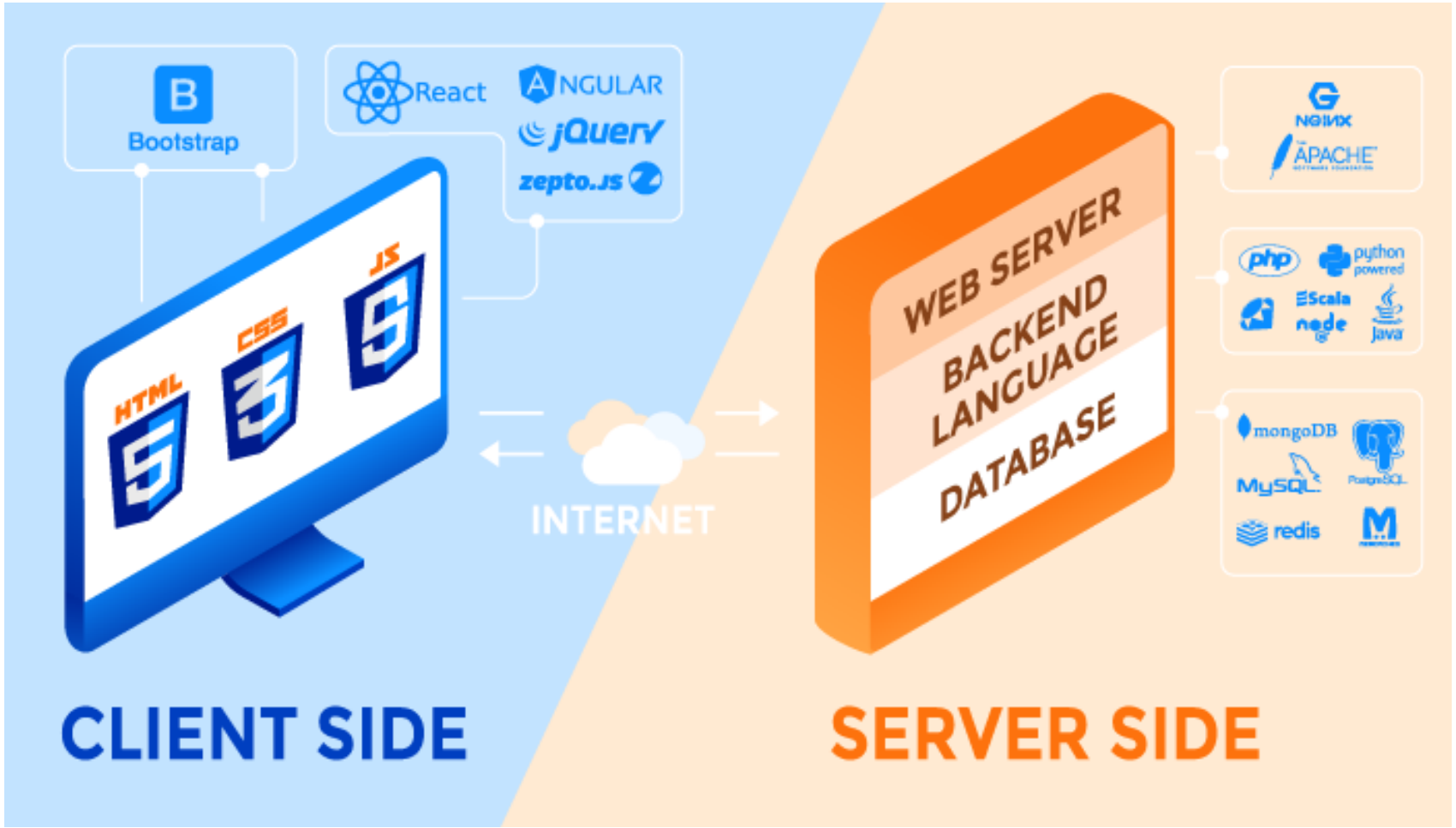
Overview Web Application



Overview Web Application

Modern

web development

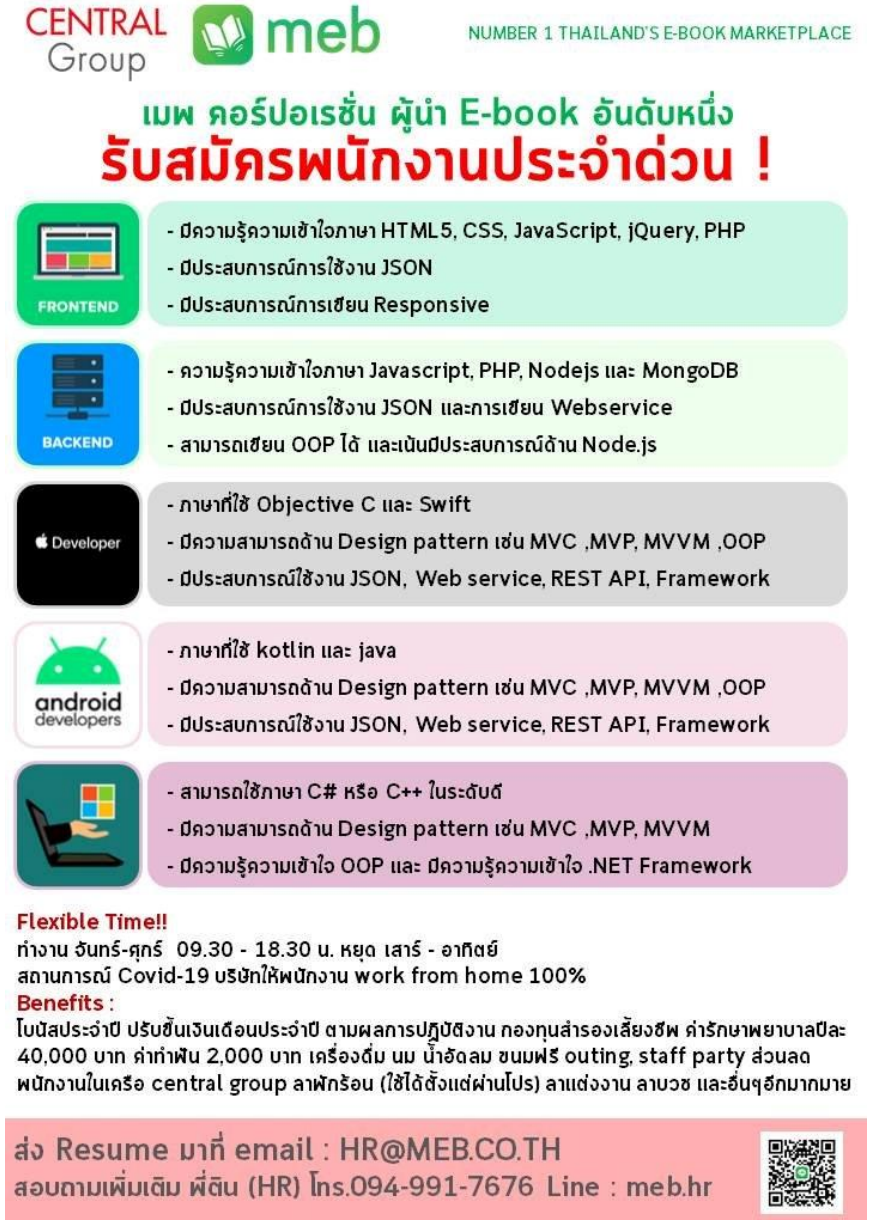


Overview Web Application

You will see it in

Job Hiring





Overview Web Application

Front-End Developer

Main responsibilities

The main responsibility of the Front-End Developer is the **User interface**.

Simply put, create things that the user sees.

Back-End Developer

A Back-End web developer is a person who can develop **server-side** software.

- Program a **server** (like using PHP, ASP, Python, or Node)
- Program a **database** (like using SQL, SQLite, or MongoDB)

Overview Web Application

Full Stack Web Developer (very popular in Thailand…)

Popular Stack

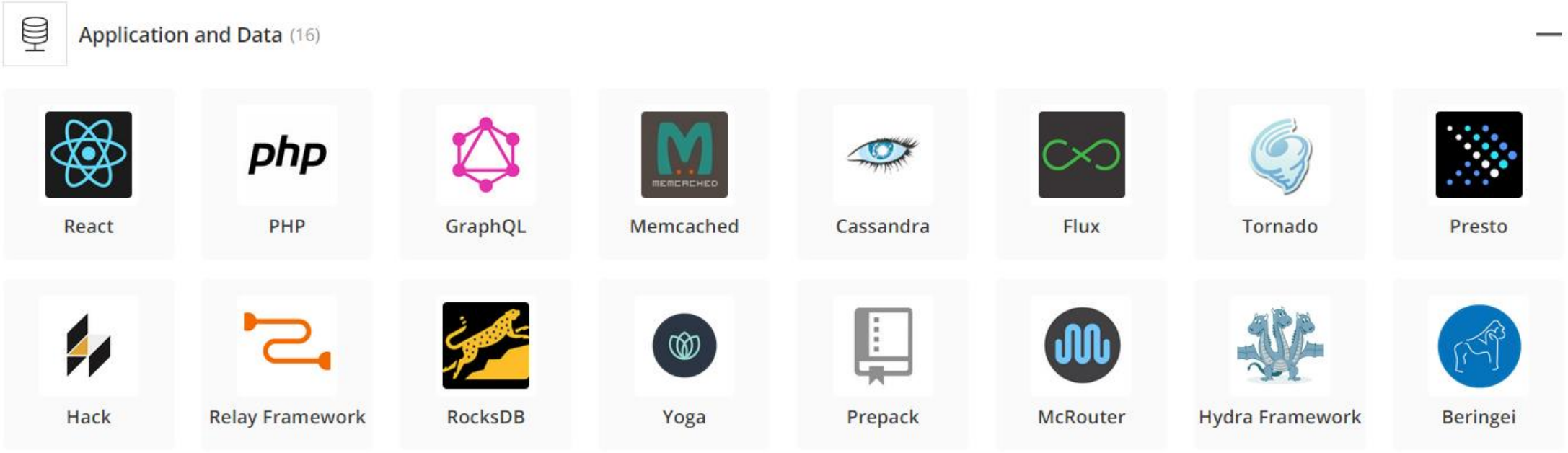
- LAMP stack: JavaScript - Linux - Apache - MySQL - PHP
- LEMP stack: JavaScript - Linux - Nginx - MySQL - PHP
- MEAN stack: JavaScript - MongoDB - Express - AngularJS - Node.js
- MERN stack: JavaScript - MongoDB - Express - React.js - Node.js
- MEVN stack: JavaScript - MongoDB - Express - Vue.js - Node.js
- Django stack: JavaScript - Python - Django - MySQL
- Ruby on Rails: JavaScript - Ruby - SQLite - Rails

Tech Stack



Facebook

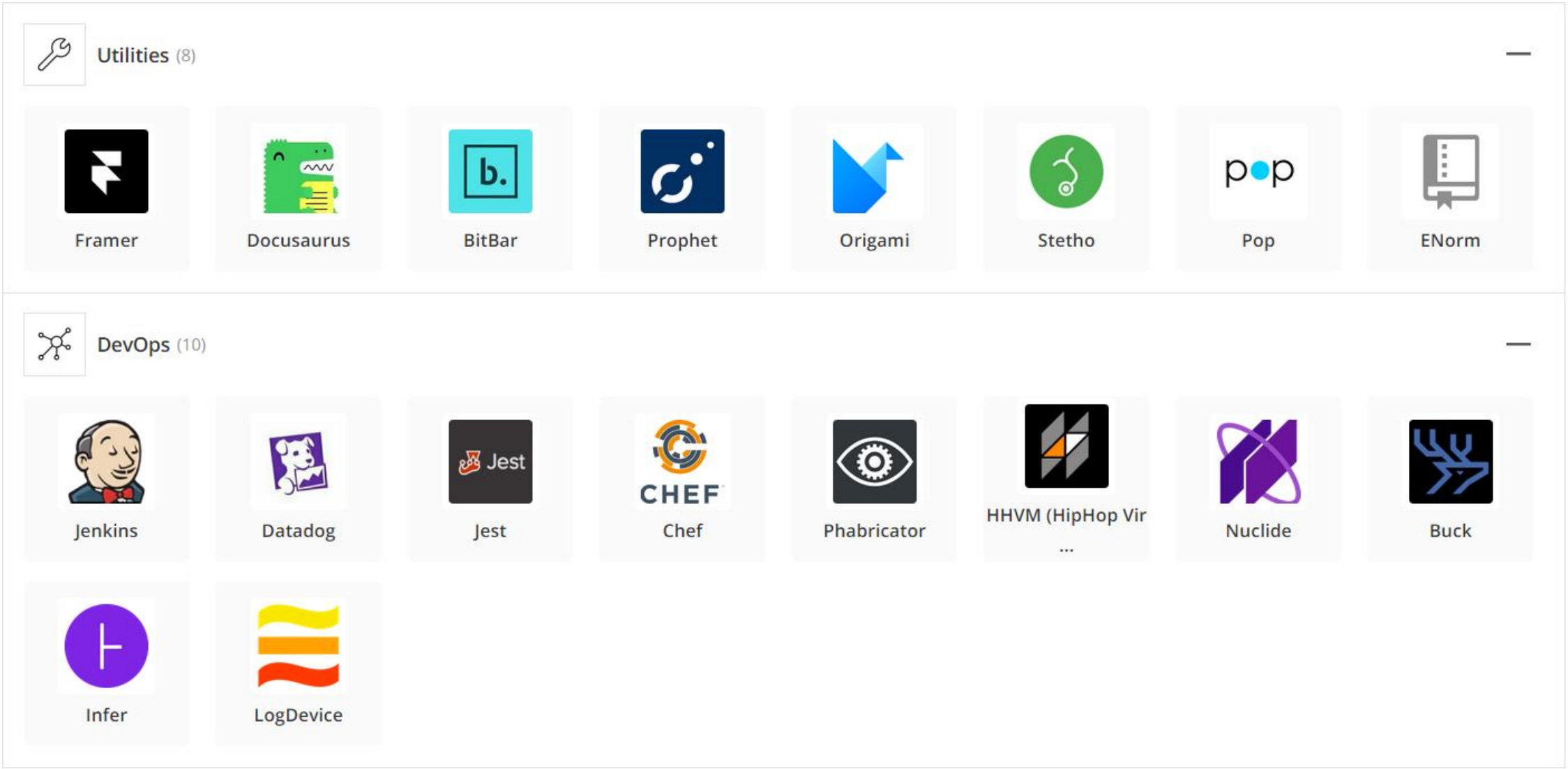
The world's largest social utility, 1.3 billion active users as of June 2014.





https://stackshare.io/stacks

Tech Stack


Facebook
The world's largest social utility, 1.3 billion active users as of June 2014.
Utilities (8)
Framer
Docusaurus
BitBar
Prophet
Origami
Stetho
Pop
ENorm
DevOps (10)
Jenkins
Datadog
Jest
Chef
Phabricator
HHVM (HipHop Vir ...
Nuclide
Buck
Infer
LogDevice

Roadmap

You need to learn the basics in both
Front-End & Back-End

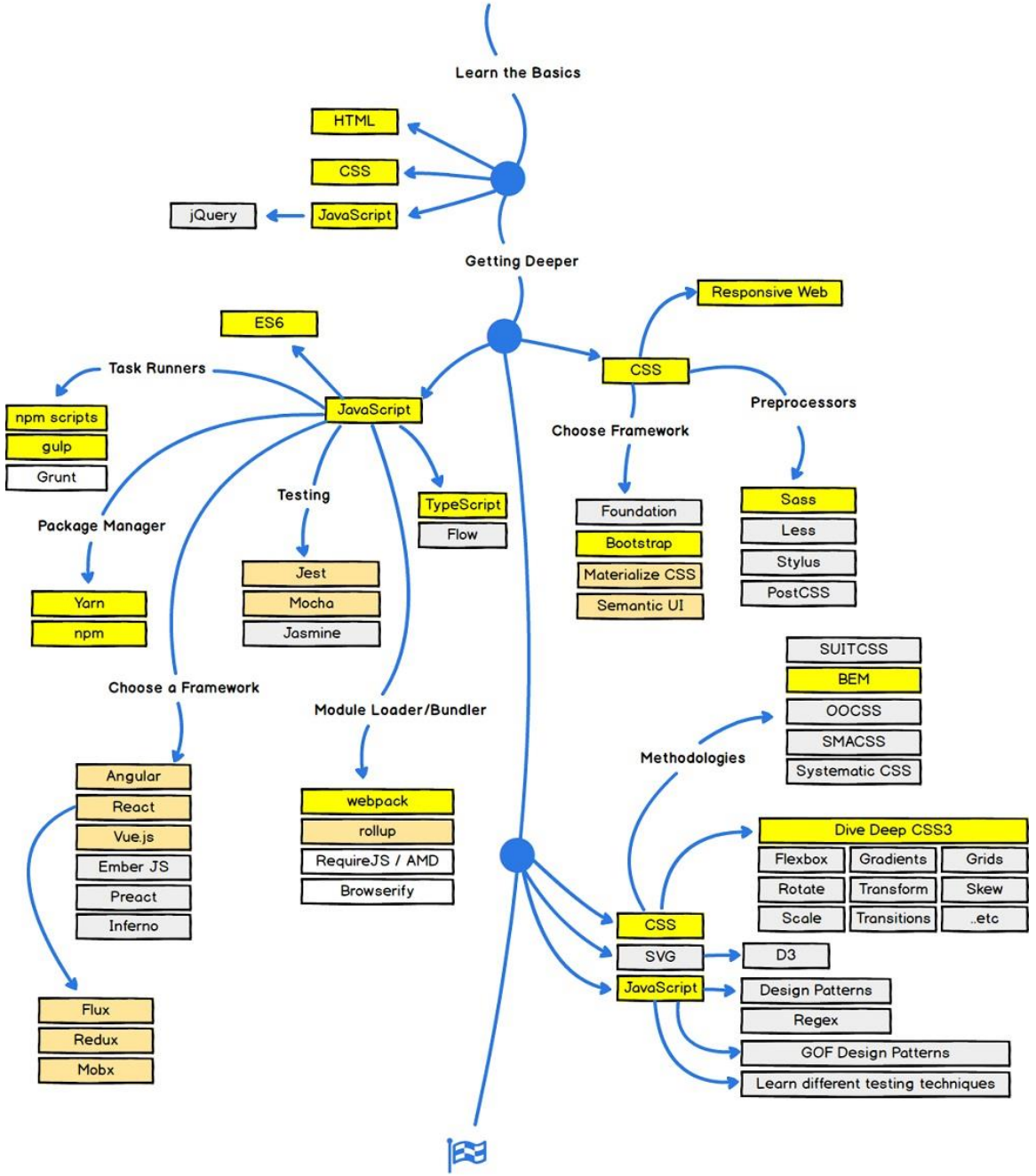
Learn the Basics

Every **Web Developer** must have a basic understanding of **HTML**, **CSS**, and **JavaScript**.

Responsive Web Design is used in all types of modern web development.

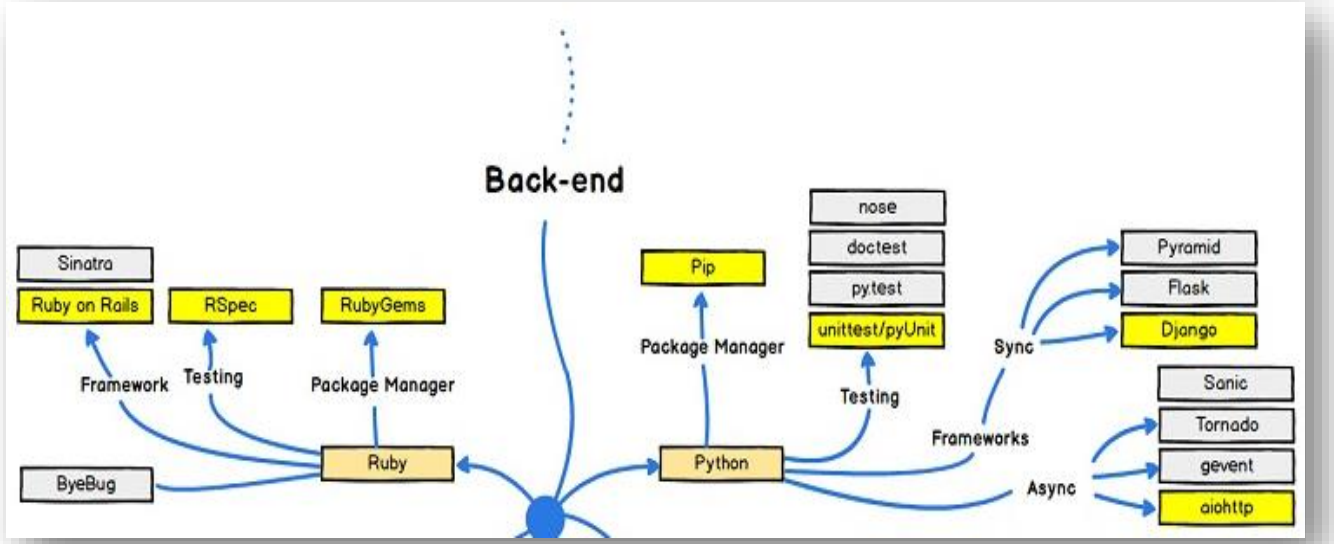
ECMAScript 5 (JavaScript 5) is supported in all modern browsers. Take a good look at it, especially the new array functions.

HTML	CSS	JavaScript
HTML	CSS	JavaScript
HTTP / XHR	CSS Responsive	ECMAScript 5

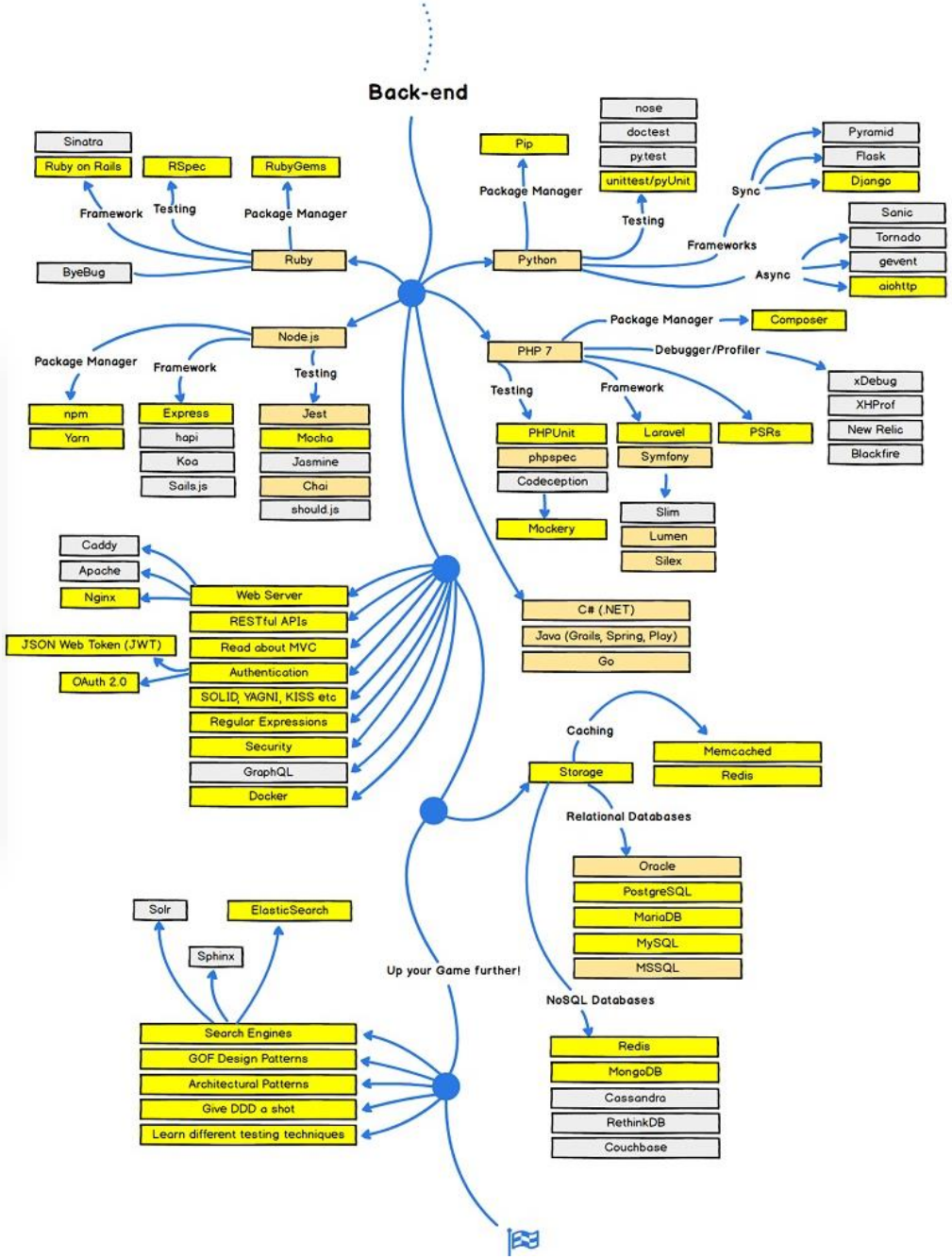


Front-End Roadmap

Roadmap



Back-End Roadmap

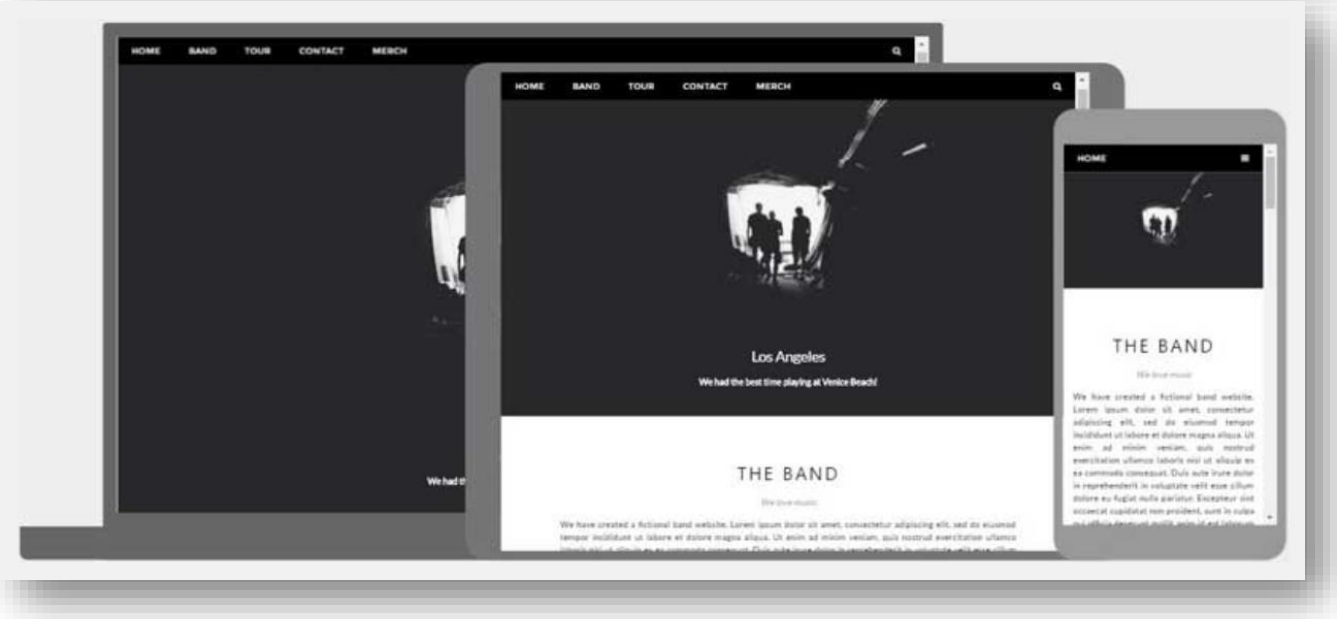


Thing you should learn by yourself



Responsive Web Design is about using HTML and CSS to automatically resize a website.

Responsive Web Design is about making a website look good on all devices (desktops, tablets, and phones):



Responsive Web Design

Distributed System

สมาชิกแต่ละคน ใช้ PC
แยก IP Address กัน



Browser (User)

ให้ใช้ความรู้จากรายวิชา Network
เพื่อทำให้ PCs เชื่อมโยงกัน
แต่ละเครื่องทำหน้าที่แตกต่างกัน



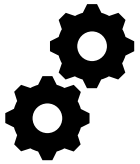
Nginx (Reverse Proxy)



Frontend
(HTML / CSS / JS)



192.168.1.10



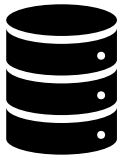
Backend (Flask)

192.168.1.20



AI Server

192.168.1.30



Database
(SQLite)

192.168.1.20



Flask is a micro web framework written in **Python**.

A **microframework** is a term used to refer to minimalistic web application frameworks. It is contrasted with full-stack frameworks. It **lacks** most of the functionality which is common to expect in a full-fledged web application framework, such as:

- ❑ Accounts, authentication, authorization, roles ➔ Flask-Login, Flask-JET-Extended
- ❑ Database abstraction via an object-relational mapping ➔ Flask-SQLAlchemy
- ❑ Input validation and input sanitation ➔ Flask-WTF

Flask

Advantages

- ❑ Easy to understand development
- ❑ Small, not rely on many library
- ❑ It is very flexible.
- ❑ Easy to test

Disadvantages

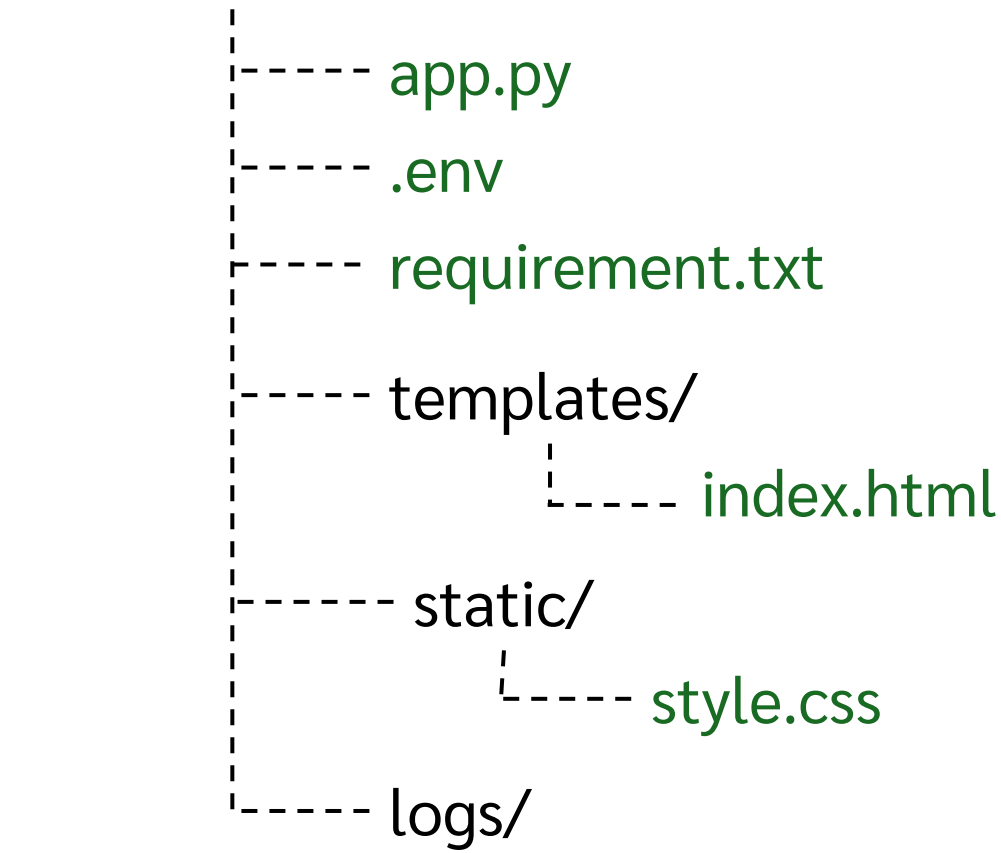
- ❑ Not suitable for big applications

What you need to known before

- ❑ HTML5
- ❑ CSS
- ❑ Python

Flask is good when you want to build simple, innovative use case to be added in existing application.

Backend – Project structure (Flask)

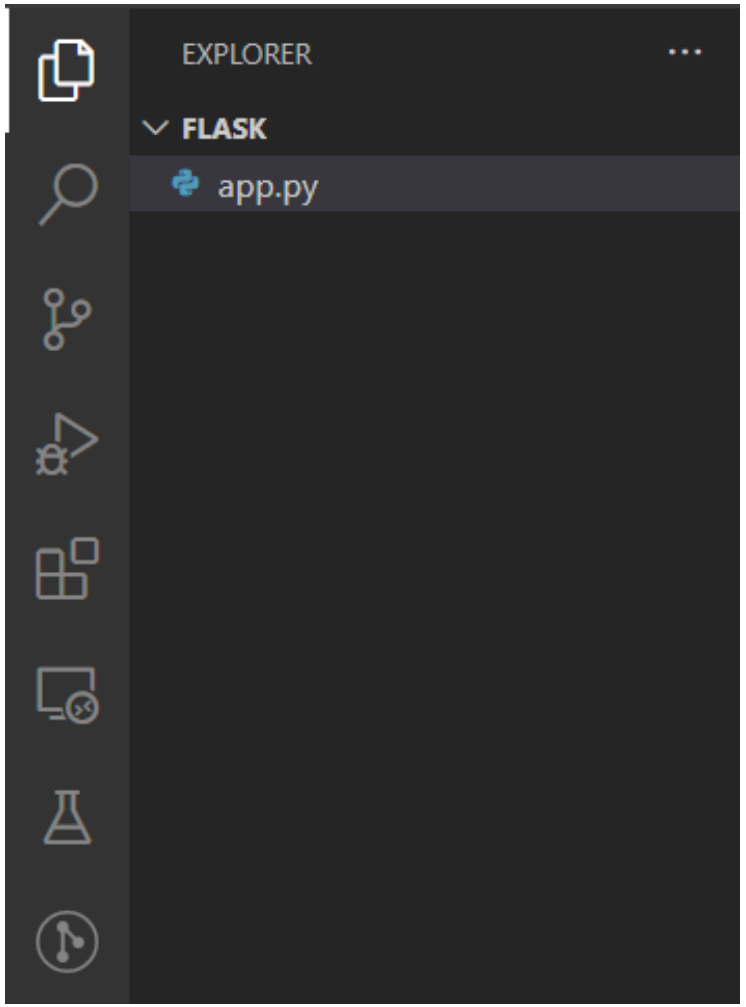


File และ Directory สำคัญ

- templates/ หน้า page
- static/ CSS, JavaScript, Image
- logs/ บันทึกการทำงาน
- app.py WebApp หลัก

Install everything you need (5 min)

Create your first Flask project




```python
from flask import Flask
app = Flask(__name__)

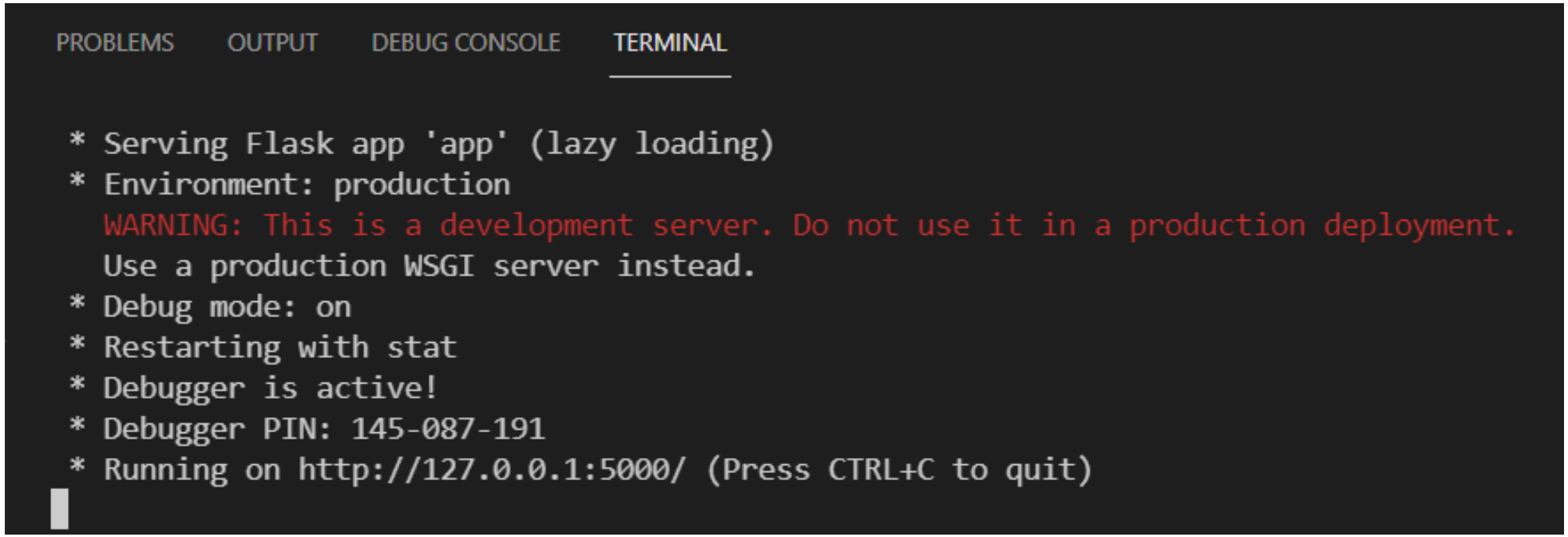
if __name__ == "__main__":

    app.run()

```

Create Flask application

Create your first Flask project

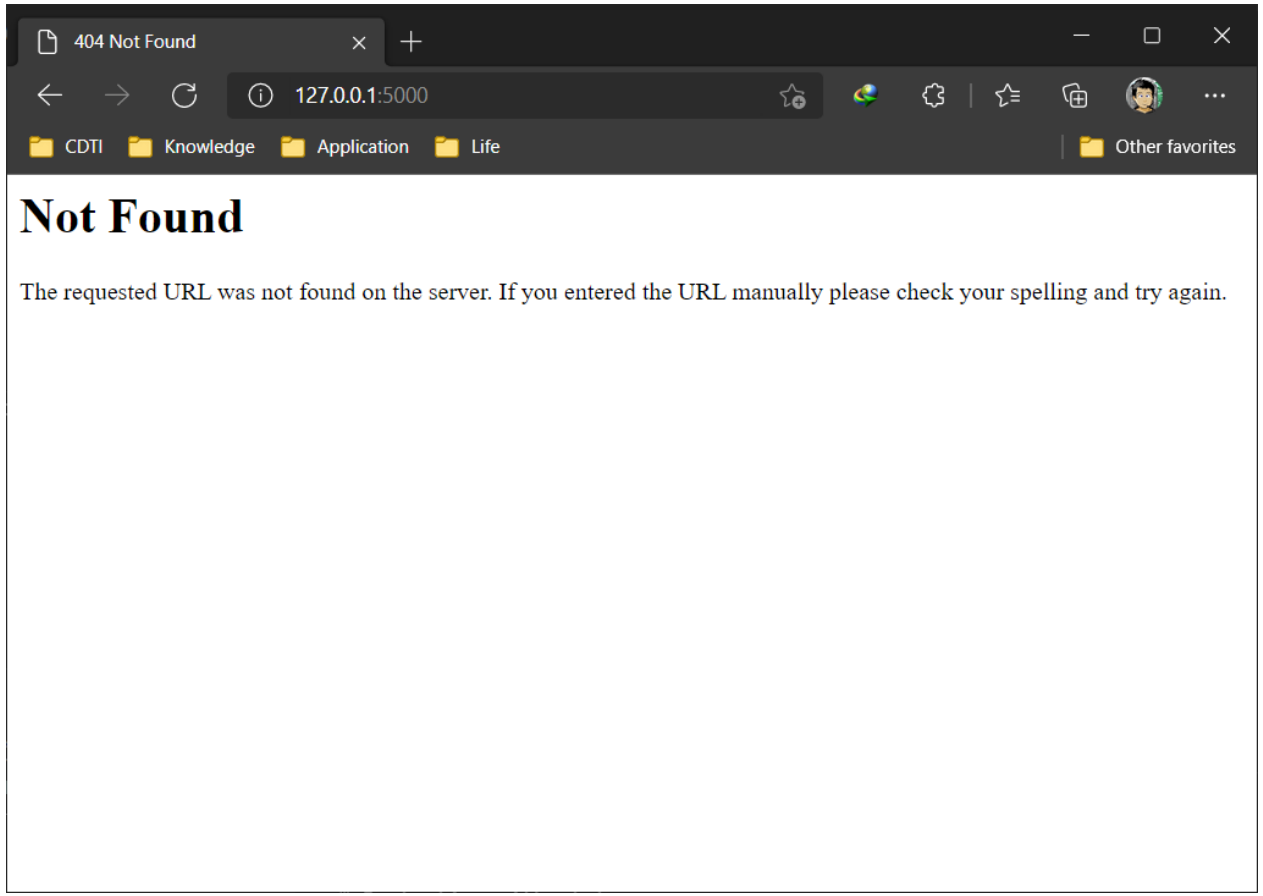


[Run], then message will show in Terminal like this

Go to the http://127.0.0.1:5000/ with your browser

(Or press [CTRL] and Click with Mouse)

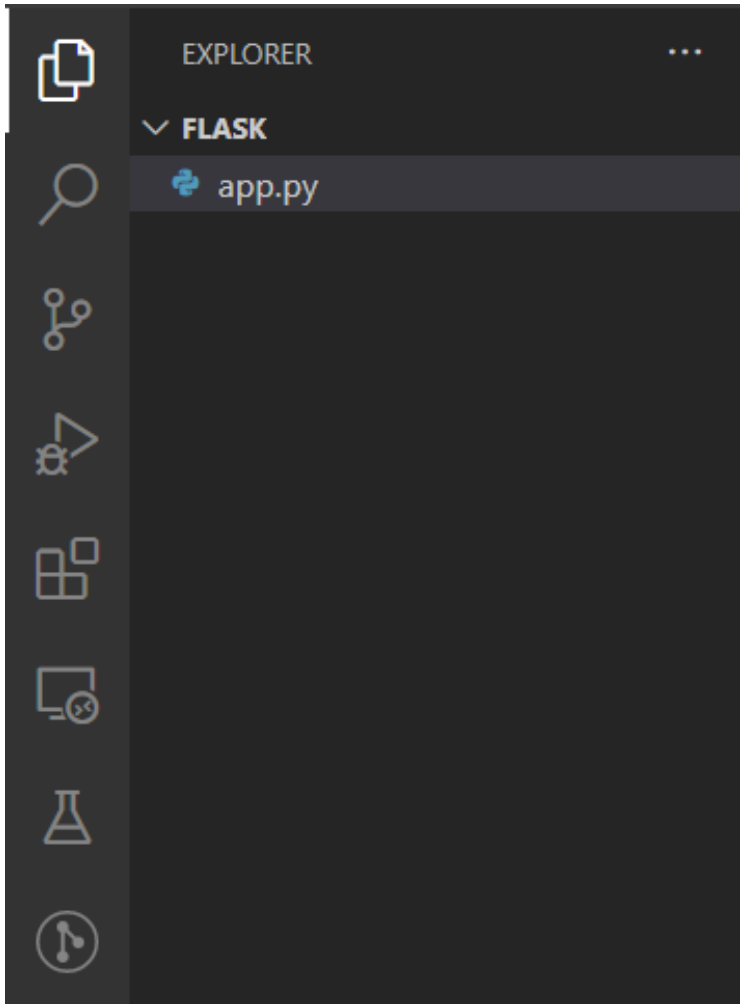
Create your first Flask project



Your browser will show error like this.

Because we didn’t specify [Route]

Create your first Flask project



app.py > ...

```python

from flask import Flask
app = Flask(__name__)

@app.route('/')
def index():
    return "<h1>Hello Flask Framework</h1>"

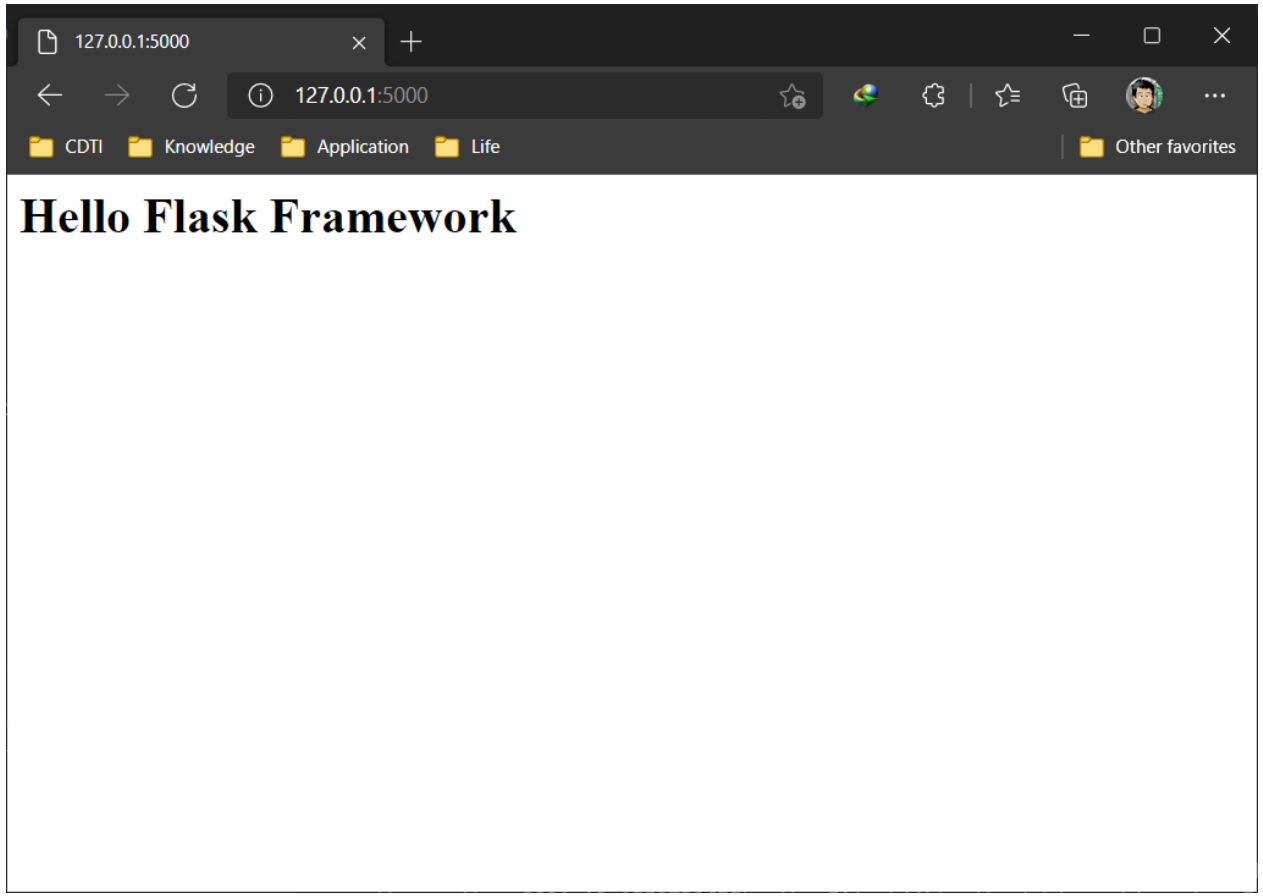
if __name__ == "__main__":

    app.run()

```

Create Flask application

Create your first Flask project



[Run]

Now, the message should show like this.

How to define [Route]

Modern web apps use a technique named **routing**. This helps the user remember the URLs.

For instance, instead of having /booking.php they see /booking/.

https://student360.cdti.ac.th/index.php?q=/modules/Planner/units.php

https://student360.cdti.ac.th/index.php?q=/modules/Planner/units.php&gibbonSchoolYearID=026&viewBy=class&gibbonCourseID=00230626

https://student360.cdti.ac.th/index.php?q=/modules/Planner/units.php&gibbonSchoolYearID=026&viewBy=class&gibbonCourseID=00230626

Variable(s)

gibbonSchoolYearID	= 026
viewBy	= class
gibbonCourseID	= 00230626

How to define [Route]

```
@app.route('/')
def index():
        return 'Hello World'
```

127.0.0.1:5000/

127.0.0.1:5000/home

127.0.0.1:5000/user

127.0.0.1:5000/user/krisada

How to define [Route]

```
@app.route(‘/’)
def index():
        return ‘Hello World’
```

app.py > ...

```python

from flask import Flask
app = Flask(__name__)

@app.route('/')
def index():
    return "<h1>Hello Flask Framework</h1>"

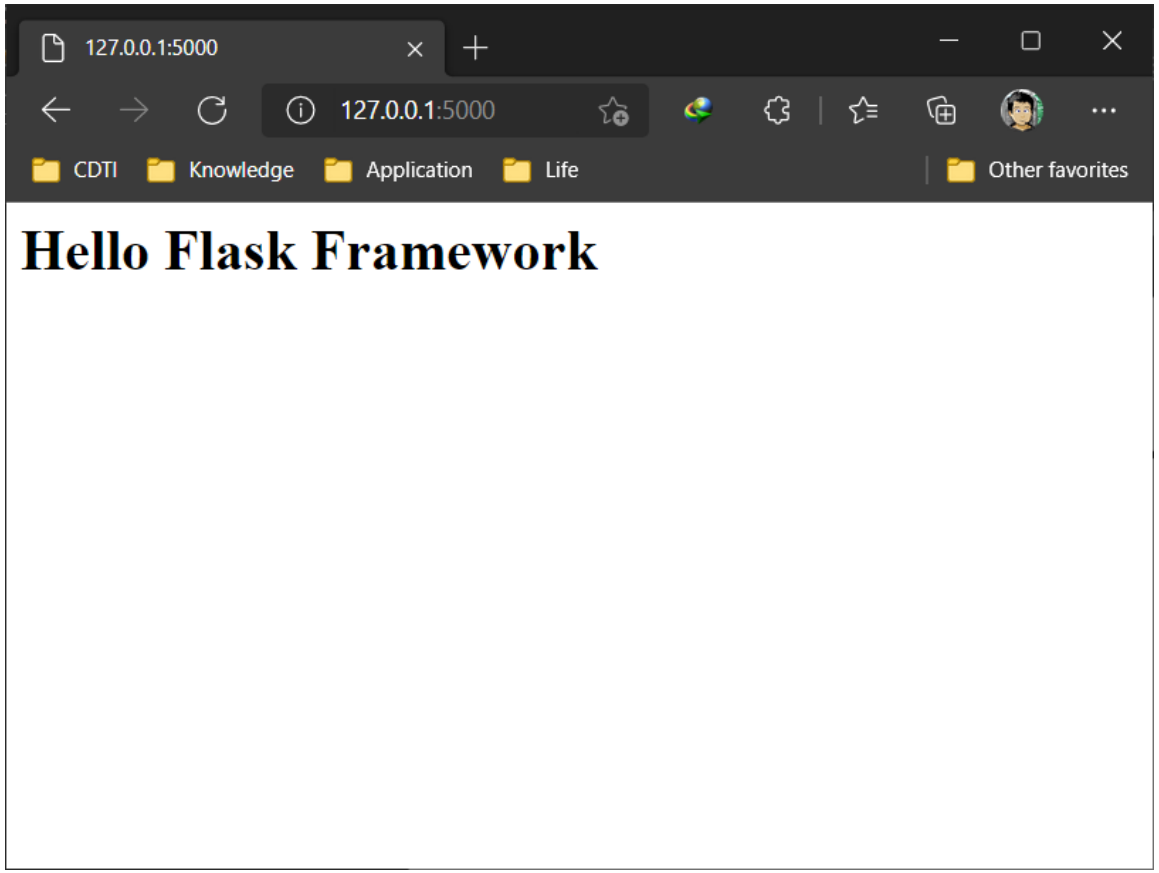
@app.route('/user')
def user():
    return "This is a User Page"

if __name__ == "__main__":

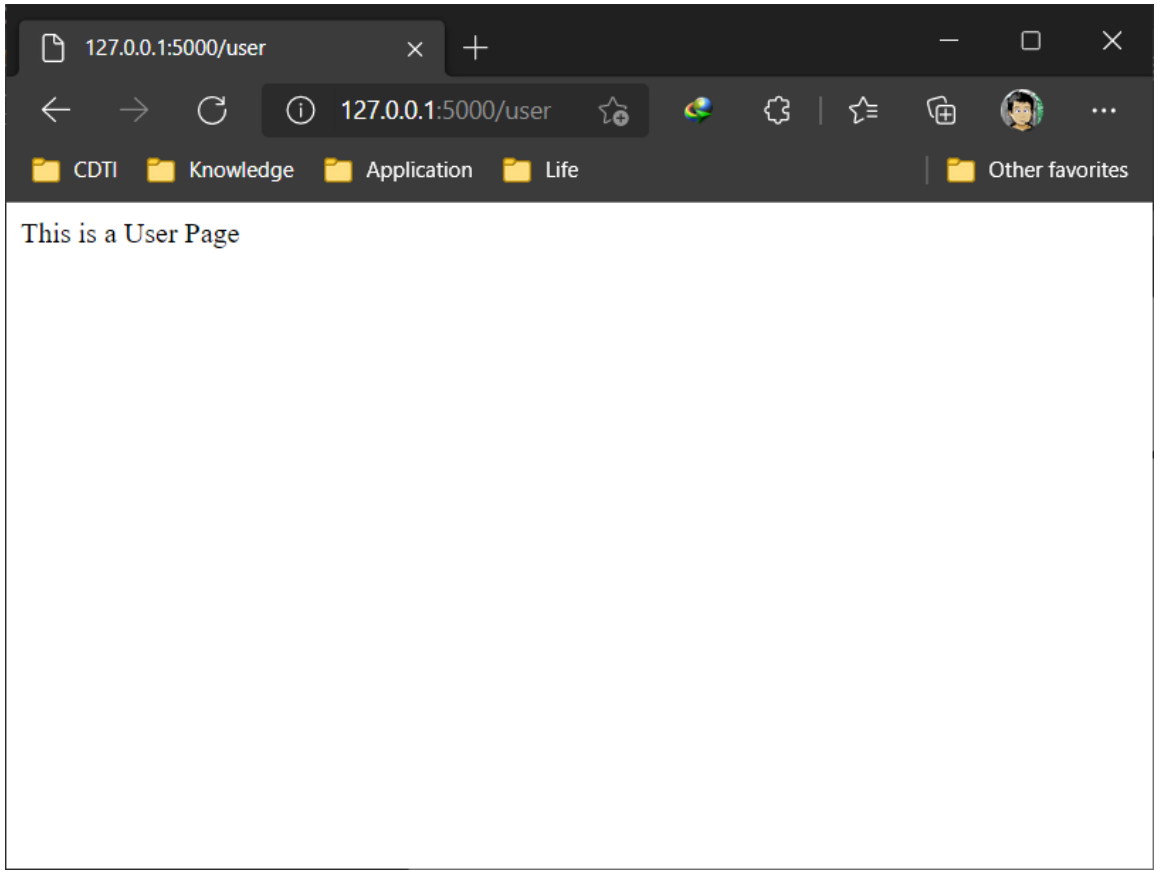
    app.run()

```

How to define [Route]



http://127.0.0.1:5000/



http://127.0.0.1:5000/user

Route with Parameter

app.py > ...

```python

from flask import Flask
app = Flask(__name__)

@app.route('/')
def index():
    return "<h1>Hello Flask Framework</h1>"

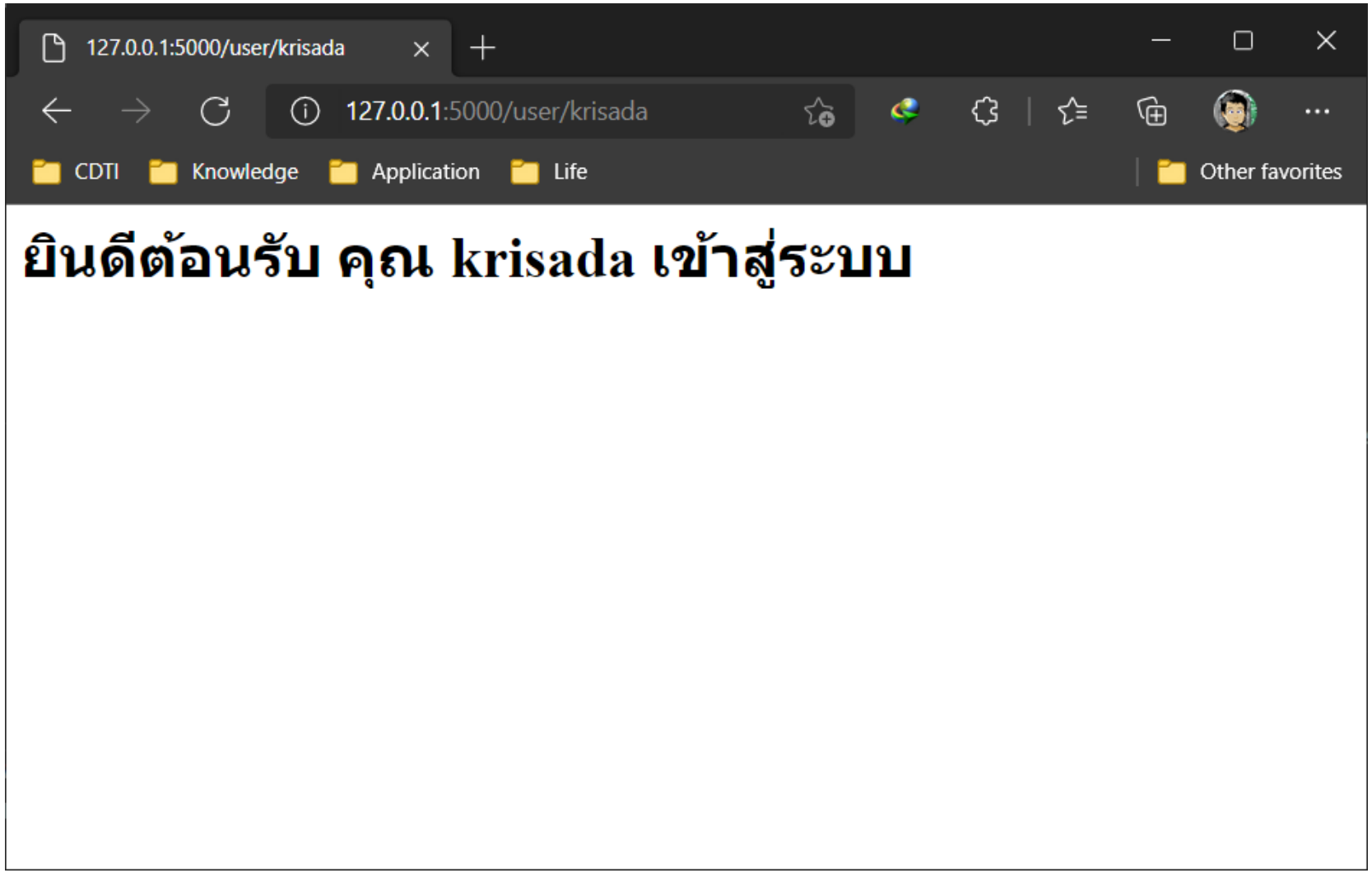
@app.route('/user/<name>')
def user(name):
    return "<h1>ยินดีต้อนรับ คุณ {} เข้าสู่ระบบ</h1>".format(name)

if __name__ == "__main__":

    app.run()

```

Route with Parameter



Route with Parameter

```python
@app.route('/user/<name>/<ID>')
def user(name, ID):
    return f"This is user page for {name} with ID {ID}"
```

ใช้ F-string ก็ได้
(formatted string literals)

app.py > ...

```python

from flask import Flask
app = Flask(__name__)

@app.route('/')
def index():
    return "<h1>Hello Flask Framework</h1>"

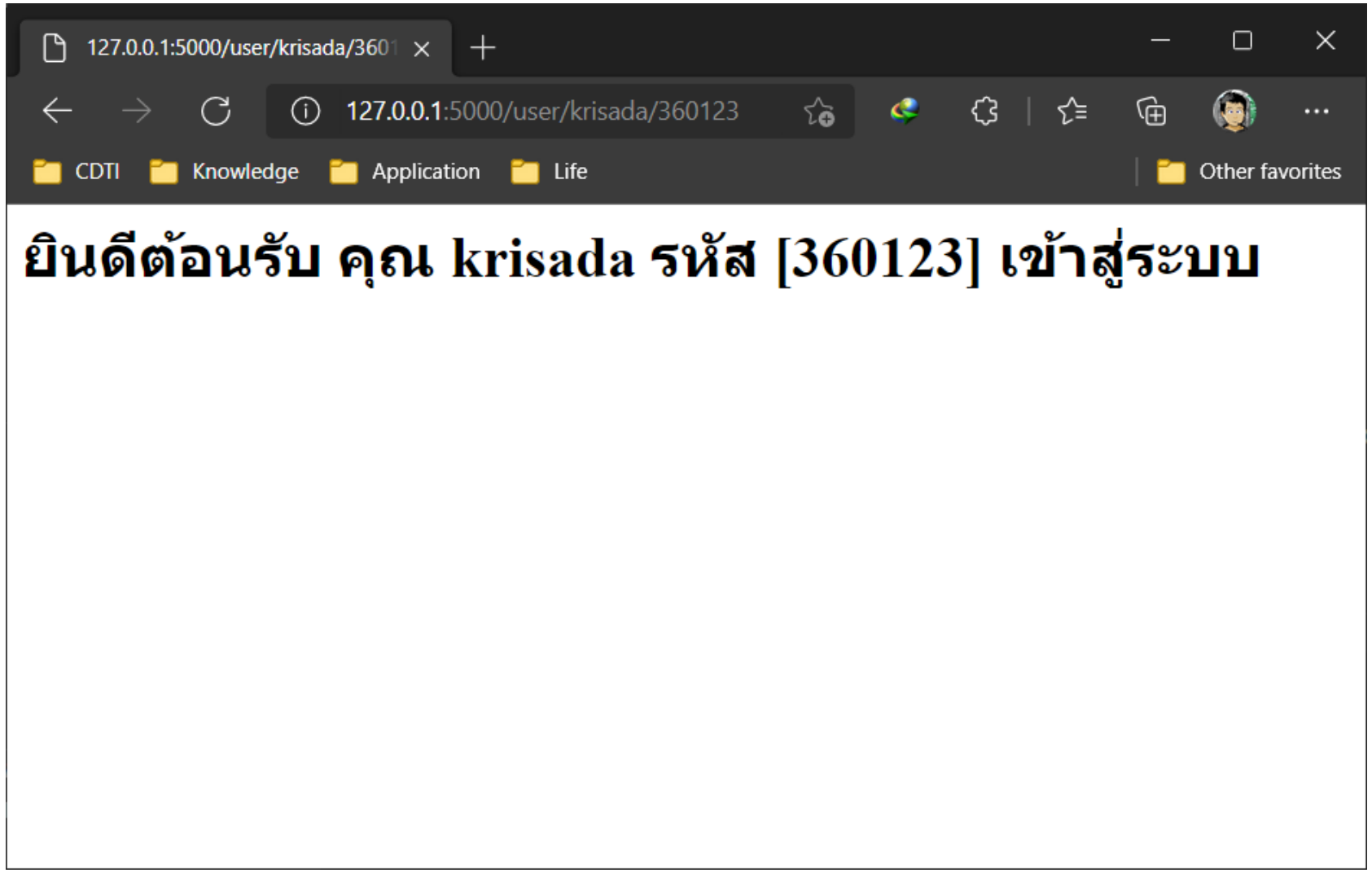
@app.route('/user/<name>/<ID>')
def user(name, ID):
    return "<h1>ยินดีต้อนรับ คุณ {} รหัส [{}] เข้าสู่ระบบ</h1>".format(name, ID)

if __name__ == "__main__":

    app.run()

```

Route with Parameter



Web Development with Debug Mode

app.py > ...

```python
from flask import Flask
app = Flask(__name__)

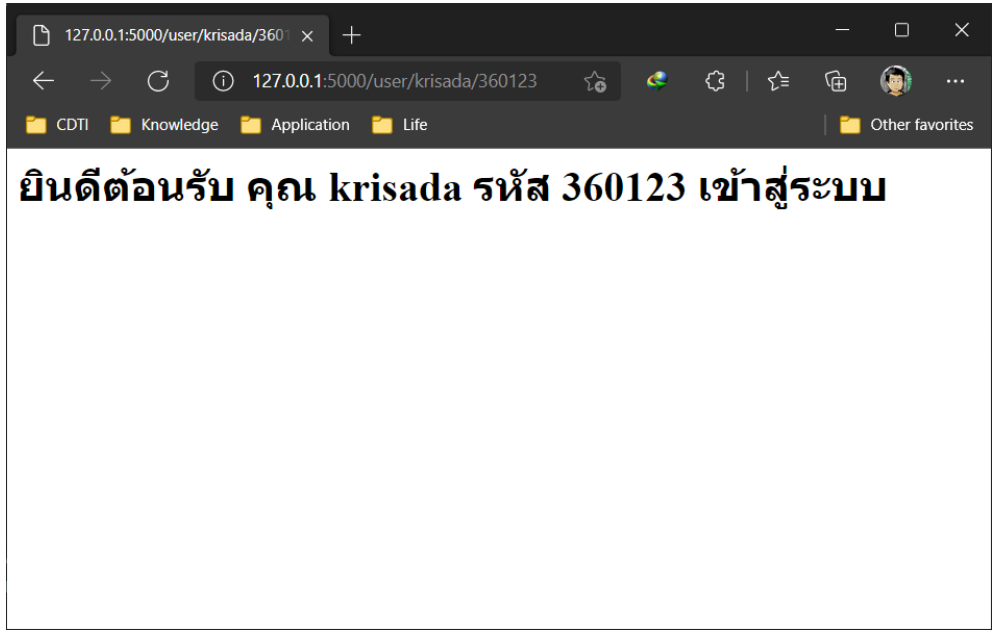
@app.route('/')
def index():
    return "<h1>Hello Flask Framework</h1>"

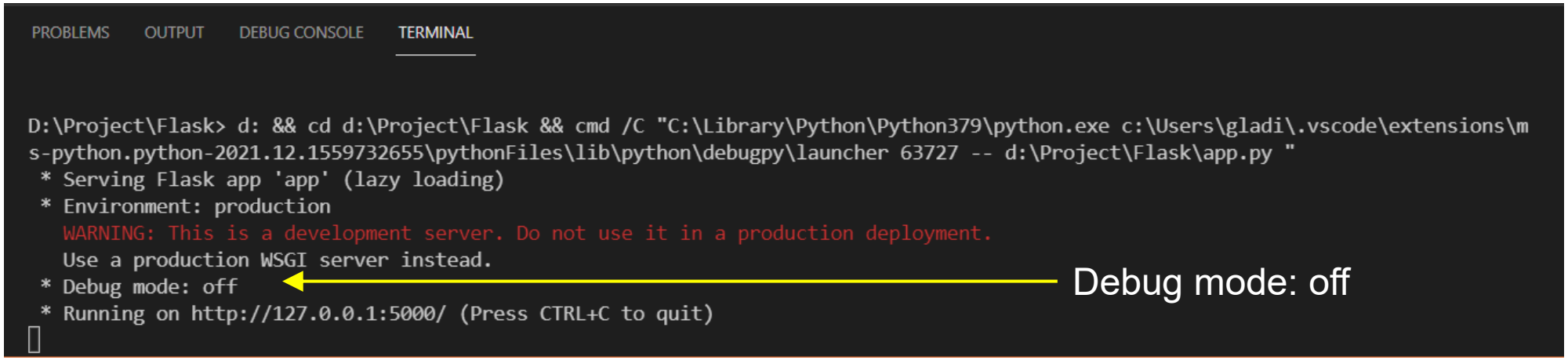
@app.route('/user/<name>/<ID>')
def user(name, ID):
    s1 = "คุณ {}".format(name)
    s2 = "รหัส {}".format(ID)
    s = "<h1>ยินดีต้อนรับ {} {} เข้าสู่ระบบ</h1>".format(s1,s2)
    return s

if __name__ == "__main__":

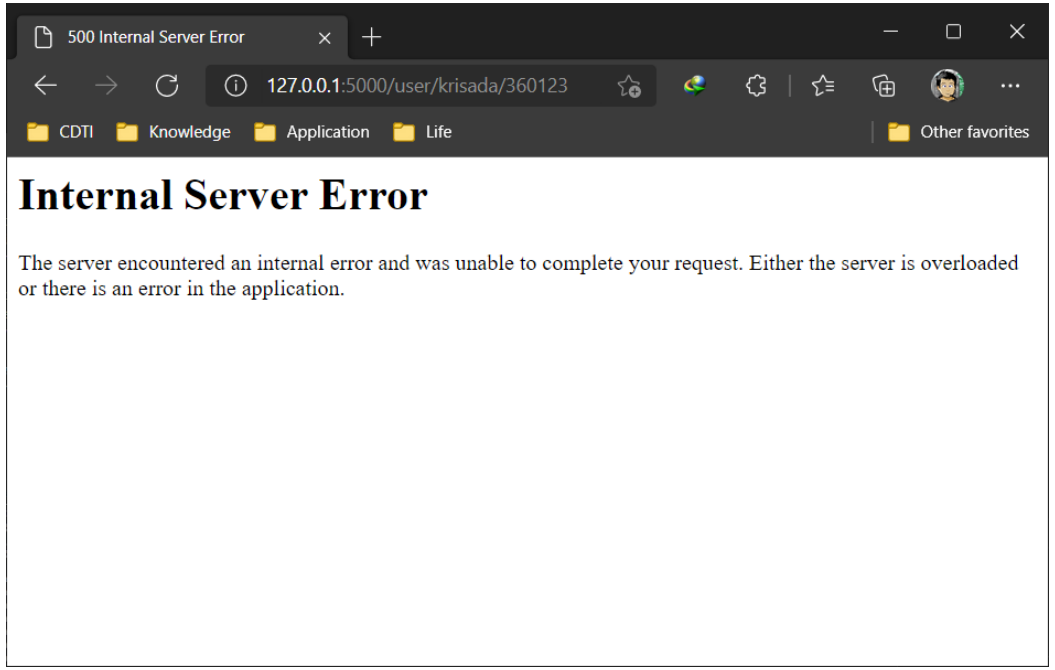
    app.run()

```





Web Development with Debug Mode



We cannot find what wrong in our code.

app.py > ...

```python
from flask import Flask
app = Flask(__name__)

@app.route('/')
def index():
    return "<h1>Hello Flask Framework</h1>"

@app.route('/user/<name>/<ID>')
def user(name, ID):
    s1 = "คุณ {}".format(name)
    s2 = "รหัส {}".format(ID[100])
    s = "<h1>ยินดีต้อนรับ {} {} เข้าสู่ระบบ</h1>".format(s1,s2)
    return s

if __name__ == "__main__":

    app.run()
```

What will happened if our code has an error?

Web Development with Debu

app.py > ...

```python
from flask import Flask
app = Flask(__name__)

@app.route('/')
def index():
    return "<h1>Hello Flask Framework</h1>"

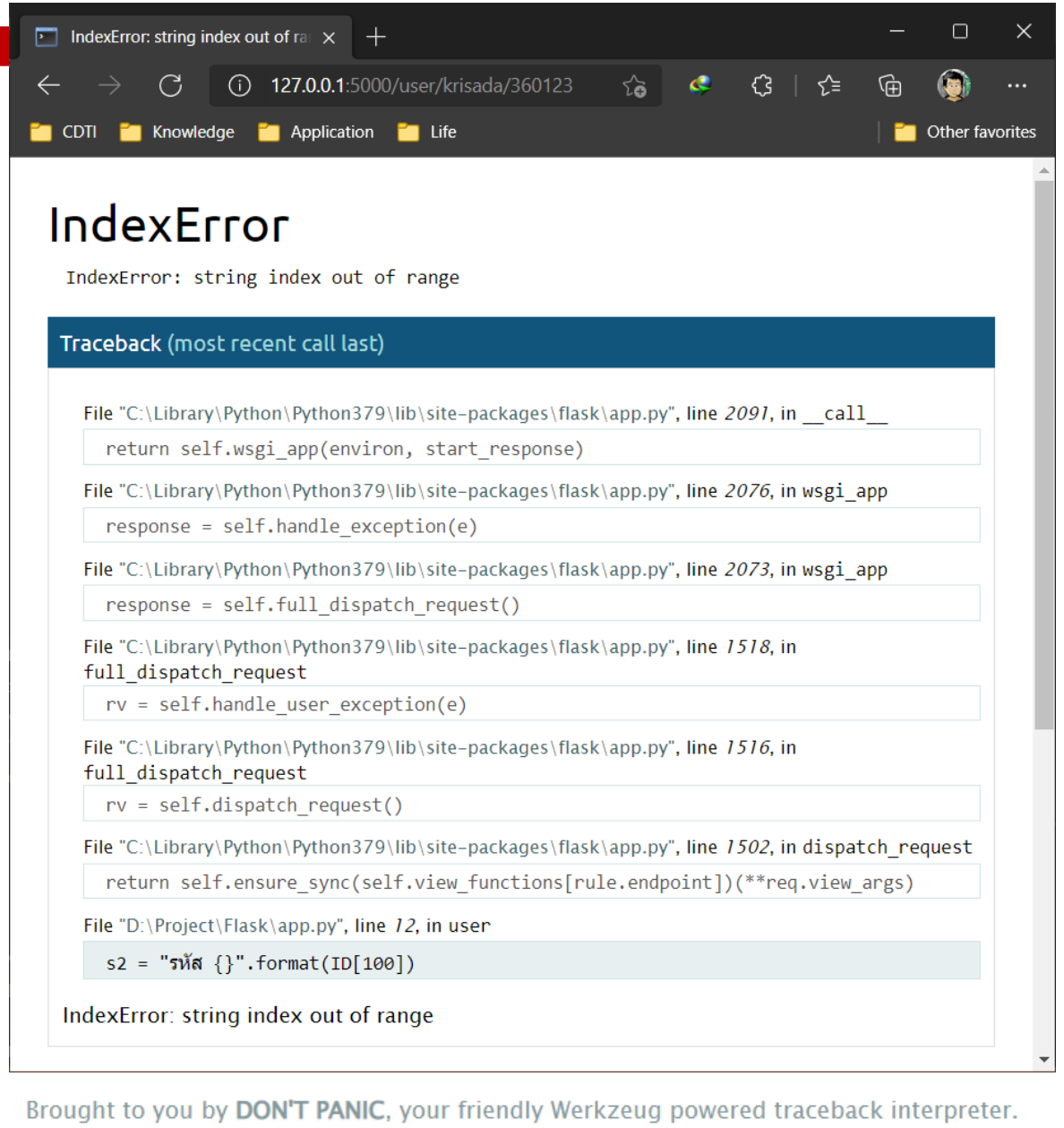
@app.route('/user/<name>/<ID>')
def user(name, ID):
    s1 = "คุณ {}".format(name)
    s2 = "รหัส {}".format(ID[100])
    s = "<h1>ยินดีต้อนรับ {} {} เข้าสู่ระบบ</h1>".format(s1,s2)
    return s

if __name__ == "__main__":

    app.run(debug=True)

```

Enable [Debug mode]



HTTP - Status Codes

S.N.	Code and Description
1	**1xx: Informational** It means the request has been received and the process is continuing.
2	**2xx: Success** It means the action was successfully received, understood, and accepted.
3	**3xx: Redirection** It means further action must be taken in order to complete the request.
4	**4xx: Client Error** It means the request contains incorrect syntax or cannot be fulfilled.
5	**5xx: Server Error** It means the server failed to fulfill an apparently valid request.

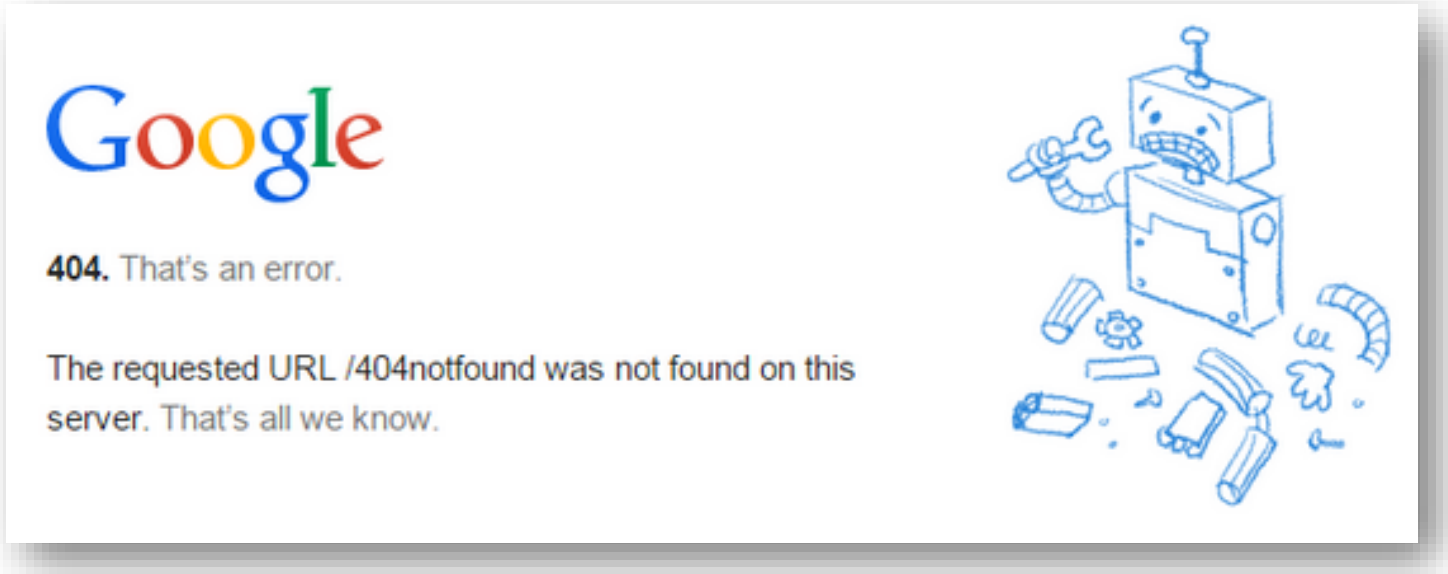
HTTP - Status Codes

200 OK

400 Bad Request

404 Not Found

500 Internal Server Error



https://www.tutorialspoint.com/http/http_status_codes.htm

Template with Jinja







Jinja is a template engine for Python.

A template engine or template processor is a library designed to combine templates with a data model to produce documents. Templates engines are often used to generate large amounts of emails, in source code preprocessing, or producing dynamic HTML pages.



หยุดโค้งก่อน ห้ามเดินผ่านตรงกลาง ให้เดินชิดซ้าย-ขวา

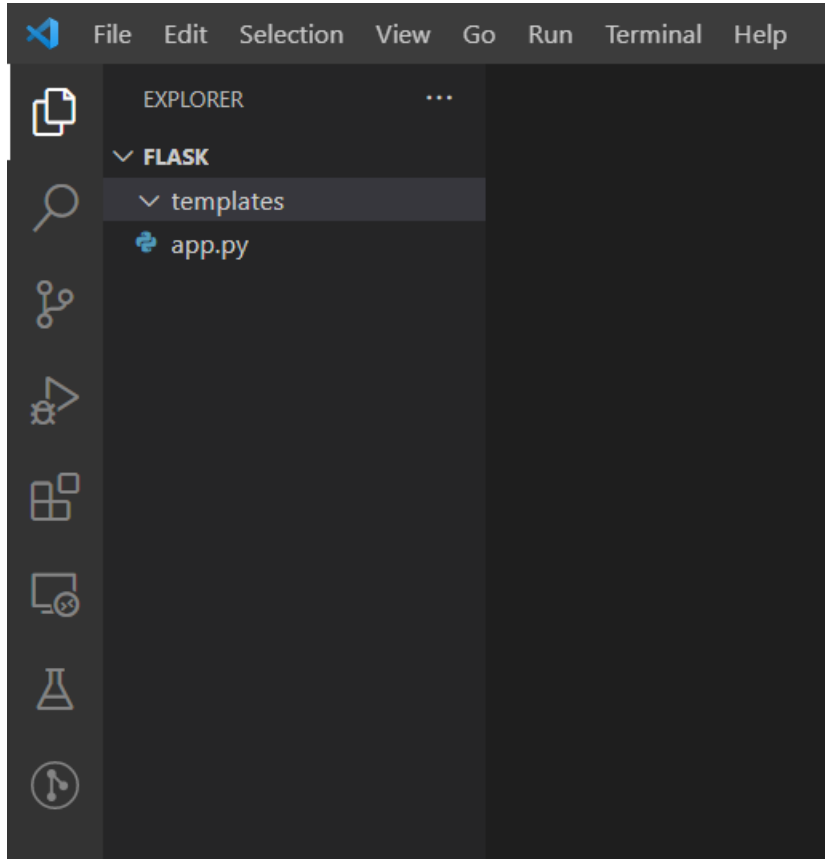
Jinja (**神社**) => Shinto shrines => ศาลเจ้า

Render

Now, we can insert python source code to HTML.

Template with Jinja

Use [render_template] to render html file with python



Create a folder name “templates”

(.html files)

app.py admin.html about.html index.html

app.py > ...

```python
from flask import Flask, render_template
app = Flask(__name__)

@app.route('/')
def index():
    return render_template("index.html")

@app.route('/about')
def about():
    return render_template("about.html")

@app.route('/admin')
def admin():
    return render_template("admin.html")

if __name__ == "__main__":

    app.run(debug=True)
```

Template with Jinja

Follow my guide then…Write a html code

Create 3 links to go to home / about / admin page

(5 min)

Template with Jinja

How to send <data> to show in <template>

Dynamic website

Data…from database / python process / etc.…

Variable in Python

```
13  @app.route('/admin')
14  def admin():
15      name = "krisada"
16      return render_template("admin.html", myname = name)
17
```

Parameter name for html file

Template with Jinja

Jinja syntax

Display

{{ Parameter }}

Flow Control

{% %}

app.py admin.html

templates > admin.html > html

```
1  <!DOCTYPE html>
2  <html lang="en">
3      <head>
4          <meta charset="UTF-8">
5          <meta name="viewport" content="width=device.width, initial-scale=1.0">
6          <title>Index Page</title>
7      </head>
8      <body>
9          <h1>หน้าสำหรับ Admin</h1>
10         <p>{{myname}}</p>
11         <ul>
12             <li><a href="/">Home</li>
13             <li><a href="/about">About</li>
14             <li><a href="/admin">Admin</li>
15         </ul>
16     </body>
17
18 </html>
```

Template with Jinja

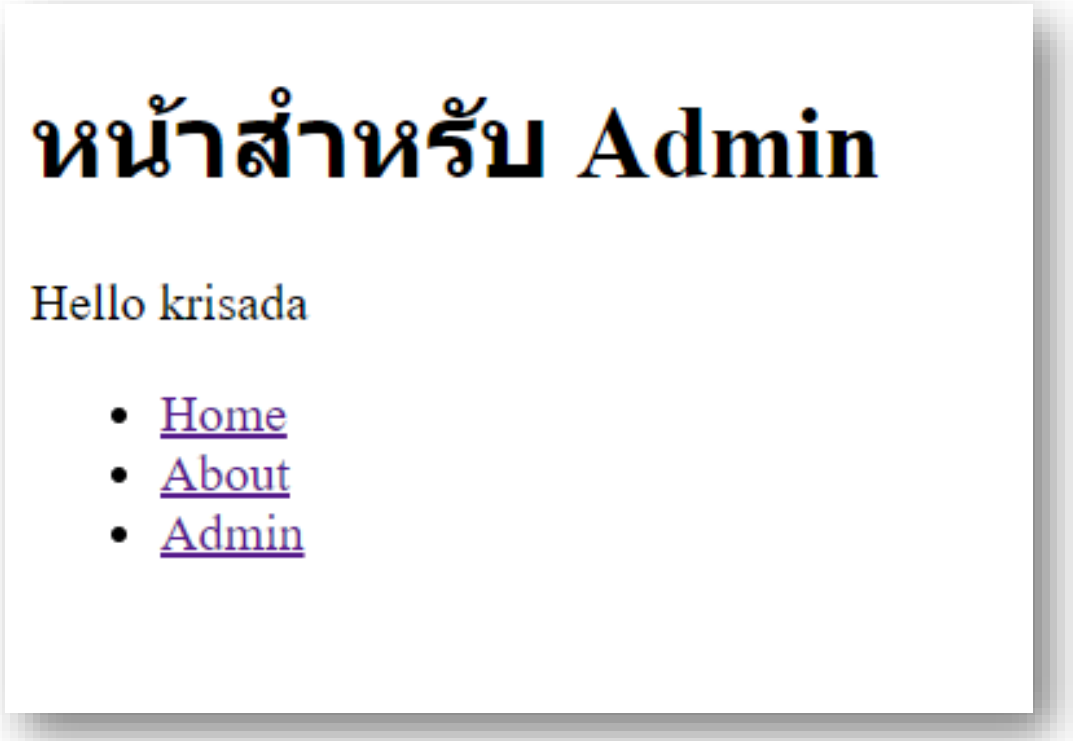
Jinja syntax

Display

{{ Parameter }}

Flow Control

{% %}



Template with Jinja

```python
@app.route('/')
def index():
    data = {"name":"Oud", "height":175, "blood":"B"}
    return render_template("index.html", mydata = data)

```

Jinja syntax

Next, we are going to send a group of data (**dictionary**) to display in Jinja

Key / Value

app.py | index.html

templates > index.html > html

```html
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device.width, initial-scale=1.0">
        <title>Index Page</title>
    </head>
    <body>
        <h1>ยินดีต้อนรับ</h1>
        <p>สวัสดีครับ</p>
        {{mydata}}<br>
        {{mydata.name}}<br>
        {{mydata.height}}<br>
        {{mydata.blood}}<br>
        <ul>
            <li><a href="/">Home</li>
            <li><a href="/about">About</li>
            <li><a href="/admin">Admin</li>
        </ul>
    </body>
</html>
```

Template with Jinja

Jinja syntax : Flow Control (IF - Else)

```
{% if condition %}
        Do this if condition is True
{% else %}
        Do this if condition is False
{% endif %}
```

Template with Jinja

```python
@app.route('/about')
def about():
    user = "K"
    return render_template("about.html", username = user)

```

templates > about.html > html

```html
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device.width, initial-scale=1.0">
        <title>Index Page</title>
    </head>
    <body>
        <h1>นี่คือหน้า About</h1>

        {% if username == "K" %}
            <h2>ยินดีต้อนรับคุณ {{username}} เข้าสู่ระบบ</h2>
        {% else %}
            <h2><a href="/">เราไม่รู้จักคุณ กรุณากดเพื่อย้อนกลับ</a></h2>
        {% endif %}

        <img src="../static/picture3.jpg" alt="Pic3">
        <ul>
            <li><a href="/">Home</li>
            <li><a href="/about">About</li>
            <li><a href="/admin">Admin</li>
        </ul>
    </body>
</html>
```

Try to change [user]

and see the difference result

Template with Jinja

Jinja syntax : Flow Control (For Loop)

```
{% for item in List %}
        Do this for each item
{% endfor %}
```

Template with Jinja

```python
@app.route('/about')
def about():
    animals = ["Cat","Dog","Rabbit"]
    return render_template("about.html", mypet = animals)

```

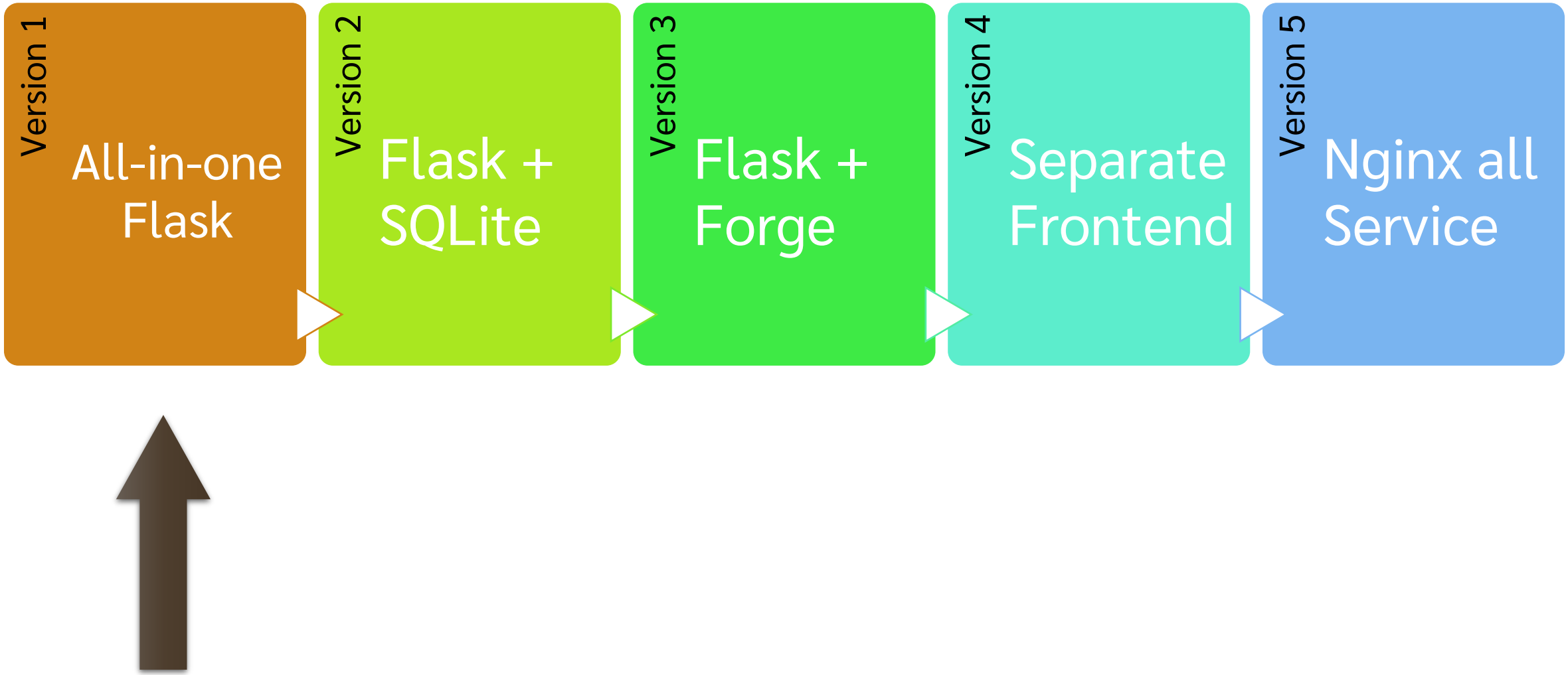
templates > about.html > html > body

```html
<!DOCTYPE html>
<html lang="en">
    <head>
        <meta charset="UTF-8">
        <meta name="viewport" content="width=device.width, initial-scale=1.0">
        <title>Index Page</title>
    </head>
    <body>
        <h1>นี่คือหน้า About</h1>

        <ul>
            {% for item in mypet %}
                <li>my pet: {{item}}</li>
            {% endfor %}
        </ul>

        <img src="../static/picture3.jpg" alt="Pic3">
        <ul>
            <li><a href="/">Home</li>
            <li><a href="/about">About</li>
            <li><a href="/admin">Admin</li>
        </ul>
    </body>
</html>
```

Let's do a project together :)


Version 1
All-in-one Flask
Version 2
Flask + SQLite
Version 3
Flask + Forge
Version 4
Separate Frontend
Version 5
Nginx all Service